Operator Graphs for Protein Intelligence

A Mathematical Language for Energy, Probability, and Computation

Haobo Wang
Levinthal Biotech, Hangzhou, China

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Preface

Protein science, statistical physics, graphical modeling, and modern machine learning repeatedly use the same words for different mathematical objects. A coupling may mean a term in a joint Hamiltonian, a conditional dependence parameter, a message map, or an experimentally measured epistatic effect. A graph may encode model factorization, computational communication, an operator-valued response, or physical contact. Softmax may normalize configurations, residue identities, source positions, memories, or latent edge states. The formulas can look nearly identical while their scientific claims remain incompatible.

This book develops a typed mathematical language for navigating those translations. Its central object is a reference-centered operator-valued graph: every sequence position carries a categorical contrast space, and every directed edge carries a map between such spaces. This construction places Potts couplings, protein-language-model responses, attention routes, latent memories, spectral summaries, and biological interventions in common coordinates without declaring them equivalent.

The purpose is not to attach physical language to every neural computation. It is to identify exactly what additional assumptions turn a score into an energy coordinate, a collection of conditionals into a joint law, a route into an interaction, a scalar network into an operator graph, and a predictive association into biological evidence. Where those assumptions fail, the book treats the failure as information rather than as a defect to conceal.

The material grew from a theory paper and retains theorem–proof precision, but its natural scope is broader. The translation dictionary, gauge geometry, static–dynamic decomposition, higher-order hierarchy, graph probability, model correspondences, spectral analysis, realizability constraints, and falsification program are parts of one framework. Presenting them as a book allows each layer to keep its proper meaning instead of forcing every distinction into the narrow arc of a single research claim.

How to read this book

Part I establishes the cross-disciplinary vocabulary and the hierarchy of claims. Part II is the theorem–proof core: it builds the operator geometry and develops static, dynamic, higher-order, and probability-exact edge objects. Part III has a mixed role. It contains exact model correspondences where their assumptions permit them, then uses the same coordinates as a structured audit of Potts models, Gaussian models, Hopfield systems, RBMs, Transformers, and the three generations of AlphaFold before turning to MSA estimators and graph spectra. In particular, the AlphaFold comparison is an architectural reading of published systems, not an equivalence theorem about their algorithms. Part IV asks which objects are realizable in trainable architectures and what evidence is required for statistical, physical, or biological interpretation.

Readers interested primarily in protein language models can begin with Chapters 1, 4, 7, 8, and 9. Readers coming from statistical physics may prefer Chapters 2, 3, 6, 7, 8, and 10. Readers designing sparse or interpretable architectures should read Chapters 3, 4, 8, 11, and 12 before treating an internal route as an identified interaction.

The front-matter object map gives a lighter first route through the notation. On a first reading, the opening prose, boxed equations, worked cases, and codas carry the conceptual argument. Proofs and mathematical interludes can be read on a second pass; they establish the exact boundaries of the claims rather than introducing a separate storyline.

Objects and Notation

The notation in this book is easiest to read as a sequence of typed transformations. Data do not begin as an interaction graph, and a spectrum is not computed until an edge object has been declared. The main path is

$$\boxed{\text{observations} \longrightarrow \text{estimator} \longrightarrow \text{typed edge operator} \longrightarrow \text{declared compression} \longrightarrow \text{spectrum or experiment.}}$$

Every arrow can lose information. The book repeatedly asks what survives each arrow and what new assumption would be required to reverse its interpretation.

The recurring objects

Symbol	Type	Question answered
$X = (X_1, \ldots, X_L)$	sequence in $[q]^L$	Which residue identity occurs at each position?
p_i	reference distribution on the alphabet at site i	Relative to which background are categorical contrasts centered?
$\mathcal{H}_i^0$	$(q-1)$-dimensional centered categorical fiber	Which source and target perturbations count as genuine contrasts?
P_i	projection onto $\mathcal{H}_i^0$	Which constant and one-body directions are removed from a table?
$G_{i\leftarrow j}$	reference-centered $q \times q$ kernel	Which source-category contrasts are coupled to which target-category contrasts?
$\mathcal{K}_{i\leftarrow j}$	map $\mathcal{H}_j^0 \to \mathcal{H}_i^0$ induced by $G_{i\leftarrow j}$	How does a centered contrast at j act on centered predictions at i?
$\mathbb{K}$	block operator on $\bigoplus_i \mathcal{H}_i^0$	How are all local maps assembled without scalarizing them?
$G_{i\leftarrow j}(C)$	categorical kernel conditional on background C	Does the effective relation change with the surrounding sequence?
$\overline{G}, \delta G$	mean operator and context-dependent residual	What is the best static approximation, and what does it leave unexplained?
Q_{ij}	pair probability law with declared marginals	Does a finite normalized pair law realize the local score geometry?
A, L_G	scalar shadow and a declared graph Laplacian	Which positional organization remains after categorical information is compressed?
ϵ_{ij}	projected experimental epistasis table	Does a matched intervention support the predicted categorical relation?

The same letter can appear with a superscript identifying its origin, such as G^{P} for a Potts estimate or G^{pLM} for a language-model response. An arrow $j \to i$ uses j as source and i as target.

Static pair energies may eventually identify the two directions, but directed responses are kept separate until reciprocity has been tested.

The distinction between G and $\mathcal{K}$ is representational, not a second scientific claim. In the declared categorical bases, $G_{i\leftarrow j}$ is the matrix kernel of $\mathcal{K}_{i\leftarrow j}$. The calligraphic symbol emphasizes the typed input and output; the plain capital emphasizes the entries that are estimated, projected, plotted, or compressed.

How the notation carries type

The typography follows a small grammar. It does not make every symbol globally unique, but it narrows the possibilities before an equation is read in detail.

Form	Usual role	Examples
Lower-case italic	indices, states, scalar functions, or vectors	i, j, a, b, t; f, u, v
Plain capital	finite table, matrix, random object, or model state	T, J, G, A, P, X
Calligraphic capital	space, set, typed map, or functional	$\mathcal{H}, \mathcal{U}, \mathcal{K}, \mathcal{E}$
Blackboard/fraktur capital	assembled system or expectation operator	$\mathbb{K}, \mathfrak{G}, \mathbb{E}$
Arrow and indices	source, target, and level of action	$G_{i\leftarrow j}$ maps source j toward target i

Decorations have stable meanings throughout the main construction:

$$\widehat{G}: \text{ sample estimate,} \qquad \overline{G}: \text{ declared background average,}$$
$$\delta G: \ G - \overline{G}, \qquad (\cdot)^*: \text{ population target or adjoint, as stated locally.}$$

Superscripts such as P, pLM, stat, and dyn name origin or role; they do not change the domain and codomain unless the text explicitly says so.

Four overloaded letters remain because they are standard in the literatures being compared. Their syntax disambiguates them. L is sequence length in $[q]^L$ and an index bound, whereas L_G or a subscripted $L_{\mathrm{sym}}, L_{\mathrm{rw}}$ is a graph Laplacian. P_i is categorical centering, bare P in the routing chapters is a row-stochastic matrix, and $P(\cdot)$ denotes a probability. Q_{ij} is a pair law, while Q_i is the Euclidean form of a categorical projector; nuisance residualizers are written R rather than adding another meaning to Q. Finally, bare G can denote a generic graph in a quoted graph-network formalism, while an indexed $G_{i\leftarrow j}$ is always a categorical kernel in this book.

Three spaces that should not be merged

The book moves among three geometries:

Geometry	Native object	Typical operation
Observation space	MSA rows, positions, and residue categories	weighting, residualization, covariance estimation
Categorical operator space	centered contrasts and maps between local fibers	gauge projection, reference transport, response audit
Position-graph space	one scalar or transported vector state per residue position	Laplacian spectrum, heat filtering, wavelets

An eigenvector across MSA rows, a categorical constant mode, and an eigenvector across residue positions are therefore different objects even when all three are called modes.

A first and a second reading

The main argument can be followed from the opening prose of each chapter, the boxed equations, the worked cases, and the codas. The theorem and proof blocks establish exactness but need not be read on the first pass. Long derivations labeled as mathematical interludes answer a narrower question and can likewise be deferred without losing the book's conceptual sequence.

On a second reading, each proposed correspondence can be checked with the same ledger:

$$\text{sample space}, \quad \text{normalization axis}, \quad \text{available context},$$
$$\text{gauge or symmetry}, \quad \text{state dynamics}, \quad \text{estimator}, \quad \text{validation endpoint}.$$

These are not seven hurdles that every useful model must clear. They are seven coordinates that prevent a valid statement in one space from being silently promoted into a stronger statement in another.

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Part I

Language and Foundations

Chapter 1

One Edge, Several Meanings

Consider two positions, i and j, in a protein family. A structural biologist may draw a line between them because their atoms lie within a chosen distance. A statistical physicist may fit a Potts model and attach a $q \times q$ coupling table to the same pair. A Transformer may send a large attention weight from j to i. A model audit may find that substituting the residue at j changes the predicted distribution at i. A double-mutant experiment may report nonadditive fitness. Each result is naturally drawn as an edge. The pictures look alike, and that visual agreement invites a stronger conclusion: perhaps all five procedures have found the same interaction.

They have not—at least not yet. The structural line is defined by geometry. The Potts table is a component of a probability model for an ensemble of homologous sequences. Attention is a route inside a computation. The substitution response is a directed input–output property of a fitted function. Experimental epistasis is a statement about an intervention and a measured biological endpoint. Agreement among these objects can be scientifically important. Shared indices (i, j) provide the comparison point; the carried objects and validation criteria determine the meaning.

This book begins with that apparently minor distinction because much of the conceptual difficulty in protein machine learning is concentrated inside it. Words such as *interaction*, *energy*, *memory*, *direct*, and *graph* travel easily between statistical physics and neural computation. The equations often travel with them, while their assumptions must be carried explicitly.

1.1 A residue pair enters five descriptions

Let $X = (X_1, \ldots, X_L)$ denote a protein sequence over an alphabet of size q. Suppose that a multiple-sequence alignment, a trained protein language model, a three-dimensional structure, and a mutational assay are all available for one family. For the pair (i, j) they provide five candidate edge objects:

The entries differ along several axes at once. Some are scalar and some are categorical tables. Some are symmetric and some directed. Some are fixed across a family and some depend on the input sequence. Some belong to a joint probability law; others belong to an algorithm or an assay. Even their zeros mean different things. A zero Potts block may express conditional factorization within a fitted model, a zero route may mean skipped computation, and a missing structural contact may only mean that one distance threshold was not crossed.

The first principle of the book is therefore simple:

$$\boxed{\text{shared endpoints do not imply a shared edge semantics.}} \tag{1.1}$$

This principle sets the standard for unification: a successful comparison must preserve the relevant

Description	Edge object	Question answered
Structure	contact indicator or geometric relation	Are the two sites close under a declared structural criterion?
Potts model	coupling table $J_{ij}(a, b)$	Which pair term helps a fitted family distribution explain sequence variation?
Attention	route coefficient $\alpha_{ij}^{(h,\ell)}(x)$	How much source-normalized weight does one head assign to this computational route?
Model response	categorical map $G_{i\leftarrow j}(C)$	How does a substitution at j change the model prediction at i in background C?
Experiment	epistasis or another intervention response	Does the joint perturbation differ from the declared additive biological null?

Figure 1.1: The same endpoints support five different edge semantics. Agreement must be established after the carried object and validation question have been declared.

types. If two edge objects are to be compared, their state spaces, normalization, conditioning information, invariances, dynamics, and validation criteria must remain visible during the comparison.

The same caution applies to familiar formulas and familiar adjectives. Softmax can normalize residue identities, source positions, memories, or complete configurations. The normalization axis determines which probability law the shared algebra defines. Likewise, an edge may be called *direct* because it is a one-step computational route, a term in a declared factorization, a spatial contact, or an experimental effect that survives a lower-order null. Chapter 2 separates these meanings systematically. Here it is enough to retain one reading rule: before interpreting a formula or an edge, name the space in which it lives and the alternative explanations that have actually been excluded.

1.2 From an edge weight to an edge operator

For categorical sequences, a scalar edge weight is a compression of a richer object. If residue b at site j is replaced by b', the effect at site i is a change across all possible target identities a. A pair relation is therefore naturally represented by a table or operator

$$G_{i\leftarrow j}(a, b), \tag{1.2}$$

mapping source substitutions to target contrasts.

This operator contains information that every scalar compression discards. Two edges can have the same Frobenius norm while favoring different amino-acid exchanges. They can have the same support while acting in different singular directions. They can generate the same contact score but make opposite predictions for a particular mutation. A graph of scalar strengths is useful, but it is a shadow of the typed relation carried by the edge.

Raw categorical tables introduce another problem. A constant term, a function of a alone, and a function of b alone can be moved between one-body and pair terms without changing the modeled law. The pair object is therefore an equivalence class until a reference geometry is declared. The reference-centered operator developed later in the book removes those lower-order directions and retains the genuinely bivariate action. Gauge fixing supplies the identified coordinates that make edge operators comparable across models and references.

1.3 One family graph, many sequence responses

The contrast between a Potts model and a protein language model supplies the running mathematical problem. In an additive pairwise Potts model, a fixed gauged block describes the response of site i to site j throughout the family model. A nonlinear protein language model generally produces a different effective response in each sequence background C:

$$G_{i\leftarrow j}(C) = \overline{G}_{i\leftarrow j} + \delta G_{i\leftarrow j}(C). \tag{1.3}$$

The mean $\overline{G}_{i\leftarrow j}$ is the best static approximation in a declared reference geometry. The residual $\delta G_{i\leftarrow j}(C)$ measures context dependence. Under a fixed product reference, its variance can be identified with higher-order functional components that contain the audited source and take values in the target contrast fiber. The target category is an output coordinate, not an assumed input to the masked model. Under correlated empirical backgrounds, additional cross terms appear and the interpretation must change.

This decomposition separates two questions that are often merged. Does the model contain a stable family-like pair response? How much of its behavior depends on the surrounding sequence? A model can answer both strongly, one strongly, or neither. Calling the entire response a "learned Potts coupling" erases the distinction.

Directionality adds another layer. The response from j to i and the response from i to j are separately measurable. They define a joint pair energy only if they satisfy a reciprocity condition after reference transport. A directed response graph is therefore a valid primary object. Symmetrization becomes a later operation, justified after its asymmetry has been measured.

1.4 The coordinate system to come

The rest of the book builds a mathematical coordinate system in which these comparisons can be made without flattening them. Each model will be described by the variables carried at its nodes, the support and type of its edges, the context available to an edge, the represented interaction order, the updated state, the normalization or conserved quantity, the estimation procedure, and the external endpoint used for validation.

The construction proceeds from local meaning to global claims. We first establish a translation language across physics, probability, and computation. We then define reference-centered categorical fibers and the operators between them. Static and dynamic responses lead to higher-order decompositions, probability-exact edge geometry, and compatibility conditions for joint energies. Potts, Gaussian, Hopfield, RBM, and Transformer models are subsequently placed in the same coordinates. Only after the edge object has been declared do we turn to estimation, spectra, architectural realizability, and biological evidence.

The ambition is to make differences between protein graphs mathematically explicit enough that genuine agreement becomes meaningful. The edge between i and j begins a sequence of questions: an edge in which space, carrying what object, conditioned on what information, invariant under which changes, and validated against which world?

Chapter 2

When Equations Change Their Meaning

Mathematical notation creates a powerful illusion of continuity. Write an exponential, divide by a sum, and a probability simplex appears. Write a quadratic form and one is tempted to call it an energy. Place coefficients between indexed objects and a graph seems to have been defined. These moves are algebraically legitimate. The difficulty begins when the semantic content of the symbols is allowed to travel farther than the derivation.

Cross-disciplinary work depends on such travel. Statistical physics, graphical modeling, and machine learning would remain isolated if every familiar operation had to be renamed in each field. The aim is to preserve enough type information for analogies and equivalence claims to remain distinguishable.

This chapter develops that discipline. The central rule is that an equation is interpreted only after its random variable, sample space, conditioning information, invariance class, and validation claim have been named. The same numerical kernel can then connect several theories without collapsing them into one.

2.1 The sample space comes before the probability

A probability vector is incomplete until we know what its coordinates enumerate. Consider

$$\mathrm{softmax}_u(s)_u = \frac{e^{s_u}}{\sum_v e^{s_v}}. \tag{2.1}$$

The expression guarantees positivity and unit mass. The meaning of that mass comes from whether u denotes a residue identity, a source position, a stored memory, an edge type, or an entire protein configuration. Each choice produces a different random object.

For a Potts conditional, $u = a \in [q]$ and the simplex describes the possible identities of one residue given the remaining sequence. For attention, $u = j \in [L]$ and the simplex distributes one unit of routing mass over source positions for a fixed target and head. In a modern Hopfield update, $u = \mu \in [M]$ indexes memories that compete for retrieval. In a latent relation model, $u = z$ may index the possible states of one candidate edge. A globally normalized Gibbs law is different again: its index ranges over complete configurations, often an exponentially large set.

These distributions can share software and calculus while retaining different normalization constraints. Source-softmax makes positions compete. Edge-state softmax resolves uncertainty about one relation and acquires a degree budget only when edges are coupled. A site conditional is normalized for each context; compatibility with one joint distribution is a further property to be established.

The normalization axis is consequently part of the mathematical type. Omitting it is comparable to omitting the units from a physical measurement: the remaining number can still be manipulated, but its comparisons become unsafe.

2.2 The exact content of the Gibbs analogy

Once a finite state space $\mathcal{Z}$ has been declared, softmax possesses an exact variational meaning. Let r_z be a score and let $E_z = -r_z$. At temperature $\tau > 0$,

$$p_z = \frac{e^{-E_z/\tau}}{\sum_{z'} e^{-E_{z'}/\tau}}. \tag{2.2}$$

The associated free energy satisfies

$$-\tau \log \sum_z e^{-E_z/\tau} = \min_{q \in \Delta(\mathcal{Z})} \left\{ \sum_z q_z E_z - \tau H(q) \right\}. \tag{2.3}$$

This is more than a metaphor. The distribution exactly balances expected energy against entropy. As τ decreases it concentrates on score maxima; as τ increases it spreads mass across alternatives. Adding the same constant to every E_z changes the free energy by that constant but leaves p unchanged, so only energy differences are observable within this coordinate.

The identity gives the exact geometry of a declared categorical choice. A physical reading adds further structure: calibrated energy units, a thermodynamic temperature, an equilibrium ensemble, and compatibility with a global Hamiltonian. Learned logits and routing temperatures supply these features only when the corresponding model and calibration establish them.

This distinction is especially sharp for graphs. Suppose each candidate edge e has a latent state z_e and an independent softmax posterior $q_e(z_e \mid x)$. The resulting law is

$$Q(G \mid x) = \prod_{e \in E_c} q_e(z_e \mid x). \tag{2.4}$$

Every factor has a local Gibbs representation. The product is a factorized graph distribution; shared degree budgets, reciprocal constraints, loop energies, connectivity, geometric exclusion, and chemical consistency require additional coupled factors or a restricted graph space.

A coupled graph ensemble instead has one normalization over admissible graphs:

$$P(G \mid x) = \frac{\exp[-\mathcal{H}_{\text{graph}}(G; x)/\tau]}{\sum_{G'} \exp[-\mathcal{H}_{\text{graph}}(G'; x)/\tau]}. \tag{2.5}$$

The two laws coincide only when the graph energy separates over edges and the admissible set adds no global coupling. Thus “each edge is Gibbs distributed” and “the graph is a Gibbs system” are different statements.

2.3 Four uses of interaction

The word *interaction* can now be separated by the object whose nonadditivity is at issue.

Computational interaction. Changing one node can alter another through a route, message, attention coefficient, residual path, or nonzero network response. This is a property of an implemented function and its execution graph.

Statistical interaction. A joint or conditional function contains a component outside the span of the declared lower-order terms. The statement depends on a reference measure, a projection or factorization, and an interaction order.

Physical interaction. A term belongs to a justified Hamiltonian or effective energy under an explicit coarse graining, symmetry class, and regime of validity. Statistical fit alone does not supply those assumptions.

Biological interaction. A joint perturbation produces a reproducible effect on structure, fitness, activity, or another biological endpoint relative to a declared null. This is an experimental statement and inherits the scope of the assay.

The four meanings can support one another without forming a chain of automatic implications. A computational route can help predict a statistical coupling. A fitted coupling can correlate with contact geometry. A contact can constrain mutational fitness. Each transition is an empirical result with its own failure modes.

2.4 Four graphs on the same protein

The corresponding graphs must also be typed.

The *route graph* records where a computation can read or send a message. In ordinary self-attention its support is usually the complete directed graph; what varies is the weight and the effective value-path operator. The *statistical support graph* records nonzero pair components in a declared probability or conditional model. The *operator graph* records the full categorical action attached to each directed pair. The *physical or biological graph* records relations defined by structural, energetic, kinetic, or intervention criteria.

These distinctions matter even when all graphs predict the same contact benchmark. Contact prediction compresses a rich pair object to a scalar label. Agreement at that level leaves substitution specificity, reciprocity, background dependence, and mediation as separate comparison questions. Agreement after scalar compression is evidence, but it is evidence about the compressed observable.

Topology adds information that no isolated edge contains. Support determines which effects are represented as direct and which require paths. Degree, components, cycles, bottlenecks, and spectral modes describe collective organization. Yet topology alone remains incomplete for categorical systems. Two graphs with identical support can carry different edge operators and make different mutational predictions. This is why the position graph will later appear as a shadow of a richer operator-valued object.

Static and dynamic graphs occupy different roles as well. A family-level interaction graph held fixed while homologous sequences are modeled resembles a quenched structure. A route graph that changes with sequence, layer, and head is an adaptive computational state. It is sometimes useful to compare the latter with an annealed graph, but input dependence alone does not produce an equilibrium measure over graph configurations.

2.5 A claim is a path with missing arrows

Cross-disciplinary overstatement often takes the form of a skipped arrow. A route is observed and called an interaction; a conditional score is observed and called a Hamiltonian; a contact correlation is observed and called a mechanism. A more durable argument makes the missing arrows explicit:

$$\begin{aligned}&\text{projected bivariate component} \xRightarrow[\text{fixed context}]{\text{endpoint exclusion}} \text{conditional response}\\ &\xRightarrow[\text{one joint law}]{\text{compatibility}} \text{distributional interaction} \xRightarrow[\text{matched endpoint}]{\text{independent intervention}} \text{biological interaction.}\end{aligned} \tag{2.6}$$

Each arrow names an additional assumption or experiment. The ladder preserves the value of every established level: a computational route can be useful before conditional semantics are known, and a statistical interaction can remain predictive while its microscopic interpretation is open.

This also changes how negative results are read. Failure of reciprocity may reject a joint Potts interpretation while revealing directed model behavior. Failure of background invariance may reject a static pair description while quantifying higher-order context. Failure to predict contacts may reject one structural interpretation without erasing an operator's role in sequence prediction.

2.6 The translation ledger

Before identifying two constructions, we will record the following coordinates:

$$\boxed{\begin{gathered}\text{object and sample space,}\quad \text{normalization axis,}\quad \text{conditioning information,}\\ \text{gauge or invariance class,}\quad \text{state dynamics,}\quad \text{estimator and validation endpoint.}\end{gathered}} \tag{2.7}$$

The ledger is deliberately asymmetric across fields. Translating a Hamiltonian into a score may discard physical assumptions; translating a score into a Hamiltonian requires adding them. A hidden feature can function as a memory without being a metastable thermodynamic state. A sparse execution graph can save computation without identifying conditional independence. A gauge-fixed pair table can isolate statistical nonadditivity without establishing biological causality.

This asymmetry is the price of honest unification. The common language developed in the next part will be strong enough to compare models because it refuses to pretend that every coordinate has the same semantics. We begin with the categorical edge itself: what remains of a pair table after constants and one-body effects have been removed, and how that remainder acts as an operator between reference-centered state spaces.

Part II

Operator Geometry of Sequence Interactions

Chapter 3

Reference-Centered Operator-Valued Graphs

A pair table appears to be one of the simplest objects in sequence modeling. Choose two positions, list the possible residue identities along the rows and columns, and attach a number to every combination. The result is finite, visible, and easy to store. Yet before it can represent an interaction, a basic ambiguity must be resolved: which part of the table belongs specifically to the pair?

Suppose a table $T(a, b)$ is changed by

$$T(a, b) \longmapsto T(a, b) + u(a) + v(b) + c. \tag{3.1}$$

In a conditional model, the added row term may be absorbed into a local field at the target; in a joint energy, row and column terms can be redistributed among one-body potentials. Different raw tables can therefore encode the same observable distribution or the same categorical response. Taking the matrix at face value confuses a parameterization with an identified pair object.

The remedy is often called gauge fixing, but the phrase can make the operation sound like a convention chosen for numerical convenience. The construction in this chapter is stronger. A reference distribution at each endpoint defines which directions count as constants and one-body effects. Double projection removes exactly those directions. What remains is the canonical bivariate component relative to that reference.

The geometric interpretation is the key. Position i carries a $(q-1)$-dimensional space of centered categorical contrasts behind its q labels. A pair interaction is a map from the source contrast space to the target contrast space. The resulting protein graph is a bundle of local categorical spaces connected by typed linear operators; scalar edge weights are compressed shadows of these maps.

This shift resolves several problems at once. It explains why raw Potts tables are not directly comparable across gauges, why reference changes require a transport law, why scalar norms lose substitution direction, and why directed model responses can be compared with static couplings without pretending that their original parameters are the same. The formal definitions below make these claims exact.

The common object must record more than a matrix attached to each pair. In particular, changing the context available to an edge or the state updated by that edge changes the model even when the displayed matrix is unchanged.

Definition 3.1 (Typed interaction system)**.** A typed interaction system is a tuple

$$\mathfrak{G} = \left(V, \{(\Omega_i, p_i, \mathcal{H}_i^0)\}_{i\in V}, \mathcal{C}, E, h, \mathcal{K}, \mathcal{O}, \mathcal{D}, \mathcal{N}, \mathcal{E}_{\mathrm{st}}, \mathcal{V}_{\mathrm{d}}\right). \tag{3.2}$$

Here V is the node set; (Ω_i, p_i) is the reference probability space at node i; $\mathcal{H}_i^0 \subseteq L_0^2(p_i)$ is the chosen centered feature fiber; $\mathcal{C}$ is the context sigma-algebra; $E(C) \subseteq V \times V$ is measurable support; $h(C) \in \bigoplus_i \mathcal{H}_i^0$ is the one-body field; and $\mathcal{K}_{i\leftarrow j}(C) : \mathcal{H}_j^0 \to \mathcal{H}_i^0$ is the edge map. The coordinate $\mathcal{O}$ records any higher-order maps, $\mathcal{D}$ the represented state and update rule, $\mathcal{N}$ the normalization or Lyapunov functional, $\mathcal{E}_{\mathrm{st}}$ the estimator and nuisance model, and $\mathcal{V}_{\mathrm{d}}$ the independent validation endpoint.

A model audit is a typed extraction map from model parameters, data, and interventions to $\mathfrak{G}$. Two models may agree in h and $\mathcal{K}$ while differing in $\mathcal{D}$, $\mathcal{N}$, or $\mathcal{E}_{\mathrm{st}}$; their agreement is coordinate-wise. A PSSM, for example, has a nonzero field h and zero off-diagonal edge operator. A Gaussian model can use centered linear features in $L_0^2(p_i)$, whereas the protein models below use the full centered categorical fiber.

3.1 Relational computation and the Graph Network interface

The Graph Network (GN) formalism isolates a reusable relational computation without prescribing the scientific semantics of its edges [33]. For a graph $G = (\mathbf{u}, V, E)$ with directed edge records (e_k, r_k, s_k), one GN block performs

$$e'_k = \phi^e(e_k, v_{r_k}, v_{s_k}, \mathbf{u}), \tag{3.3}$$

$$\bar{e}'_i = \rho^{e\to v}\left(\{e'_k : r_k = i\}\right), \qquad v'_i = \phi^v(v_i, \bar{e}'_i, \mathbf{u}), \tag{3.4}$$

$$\bar{e}' = \rho^{e\to u}(E'), \qquad \bar{v}' = \rho^{v\to u}(V'), \qquad \mathbf{u}' = \phi^u(\mathbf{u}, \bar{e}', \bar{v}'). \tag{3.5}$$

The learnable maps ϕ specify relation, node, and global updates. The reducers ρ are invariant to an arbitrary enumeration of their input sets, making the complete block equivariant to node relabeling when the graph and features are relabeled together. Parameter sharing permits the same local rule to act on new graph sizes and new combinations of entities. This is the architectural basis for combinatorial generalization, not a theorem that such generalization will occur outside the training distribution.

The typed system in Eq. 3.2 refines this interface. Its support $E(C)$ records the admissible senders and receivers; $\mathcal{K}_{i\leftarrow j}(C)$ types the edge update as a map between declared fibers; and $\mathcal{D}$ records the aggregation and node-state update. The remaining coordinates answer questions that a generic GN block intentionally leaves open: whether an edge is a route or an identified pair component, whether the updates arise from one joint probability law, which parameter redundancies preserve observables, how the edge object is estimated, and what endpoint validates it. Message-passing networks, Potts fields, and Transformers can therefore share a relational computation diagram while occupying different probabilistic and inferential coordinates. The unification is an interface with typed translations, not an assertion of semantic identity.

3.2 Node fibers and reference geometry

The word *fiber* carries little mystery here. At every residue position, take all numerical functions of residue identity and remove the constant function. The remaining $(q-1)$ directions encode comparisons such as "charged rather than hydrophobic" or any other centered categorical contrast. Calling these local spaces fibers keeps two indices separate: position identifies the node, while a vector inside the fiber identifies a pattern across residue categories.

Let positions be indexed by $[L] = \{1, \ldots, L\}$ and categorical states by $[q] = \{1, \ldots, q\}$. Fix a full-support reference distribution $p_i \in \Delta_{q-1}$ at each position and write $D_i = \mathrm{diag}(p_i)$. The node fiber at position i is

$$\mathcal{H}_i = L^2(p_i), \qquad \langle f, g\rangle_{p_i} = f^\top D_i g, \tag{3.6}$$

with centered subspace

$$\mathcal{H}_i^0 = \{f : p_i^\top f = 0\}. \tag{3.7}$$

The total categorical tangent space is the Hilbert direct sum

$$\mathcal{H}^0 = \bigoplus_{i=1}^{L} \mathcal{H}_i^0. \tag{3.8}$$

The position index is the graph base, while amino-acid contrasts form the fiber over each node. This distinction prevents a position graph from being confused with a flat matrix of categorical coordinates.

Define the weighted zero-mean projector

$$P_i = I - \mathbf{1}p_i^\top. \tag{3.9}$$

It satisfies $P_i^2 = P_i$ and $P_i^\top D_i = D_i P_i$. In whitened coordinates its Euclidean form is

$$Q_i = D_i^{1/2} P_i D_i^{-1/2} = I - \sqrt{p_i}\sqrt{p_i}^\top. \tag{3.10}$$

Thus the removed constant direction depends on the declared categorical reference. A uniform reference, a family marginal, and a model marginal define different geometries and should not be interchanged silently. Operationally, P_i takes any residue score vector and subtracts its p_i-weighted mean. It computes a centered contrast. Interaction estimation, biological choice of reference, and physical calibration enter at later stages.

3.3 The canonical bivariate component

For any fixed pair table $T_{ij} \in \mathbb{R}^{q\times q}$, define

$$\boxed{G_{ij} = P_i T_{ij} P_j^\top.} \tag{3.11}$$

Read this operation row by row and column by column: the left projection removes target-only variation, the right projection removes source-only variation, and the surviving block changes only when both endpoints vary. The theorem below establishes that this remainder is unique in the chosen reference geometry. Uniqueness is a statement about the decomposition of the table, not about how the table was generated.

Theorem 3.2 (Canonical categorical decomposition)**.** *Equip pair tables with*

$$\langle A, B\rangle_{p_i,p_j} = \sum_{a,b} p_i(a)p_j(b)A(a,b)B(a,b). \tag{3.12}$$

Every table has a unique orthogonal decomposition

$$T_{ij}(a,b) = c + u(a) + v(b) + G_{ij}(a,b), \tag{3.13}$$

where $p_i^\top u = 0$, $p_j^\top v = 0$, and

$$p_i^\top G_{ij} = 0, \qquad G_{ij}p_j = 0. \tag{3.14}$$

The bivariate component is exactly Eq. 3.11.

Proof. The orthogonal decomposition $L^2(p_i) = \mathrm{span}\{\mathbf{1}\} \oplus \mathcal{H}_i^0$ and its analogue at j induce four orthogonal tensor-product subspaces: constant, i-only, j-only, and centered bivariate. Applying P_i on the left and P_j^T on the right selects the last subspace. Existence and uniqueness follow from orthogonal projection. □

Terminology and the binary Ising case. Here *gauge* refers only to the additive non-identifiability of a redundant Potts pair-table parameterization. We call G_{ij} *reference-centered*, rather than "gauged," to avoid confusion with the spin-reversal gauge of spin glasses or with a $\mathbb{Z}_2$ lattice gauge theory; neither of those symmetries is used here. For two states with uniform reference, encode the categories by $s_i, s_j \in \{-1, +1\}$. Every table has the unique expansion

$$T(s_i, s_j) = c + h_i s_i + h_j s_j + J_{ij} s_i s_j, \qquad J_{ij} = \frac{T_{++} + T_{--} - T_{+-} - T_{-+}}{4}. \tag{3.15}$$

Double projection removes c, h_i, and h_j and leaves $G_{ij}(s_i, s_j) = J_{ij} s_i s_j$. Thus the $q = 2$ categorical construction extracts the single Ising coupling coordinate from a redundant binary table. The "gauge" here remains the additive Potts parameter redundancy, distinct from an Ising gauge field.

A three-state table, taken apart

The decomposition becomes concrete in the smallest example that retains more than one categorical contrast. Let both endpoints have three states and use uniform references

$$p_i = p_j = \frac{1}{3}(1, 1, 1)^\mathsf{T}. \tag{3.16}$$

Consider the raw pair table

$$T = \begin{pmatrix} 10 & 5 & 3 \\ 5 & 5 & 2 \\ 6 & 5 & 4 \end{pmatrix}. \tag{3.17}$$

Its grand mean is $c = 5$. Its row means are $(6, 4, 5)$, so the centered target-only term is

$$u = (1, -1, 0)^\mathsf{T}. \tag{3.18}$$

Its column means are $(7, 5, 3)$, giving the centered source-only term

$$v = (2, 0, -2)^\mathsf{T}. \tag{3.19}$$

Subtracting these three lower-order pieces leaves

$$G = T - c\mathbf{1}\mathbf{1}^\mathsf{T} - u\mathbf{1}^\mathsf{T} - \mathbf{1}v^\mathsf{T} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}. \tag{3.20}$$

Every row and column of G sums to zero. Equivalently,

$$p_i^\mathsf{T} G = 0, \qquad G p_j = 0. \tag{3.21}$$

The large entry $T_{11} = 10$ is therefore not itself the interaction. It contains the grand baseline, a target propensity, a source propensity, and the bivariate residual. In this example those contributions are $5 + 1 + 2 + 2$.

Now alter the raw table by adding any new row and column effects,

$$T'(a,b) = T(a,b) + \widetilde{u}(a) + \widetilde{v}(b) + \widetilde{c}. \tag{3.22}$$

The numerical entries and their ordinary Frobenius norm can change substantially, but double projection returns the same G. The identified interaction is therefore the quotient class of T, while Eq. 3.20 is its representative in the uniform reference.

The edge becomes an operator when it acts on a source contrast. Take

$$f = (1, -1, 0)^\mathsf{T}, \qquad p_j^\mathsf{T} f = 0. \tag{3.23}$$

Then

$$\mathcal{K}_{i\leftarrow j} f = G D_j f = \frac{1}{3} G f = \begin{pmatrix} 1 \\ -2/3 \\ -1/3 \end{pmatrix}, \tag{3.24}$$

whose target mean is again zero. The source contrast does not merely activate an edge of strength $\|G\|$; it produces a particular target contrast. Another source direction produces another response.

For uniform references the whitened edge matrix is $\widetilde{G} = G/3$. Its two nonzero singular values are 1 and 1/3, and its Hilbert–Schmidt norm is

$$\|\mathcal{K}_{i\leftarrow j}\|_{\mathrm{HS}} = \frac{\sqrt{10}}{3}. \tag{3.25}$$

The scalar norm retains their combined magnitude but discards both singular directions. This one calculation contains the sequence of transformations used throughout the book:

$$\boxed{\text{raw table} \longrightarrow \text{gauge class} \longrightarrow \text{reference representative} \longrightarrow \text{fiber operator} \longrightarrow \text{scalar shadow.}} \tag{3.26}$$

Only the first four objects retain the typed substitution action. The final scalar is useful only after the loss has been declared.

This is a functional-ANOVA statement relative to $p_i p_j^\mathsf{T}$ [1]. It identifies a mathematical component, not a physical mechanism. In particular, nonzero G_{ij} identifies a mathematical component; direct contact, causal epistasis, and phylogenetic independence are separate empirical properties.

The underlying interaction is a gauge class rather than one selected representative. Define the additive subspace

$$\mathcal{A}_{ij} = \{T : T(a,b) = c + u(a) + v(b)\} \tag{3.27}$$

and the quotient $\mathcal{Q}_{ij} = \mathbb{R}^{q\times q}/\mathcal{A}_{ij}$. Every full-support pair of references selects one representative of the same $(q-1)^2$-dimensional class.

Proposition 3.3 (Gauge quotient and reference transport)**.** *For fixed full-support references (p_i, p_j), two tables have the same projected block if and only if they define the same element of $\mathcal{Q}_{ij}$. For second full-support references (r_i, r_j), write $P_i^{[r]} = I - \mathbf{1} r_i^\mathsf{T}$ and similarly at j. The canonical representatives obey*

$$\boxed{G_{ij}^{[r]} = P_i^{[r]} G_{ij}^{[p]} P_j^{[r]\mathsf{T}}.} \tag{3.28}$$

Proof. Theorem 3.2 identifies the kernel of $T \mapsto P_i T P_j^\mathsf{T}$ with $\mathcal{A}_{ij}$, proving the quotient statement. The projectors for two references satisfy $P_i^{[r]} P_i^{[p]} = P_i^{[r]}$ and $P_j^{[p]\mathsf{T}} P_j^{[r]\mathsf{T}} = P_j^{[r]\mathsf{T}}$. Applying these identities to $G_{ij}^{[p]} = P_i^{[p]} T_{ij} P_j^{[p]\mathsf{T}}$ gives $P_i^{[r]} T_{ij} P_j^{[r]\mathsf{T}} = G_{ij}^{[r]}$. □

Reference-invariance of zero context dependence. Let a family of raw tables $T(C)$ be evaluated under one fixed background law, and let $G^{[p]}(C)$ and $G^{[r]}(C)$ be its representatives under two full-support reference pairs. If

$$V_{\mathrm{dyn}}^{[p]} = \mathbb{E}_C \left\| G^{[p]}(C) - \mathbb{E}_C G^{[p]}(C) \right\|_{p_i,p_j}^2, \tag{3.29}$$

and $V_{\mathrm{dyn}}^{[r]}$ is defined analogously, then

$$\boxed{V_{\mathrm{dyn}}^{[p]} = 0 \quad \Longleftrightarrow \quad V_{\mathrm{dyn}}^{[r]} = 0.} \tag{3.30}$$

Indeed, Eq. 3.28 maps a constant representative to a constant representative, and applying the same transport law in the reverse direction gives the converse. Thus absence of context dependence is reference independent, although every positive variance, norm, singular value, and dynamic fraction depends quantitatively on the chosen reference geometry. When the denominator in Eq. 4.22 is nonzero, the same equivalence holds with $V_{\mathrm{dyn}} = 0$ replaced by $\rho^{\mathrm{dyn}} = 0$.

Thus the gauge class is reference independent, while its centered representative, Hilbert norm, and singular geometry are reference dependent. Comparisons across families or contexts must either use one fixed reference or transport every representative to a declared anchor by Eq. 3.28.

3.4 From a table to an edge operator

A centered block becomes an edge only after its input and output are typed. The next equation feeds a centered substitution pattern at j into the block and returns a centered response pattern at i. This is what the operator computes. Physical action and reciprocity are subsequent empirical and compatibility questions.

A gauged block induces the directed map

$$(\mathcal{K}_{i\leftarrow j} f)(a) = \sum_b p_j(b) G_{i\leftarrow j}(a,b) f(b) = \big(G_{i\leftarrow j} D_j f\big)(a). \tag{3.31}$$

$$\boxed{\text{source identity at } j \longrightarrow \text{source contrast } f \in \mathcal{H}_j^0 \xrightarrow{\mathcal{K}_{i\leftarrow j}} \text{target contrast in } \mathcal{H}_i^0 \longrightarrow \text{change in target preference.}}$$

Figure 3.1: An edge is a typed map between local contrast spaces. Its scalar existence flag leaves the source-substitution and target-response directions unresolved.

Proposition 3.4 (Gauged tables are fiber maps)**.** *Equation 3.31 defines a linear map $\mathcal{K}_{i\leftarrow j} : \mathcal{H}_j^0 \to \mathcal{H}_i^0$ that annihilates the constant source mode. In whitened coordinates its matrix is*

$$\widetilde{G}_{i\leftarrow j} = D_i^{1/2} G_{i\leftarrow j} D_j^{1/2}, \tag{3.32}$$

and

$$\|\mathcal{K}_{i\leftarrow j}\|_{\mathrm{HS}} = \|\widetilde{G}_{i\leftarrow j}\|_F. \tag{3.33}$$

Proof. The left condition in Eq. 3.14 centers the output, while $G_{i\leftarrow j} p_j = 0$ gives $\mathcal{K}_{i\leftarrow j}\mathbf{1} = 0$. Conjugation by the isometries $f \mapsto D_i^{1/2} f$ and $f \mapsto D_j^{1/2} f$ gives the whitened matrix and preserves the Hilbert–Schmidt norm. □

For routing coefficients z_{ij}, the operator-valued graph is the block operator

$$(\mathbb{K}_z f)_i = \sum_{j \neq i} z_{ij} \mathcal{K}_{i \leftarrow j} f_j. \tag{3.34}$$

Its support, edge rank, and directionality are explicit. If $z_{ij} = z_{ji}$ and $G_{j \leftarrow i} = G_{i \leftarrow j}^{\mathsf{T}}$ under compatible references, $\mathbb{K}_z$ is self-adjoint. Predictive sequence models need not satisfy this reciprocity and may therefore yield genuinely directed block operators.

Coda: the graph after gauge

The object obtained in this chapter is an equivalence class made concrete by a declared reference, then interpreted as a map between centered categorical fibers. Raw coupling matrices and scalar contact scores are respectively an input representation and a later compression of this object. This is the first point at which edges from different constructions can enter a common coordinate system while retaining their type.

Three distinctions should remain visible. Gauge projection identifies the bivariate part of a table; endpoint invariance is a separate response property. Reference transport compares two coordinate choices; population comparability depends on how the empirical references were obtained. Finally, an operator attached to an edge describes categorical action. Static, dynamic, probabilistic, computational, and physical readings require additional coordinates.

The physical reading is precise but conditional. If T_{ij} is a term in one compatible joint Potts Hamiltonian, then adding $u(a) + v(b) + c$ redistributes energy between the pair table, the two one-body fields, and the global zero without changing the Gibbs law. The quotient therefore isolates the part of an effective pair energy orthogonal to every reassignment into single-site terms. It generalizes the familiar freedom to choose an energy zero: the constant fixes the zero, while the row and column terms fix how one-body and pair contributions are apportioned.

That statement concerns an effective Hamiltonian on sequence categories. A microscopic-force interpretation of $G_{ij}(a, b)$ additionally requires a declared ensemble, a coarse-graining map from molecular configurations to residue identities, calibrated energy units or a specified $k_{\mathrm{B}}T$, and evidence that the inferred sequence law is the relevant equilibrium or evolutionary law. In its native scope, G_{ij} remains an exact bivariate contrast operator: the part of the table invariant to one-body reparameterization, and hence the appropriate object for comparison across models.

Those questions require a response experiment. The next chapter therefore leaves the table as a parameter and asks what happens when one residue is actually varied while another prediction is observed. In an additive Potts model the answer is fixed. In a deep sequence model the answer can move with the surrounding sequence, turning one edge operator into a field of operators over backgrounds.

Chapter 4

Static and Dynamic Sequence Interactions

A static interaction graph describes a family by assigning one pair object to each edge. A deep sequence model does something more elusive. Its effective relation between two positions can change when a third residue changes, when the sequence enters another family-like background, or when information is rerouted through a different hidden path. The model parameters are fixed, but the response graph is not.

To see the distinction, hold all positions except i and j in a background C. Substitute the source residue at j, observe the model's categorical prediction at i, and project away target- and source-only components. The resulting operator

$$G_{i\leftarrow j}(C) \tag{4.1}$$

is a section of the model's behavior through that background. Repeating the audit over C does not estimate the same quantity with noise; it reveals whether the quantity itself changes.

This creates a clean comparison. An additive pairwise Potts conditional produces the same gauged response for every background, so its dynamic variance is zero. A nonlinear protein language model generally yields a distribution of edge operators. Their mean is the best static approximation in the chosen reference geometry, while their fluctuation measures the failure of a single static edge to describe the model.

The fluctuation has a precise higher-order interpretation. Under a product reference it can be tied exactly to higher-order source-coordinate components valued in the target contrast fiber. Under a correlated background law, orthogonality can fail and the same variance can contain cross terms. The reference distribution is therefore part of both the scientific statement and the estimator.

Two further questions run through the chapter. First, when does a directed response deserve the conditional meaning of a pair interaction? If the generator of the edge reads the endpoint being varied, the edge itself moves during the intervention. Second, when do the two directed responses $j \to i$ and $i \to j$ integrate into one undirected pair energy? That upgrade requires reciprocity after the relevant gauges have been aligned. Context dependence and directionality are therefore measurable coordinates, not inconveniences to be symmetrized away at the outset.

4.1 Potts models as static response fields

A pairwise Potts model has energy

$$E_{\mathrm{P}}(x) = -\sum_i h_i(x_i) - \frac{1}{2}\sum_{i\neq j} J_{ij}(x_i, x_j), \tag{4.2}$$

and single-site conditional

$$P_{\mathrm{P}}(X_i = a \mid x_{-i}) = \mathrm{softmax}_{a\in[q]}\left[h_i(a) + \sum_{j\neq i} J_{ij}(a, x_j)\right]. \tag{4.3}$$

The tables J_{ij} are fixed after fitting and are shared by all sequences in the family. Their canonical edge blocks are

$$G^{\mathrm{P}}_{ij} = P_i J_{ij} P_j^{\top}. \tag{4.4}$$

This is the sense in which a Potts interaction graph is static. The conditional probabilities still change with x_{-i}, but the rule mapping a source category to a target logit contrast does not. Static refers to the shared response law, not to constancy of the model output.

4.2 Categorical response of a sequence model

Let $c = x_{[L]\setminus\{i,j\}}$ be a sequence background excluding the endpoints. Evaluate a sequence model with target i withheld and source j set to category b, and denote its target logit by

$$T^{c}_{i\leftarrow j}(a, b) = \ell_i(a \mid X_j = b, C_{ij} = c). \tag{4.5}$$

The reference-centered categorical response is

$$G_{i\leftarrow j}(c) = P_i T^{c}_{i\leftarrow j} P_j^{\top}. \tag{4.6}$$

Equivalently, one may subtract any baseline column from T^c before projection. This gives the finite substitution table

$$R^{c}_{i\leftarrow j}(a, b) = \ell_i(a \mid b, c) - \ell_i(a \mid b_0, c), \tag{4.7}$$

with $P_i R^c P_j^{\top} = G_{i\leftarrow j}(c)$. The construction is sometimes called a categorical Jacobian. More precisely, it is a finite intervention, with an infinitesimal derivative as its local limit.

Proposition 4.1 (Exact recovery of an additive coupling)**.** *Suppose the target logits are additive,*

$$\ell_i(a \mid x_{-i}) = h_i(a) + \sum_{k\neq i} J_{i\leftarrow k}(a, x_k). \tag{4.8}$$

Then for every background c,

$$G_{i\leftarrow j}(c) = P_i J_{i\leftarrow j} P_j^{\top}. \tag{4.9}$$

Thus the gauged response is background independent and exactly recovers the canonical pair coupling.

Proof. In Eq. 4.5, the field $h_i(a)$ and all source terms with $k \neq j$ depend on a and c but not on b. They are annihilated by right projection. The remaining table is $J_{i\leftarrow j}(a, b)$, and left projection removes its target-only gauge component. □

For a nonlinear or higher-order conditional, the same response has the schematic expansion

$$G_{i\leftarrow j}(c) = G^{(2)}_{ij} + \sum_{k\notin\{i,j\}} G^{(3)}_{i,j,k}(c_k) + \sum_{\substack{k<l\\ k,l\notin\{i,j\}}} G^{(4)}_{i,j,k,l}(c_k, c_l) + \cdots . \tag{4.10}$$

It is therefore a well-defined *effective* edge at background c, assembled from all statistical and computational paths through which changing x_j changes the target preference at i.

4.3 The static–dynamic decomposition

Fix anchor references p_i, p_j and let C be distributed according to a declared family, model, or evaluation distribution μ_{ij}. The gauged blocks form one finite-dimensional Hilbert space with norm

$$\|G\|^2_{p_i,p_j} = \sum_{a,b} p_i(a)p_j(b)G(a,b)^2. \tag{4.11}$$

Assume $\mathbb{E}_C\|G_{i\leftarrow j}(C)\|^2_{p_i,p_j} < \infty$ and define

$$\overline{G}_{i\leftarrow j} = \mathbb{E}_C G_{i\leftarrow j}(C), \qquad \delta G_{i\leftarrow j}(C) = G_{i\leftarrow j}(C) - \overline{G}_{i\leftarrow j}. \tag{4.12}$$

Theorem 4.2 (Best static approximation and dynamic variance)**.** *The mean block $\overline{G}_{i\leftarrow j}$ is gauged and is the unique minimizer*

$$\overline{G}_{i\leftarrow j} = \arg \min_{A:\, p_i^\top A=0,\, Ap_j=0} \mathbb{E}_C\|G_{i\leftarrow j}(C) - A\|^2_{p_i,p_j}. \tag{4.13}$$

Moreover,

$$\mathbb{E}_C\|G_{i\leftarrow j}(C)\|^2_{p_i,p_j} = \|\overline{G}_{i\leftarrow j}\|^2_{p_i,p_j} + \mathbb{E}_C\|\delta G_{i\leftarrow j}(C)\|^2_{p_i,p_j}. \tag{4.14}$$

Proof. The gauged subspace is linear, so expectation preserves both zero-marginal constraints. Expanding $G(C)-A = \delta G(C)+(\overline{G}-A)$, the cross term vanishes because $\mathbb{E}_C\delta G(C) = 0$. The resulting Pythagorean identity is minimized uniquely at $A = \overline{G}$ and gives Eq. 4.14 when $A = 0$. □

The orthogonality in Theorem 4.2 is mean–residual orthogonality in $L^2(\mu_{ij})$. It holds for any declared background law μ_{ij}, including a correlated empirical MSA distribution, because the residual is centered under that same law. Independence enters only when the residual is further attributed to mutually orthogonal interaction orders, as in Corollary 5.2 below. Thus correlation changes the order-by-order interpretation of dynamic variance, while the static–dynamic Pythagorean identity remains exact.

Static component $\overline{G}_{i\leftarrow j}$	Dynamic component $\delta G_{i\leftarrow j}(C)$
Best single operator under the declared background law	Background-specific departure from that operator
Persists after averaging over C	Has mean zero over the same C
Comparable with a family-level static field	Measures the failure of one static edge to describe the response
Physical interpretation requires calibration	Origin may combine several mechanisms

Figure 4.1: The static–dynamic split separates a reusable mean relation from context dependence; biological importance is an independent validation coordinate.

A two-background calculation

The static–dynamic identity can be seen without fitting a large model. Continue with uniform three-state references and let the persistent pair component be

$$G_0 = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}. \tag{4.15}$$

Introduce a second centered direction

$$H = \begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix}. \tag{4.16}$$

Both matrices have zero row and column means, and they are orthogonal under the uniform weighted inner product. Let a third position k supply a binary background contrast $C \in \{-1, +1\}$ with equal probability, and define the effective edge

$$G_{i\leftarrow j}(C) = G_0 + \lambda C H. \tag{4.17}$$

The response of the pair (i, j) now changes when position k changes. Its mean and residual are

$$\overline{G}_{i\leftarrow j} = G_0, \qquad \delta G_{i\leftarrow j}(C) = \lambda C H. \tag{4.18}$$

Direct calculation gives

$$\|G_0\|^2_{p_i,p_j} = \frac{10}{9}, \qquad \|H\|^2_{p_i,p_j} = \frac{6}{9} = \frac{2}{3}. \tag{4.19}$$

Therefore

$$\mathbb{E}_C\|G(C)\|^2_{p_i,p_j} = \frac{10}{9} + \frac{2\lambda^2}{3}, \tag{4.20}$$

and the dynamic fraction is

$$\rho^{\mathrm{dyn}}_{i\leftarrow j} = \frac{6\lambda^2}{10 + 6\lambda^2}. \tag{4.21}$$

At $\lambda = 0$ the response is exactly static. At $\lambda = 1$, three eighths of its squared mass is dynamic. As $|\lambda|$ grows, the pair is increasingly described by its background modulation rather than by one persistent block.

The construction also displays interaction order. The term CH jointly involves the audited target i, source j, and background position k. It is stored observationally as a background-dependent edge, but functionally it is a third-order component. A scalar graph can record the variance of this edge across backgrounds; it cannot by itself say which categorical direction H carries or which position produced the modulation.

Finally, the calculation assumes that C is held fixed while the endpoint categories are varied. If the coefficient were emitted by a gate that also reads X_j, then enumerating the source state b would change both the message and the gate selecting the message. Double projection would still center the resulting table, but Eq. 4.17 would no longer describe one fixed conditional pair operator. This is the practical reason endpoint exclusion appears as a separate requirement rather than as a consequence of gauge fixing.

This theorem supplies an operational definition of static and dynamic interaction. Define

$$\rho^{\mathrm{dyn}}_{i\leftarrow j} = \frac{\mathbb{E}_C\|\delta G_{i\leftarrow j}(C)\|^2_{p_i,p_j}}{\mathbb{E}_C\|G_{i\leftarrow j}(C)\|^2_{p_i,p_j}}, \qquad 0 \le \rho^{\mathrm{dyn}}_{i\leftarrow j} \le 1, \tag{4.22}$$

whenever the denominator is nonzero. In finite-precision estimation, the ratio should be reported only when

$$\mathbb{E}_C \|G_{i\leftarrow j}(C)\|^2_{p_i,p_j} > \tau_G, \tag{4.23}$$

for a preregistered absolute or null-calibrated threshold τ_G; otherwise it is undefined, not zero or one. This prevents a ratio of two numerical underflow terms on a null edge from appearing as strong dynamic response. An additive Potts conditional has $\rho^{\mathrm{dyn}}_{i\leftarrow j} = 0$. A positive value measures background dependence of the effective response. Higher-order epistasis, remote homologous patterns, and internal circuits are candidate sources to be separated by further interventions.

Two background laws expose a genuine interpretation–geometry tradeoff. With the same endpoint anchor references, define

$$\rho^{\mathrm{dyn,prod}}_{i\leftarrow j} = \rho^{\mathrm{dyn}}_{i\leftarrow j}\Big|_{\mu_C = \bigotimes_{k\notin\{i,j\}} p_k}, \qquad \rho^{\mathrm{dyn,emp}}_{i\leftarrow j} = \rho^{\mathrm{dyn}}_{i\leftarrow j}\Big|_{\mu_C = \mu_C^{\mathrm{emp}}}. \tag{4.24}$$

Only the product-background quantity has the exact order-by-order ANOVA interpretation in Corollary 5.2; its independently assembled backgrounds may, however, be far outside the natural sequence manifold. The empirical quantity evaluates natural or curated backgrounds and is therefore closer to the model's training support, although exhaustive endpoint substitutions can still leave that support. Correlations destroy order orthogonality, so the empirical dynamic variance remains a valid mean–residual quantity but not a sum of pure higher-order masses. Both should be reported, together with

$$\Delta\rho^{\mathrm{geom}}_{i\leftarrow j} = \rho^{\mathrm{dyn,emp}}_{i\leftarrow j} - \rho^{\mathrm{dyn,prod}}_{i\leftarrow j}, \tag{4.25}$$

as a diagnostic of sensitivity to the evaluation geometry rather than as a signed amount of biological interaction.

The distribution μ_{ij} is part of the claim. Averaging over one protein family, all natural proteins, or model-generated sequences produces different static components. A pLM can therefore have learned an amortized dynamic rule, while an MSA estimates a separate family-specific statistical model against which the response mean can be tested. Equality between an MSA-fitted Potts block and the pLM mean is an empirical transfer hypothesis in their common coordinates. The family distribution supplies the background law under which that hypothesis is evaluated.

4.4 Conditional meaning requires endpoint exclusion

Equation 4.6 is a strict conditional component only if the table-generating context remains fixed while the endpoint states are varied. Let $\mathcal{C}_{ij} = \sigma\{X_k : k \notin \{i,j\}\}$ and allow the reference, raw table, and any route decision to depend on this cavity.

Proposition 4.3 (Conditional gauge under an endpoint-excluding audit)**.** *Require the audit protocol to construct $T_{i\leftarrow j}(C)$, $p_i(C)$, $p_j(C)$, and every route decision from an endpoint-excluding representation measurable with respect to $\mathcal{C}_{ij}$; in particular, it may not reuse a hidden state that previously observed x_i or x_j. If the conditional references have full support, and the enumerated source category b enters only in the final endpoint evaluation used to assemble the table, then, almost surely in C,*

$$G_{i\leftarrow j}(C) = P_i(C) T_{i\leftarrow j}(C) P_j(C)^{\mathsf{T}} \tag{4.26}$$

is the unique bivariate contrast relative to $p_i(\cdot \mid C) p_j(\cdot \mid C)$.

Proof. The protocol condition ensures that, after conditioning on $\mathcal{C}_{ij}$, the table, route, and reference measures are fixed while (X_i, X_j) vary, so Theorem 3.2 applies on each conditional fiber. □

If a hidden representation used to emit the table has already observed x_i or x_j, projection still gives a locally centered table, but not a global conditional decomposition. This caveat is especially important when interpreting ordinary attention heads: masking the target and explicitly enumerating source substitutions defines a cleaner response audit than double centering an attention-derived table computed from the unmodified endpoints.

When the conditional references in Proposition 4.3 depend on C, the blocks inhabit moving fibers. Transport them into common anchor references $(\bar{p}_i, \bar{p}_j)$ before applying Eq. 4.12:

$$G^{\mathrm{anc}}_{i\leftarrow j}(C) = P_i^{[\bar{p}]} G^{[p(C)]}_{i\leftarrow j}(C) P_j^{[\bar{p}]\mathsf{T}}; \tag{4.27}$$

then apply the static–dynamic decomposition to G^{anc}. This convention separates a change of coordinates from genuine background dependence of the gauge class.

4.5 When directed responses define a joint pair energy

A masked or autoregressive predictor naturally produces directed blocks $G_{i\leftarrow j}$, whereas a pairwise Hamiltonian uses one reciprocal table for the unordered pair $\{i, j\}$. This distinction is another independent coordinate of the framework.

Proposition 4.4 (Reciprocity criterion for additive conditionals)**.** *Consider full-support additive categorical conditionals in one common reference-centered parameterization. Their pairwise interaction branch is compatible with a joint Potts energy if and only if*

$$G_{i\leftarrow j} = G^{\mathsf{T}}_{j\leftarrow i} \qquad \textit{for every } i \neq j. \tag{4.28}$$

When this holds, the common block defines the gauged coupling of the joint energy; one-body fields are then fixed by the corresponding conditional biases up to a global additive constant.

Proof. Necessity follows because one joint pair table contributes $J_{ij}(a, b)$ to the conditional at i and $J_{ij}(a, b)$, with transposed endpoint order, to the conditional at j. Gauge projection therefore gives Eq. 4.28. Conversely, reciprocal gauged blocks can be inserted as the pair tables of Eq. 4.2; their single-site conditionals recover the specified pair branches, while endpoint-only differences are absorbed into the site fields. □

For arbitrary directed responses define the reciprocal and nonintegrable parts

$$G^{\mathrm{rec}}_{ij} = \frac{1}{2}\left(G_{i\leftarrow j} + G^{\mathsf{T}}_{j\leftarrow i}\right), \tag{4.29}$$

$$G^{\mathrm{irr}}_{ij} = \frac{1}{2}\left(G_{i\leftarrow j} - G^{\mathsf{T}}_{j\leftarrow i}\right). \tag{4.30}$$

Only the first is directly compatible with an undirected pair energy. The second measures failure of pairwise joint integrability in the chosen response coordinates. A thermodynamic irreversibility interpretation additionally requires a dynamical model.

Coda: a graph can be fixed in parameters and dynamic in response

The static–dynamic decomposition changes the comparison between family models and protein Transformers. Their difference is no longer summarized by saying that one is generative and the other contextual. Both can be interrogated through the same categorical response operator. The

Potts model occupies the background-invariant limit; the nonlinear model supplies a mean operator and a measurable field of deviations.

This common coordinate does not erase the distinction between the models. A mean pLM response is an estimator built from interventions on a trained function, not a parameter fitted from an MSA. A small dynamic residual supports a static approximation under the audited background law, not a universal physical coupling. A large residual is evidence against that approximation and a clue to higher-order organization, but its decomposition depends on the reference geometry.

The next question is what kind of complexity the residual represents. Graph support, categorical rank, and interaction order are often compressed into one informal notion of complexity. They are independent. A single edge may carry a full-rank operator; a dense graph may use only a shared low-dimensional family of maps; and nonlinear composition can generate higher-order terms even when every local message was separately pair-centered. The next chapter separates these axes.

Chapter 5

Interaction Order, Rank, and Scalar Compression

Several different simplifications are routinely called "sparse interaction." A model may use few position pairs, compress the categorical action on each pair, truncate all dependence above order two, or reduce every edge to one scalar score. These operations save different resources and discard different information. None implies the others.

Graph support answers whether a position pair is represented. Categorical rank answers how many independent substitution-contrast directions its edge operator carries. Interaction order asks how many sequence coordinates participate irreducibly in one functional component. A sparse position graph can attach a full-rank $q \times q$ operator to each surviving edge. A complete graph can reuse a one-dimensional message family. A network built from pairwise messages can generate higher-order dependence after nonlinear aggregation.

These counterexamples are not pathological. They identify three distinct approximation budgets:

$$\text{number of edges}, \qquad \text{rank per edge}, \qquad \text{maximum interaction order}. \tag{5.1}$$

A claim of compression should say which budget is controlled and how the discarded component is measured.

The chapter then confronts the most common compression: replacing an operator G_{ij} by a scalar such as its norm. This step is often necessary for contact prediction, visualization, or ordinary graph spectra. It is also irreversible. Sign, singular directions, and the identity of favored substitutions disappear. Projection and scalarization do not commute: a large raw norm can come entirely from one-body structure that vanishes in the canonical pair gauge.

Scalar graphs are therefore not rejected here. They are located. A scalar shadow answers a declared question about the magnitude of an already identified operator. Trouble begins only when properties of the shadow are read back as though no categorical information had been discarded.

5.1 From operator-valued graphs to hypergraphs

A graph retains target–source maps $\mathcal{K}_{i\leftarrow j} : \mathcal{H}_j^0 \to \mathcal{H}_i^0$. A general conditional model also contains hyperedge maps

$$\mathcal{K}_{i\leftarrow A} : \bigotimes_{j\in A} \mathcal{H}_j^0 \longrightarrow \mathcal{H}_i^0, \qquad |A| \geq 2. \tag{5.2}$$

For a fixed background outside $\{i\} \cup N$, let E_u average over x_u under a positive product reference and let $P_u = I - E_u$. For $B \subseteq \{i\} \cup N$, define

$$\Pi_B = \left(\prod_{u \in B} P_u \right) \left(\prod_{v \notin B} E_v \right). \tag{5.3}$$

Theorem 5.1 (Interaction hierarchy and pair-truncation error)**.** *For $T \in L^2(\bigotimes_{u \in \{i\} \cup N} p_u)$, the operators $\{\Pi_B\}$ are mutually orthogonal projections that sum to the identity, and $T = \sum_B \Pi_B T$ uniquely. Let*

$$\mathcal{I}_i = \sum_{\emptyset \neq A \subseteq N} T_{i,A}, \qquad T_{i,A} = \Pi_{\{i\} \cup A} T. \tag{5.4}$$

For the pairwise truncation $\mathcal{I}_i^{(1)} = \sum_{j \in N} T_{i,\{j\}}$,

$$\|\mathcal{I}_i - \mathcal{I}_i^{(1)}\|^2 = \sum_{A \subseteq N \colon |A| \geq 2} \|T_{i,A}\|^2. \tag{5.5}$$

Proof. The coordinate-wise averaging operators are commuting self-adjoint idempotents, and $P_u = I - E_u$ is the orthogonal complement of E_u. Expanding $\prod_u (E_u + P_u) = I$ gives the decomposition. Distinct components contain a factor $E_u P_u = 0$ and are orthogonal. Equation 5.5 is Pythagoras applied to the omitted terms. □

Corollary 5.2 (Dynamic variance equals higher-order response mass)**.** *Let $N \subseteq [L] \setminus \{i\}$ contain j and define the centered, vector-valued target score*

$$\Phi(x_N) = P_i \ell_i(\cdot \mid x_N) \in \mathcal{H}_i^0, \qquad \Phi \in L^2\left(\bigotimes_{u \in N} p_u; \mathcal{H}_i^0 \right). \tag{5.6}$$

For $B \subseteq N$, let $\Pi_B^N = (\prod_{u \in B} P_u)(\prod_{v \in N \setminus B} E_v)$ act on the source coordinates of this Hilbert-valued function, and write $\Phi_B = \Pi_B^N \Phi$. Under the product reference, the audited response satisfies

$$G_{i \leftarrow j}(C) = \sum_{B \subseteq N \colon j \in B} \Phi_B, \tag{5.7}$$

$$\overline{G}_{i \leftarrow j} = \Phi_{\{j\}}, \tag{5.8}$$

$$\mathbb{E}_C \|\delta G_{i \leftarrow j}(C)\|^2_{p_i, p_j} = \sum_{B \subseteq N \colon B \supsetneq \{j\}} \|\Phi_B\|^2. \tag{5.9}$$

Thus dynamic response variance is exactly the total functional-ANOVA mass of components containing the audited source and at least one additional source coordinate. The target category remains an output coordinate in $\mathcal{H}_i^0$; it is not assumed to be an observed model input. The norms on the right are the Bochner L^2 norms over the indicated source coordinates and the target fiber.

Proof. The commuting source-coordinate projections extend to the Bochner space in Eq. 5.6 and remain mutually orthogonal. Applying P_j retains exactly the components whose source set contains j; P_i has already centered the output fiber. Averaging over $C = X_{N \setminus \{j\}}$ retains only $\Phi_{\{j\}}$. Orthogonality of the remaining Hilbert-valued components gives the squared-norm identity. □

The theorem gives Eq. 4.10 a precise interpretation. A dynamic edge response is a section through the higher-order expansion with all other coordinates fixed. Its background variance is therefore an exact higher-order mass under the product reference. Under a correlated empirical background distribution the components can lose orthogonality and acquire cross terms; the product reference or a declared conditional geometry is part of the theorem. Even in the orthogonal case, resolving individual hyperedges requires a designed multi-substitution experiment beyond their total variance.

Remark 5.3 (Mean orthogonality versus order orthogonality). Theorem 4.2 and Corollary 5.2 use two different orthogonalities. The first centers one random operator around its mean and requires no coordinate independence. The second decomposes that operator by subsets of sequence coordinates and uses a product reference. If the same order-indexed components are evaluated under a correlated law μ, then in general

$$\begin{aligned}\mathbb{E}_\mu \left\| \sum_{B \supsetneq \{j\}} \Phi_B(C_B) \right\|^2 = & \sum_{B \supsetneq \{j\}} \mathbb{E}_\mu \|\Phi_B(C_B)\|^2 \\ & + \sum_{\substack{B,D \supsetneq \{j\} \\ B \neq D}} \mathbb{E}_\mu \langle \Phi_B(C_B), \Phi_D(C_D) \rangle,\end{aligned} \tag{5.10}$$

and the final sum records cross terms induced by the correlated reference geometry. Dynamic variance remains measurable, but it is no longer identifiable with a simple sum of order-specific ANOVA masses.

Proposition 5.4 (Nonlinear composition can create higher order)**.** *Separately centered pair messages can generate higher-order components after nonlinear aggregation.*

Proof. For independent Rademacher sources $x_j, x_k \in \{-1, 1\}$ under the uniform product reference and a nonzero target contrast $P_i u$,

$$(P_i u)(a)(x_j + x_k)^2 = 2(P_i u)(a) + 2(P_i u)(a) x_j x_k. \tag{5.11}$$

After assigning the first term to the target-only component, the second is a nonzero target–source–source interaction. □

Consequently, pairwise first-layer messages can produce a globally higher-order deep network. Its final response must be decomposed at the output, or its higher-order residual measured by multi-background and multi-substitution audits.

5.2 Support, categorical rank, and order are independent

For a nonzero whitened block, write

$$\widetilde{G}_{ij} = \sum_{r=1}^{q-1} \sigma_{ijr} u_{ijr} v_{ijr}^\top. \tag{5.12}$$

Graph support counts which position pairs are represented. Categorical rank describes the number of substitution-contrast directions retained on one edge. Interaction order distinguishes edges from hyperedges. These are independent coordinates: a sparse graph may have full-rank categorical blocks, while a dense graph may share only a low-dimensional message family.

For a relative tolerance ϵ, an error-controlled edge rank is

$$r_{ij}(\epsilon) = \min\left\{ r : \frac{\sum_{s>r} \sigma_{ijs}^2}{\sum_s \sigma_{ijs}^2} \le \epsilon \right\}. \tag{5.13}$$

Low rank is therefore a measurable compression hypothesis independent of graph sparsity.

5.3 The scalar position graph is a shadow

The common scalar compression is

$$M_{ij} = \|G_{ij}\|_{p_i,p_j} = \|D_i^{1/2} G_{ij} D_j^{1/2}\|_F. \tag{5.14}$$

Proposition 5.5 (Loss and order of scalar compression)**.** *The map $G_{ij} \mapsto M_{ij}$ discards sign, singular directions, and categorical action. In addition, projection must precede the norm: for any nonzero $v \in \mathbb{R}^q$, the one-body table $T = \mathbf{1}v^\mathsf{T}$ satisfies*

$$P_i T P_j^\mathsf{T} = 0, \qquad \|T\|_{p_i,p_j} = \|v\|_{p_j} > 0. \tag{5.15}$$

Proof. Norms are invariant under sign and orthogonal changes of singular vectors, so the compression is many-to-one. The final statement follows from $P_i\mathbf{1} = 0$ and direct evaluation of the weighted norm: $\sum_{a,b} p_i(a)p_j(b)v(b)^2 = \sum_b p_j(b)v(b)^2$. □

Two operators with the same scalar shadow

Return to uniform three-state references and consider two gauged blocks

$$G_A = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \qquad G_B = \sqrt{\frac{5}{3}} \begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix}. \tag{5.16}$$

Both have zero row and column means. Under the uniform references used here, their ordinary squared Frobenius norms are both 10, and hence their weighted squared norms are both $10/9$. Their weighted scalar shadows are therefore identical:

$$M_A = M_B = \frac{\sqrt{10}}{3}. \tag{5.17}$$

The operators are nevertheless different. The two nonzero singular values of the whitened G_A are 1 and $1/3$, whereas those of the whitened G_B are both $\sqrt{5}/3$. One concentrates more strength in a leading substitution direction; the other distributes its strength equally across the two-dimensional contrast fiber.

Their action on the same source contrast $f = (1, -1, 0)^\mathsf{T}$ makes the difference explicit:

$$\mathcal{K}_A f = \begin{pmatrix} 1 \\ -2/3 \\ -1/3 \end{pmatrix}, \qquad \mathcal{K}_B f = \frac{1}{3}\sqrt{\frac{5}{3}} \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}. \tag{5.18}$$

The first operator favors the first target contrast; the second suppresses it and favors the second. A contact map built only from M_{ij} assigns the two edges exactly the same value.

This is not an exotic collision. Sign reversal, rotation of singular vectors, and redistribution among singular values all create further operators with the same norm. Scalarization is suitable for asking where strong pair structure is located. It cannot answer which substitutions are compatible, compensatory, or antagonistic. Those are operator-level questions.

This proposition explains why unprojected coupling norms can contain a large conservation- or entropy-correlated mode [16]. Removing such a mode may improve contact ranking; an independent null model determines whether the removed component merits a nuisance interpretation.

For dynamic responses, Eq. 4.14 first yields two directed scalar graphs,

$$M_{i\leftarrow j}^{\text{static}} = \|\overline{G}_{i\leftarrow j}\|_{p_i,p_j}, \qquad M_{i\leftarrow j}^{\text{dynamic}} = \left(\mathbb{E}_C\|\delta G_{i\leftarrow j}(C)\|_{p_i,p_j}^2\right)^{1/2}. \tag{5.19}$$

The first ranks persistent family-level response; the second ranks background sensitivity. Their different biological predictions call for separate reporting. Any undirected contact shadow additionally requires a declared symmetrization of the two directions; reciprocity should be tested before that symmetrization is interpreted as an energy coupling. Chapter 10 shows that scalar, signed, and connection Laplacians built from these shadows or blocks retain different amounts of the original categorical geometry.

Coda: compression is a scientific choice

An operator graph supports several non-equivalent compressions. Support thresholding asks which edges survive. Low-rank approximation asks which substitution directions survive on each edge. Pair truncation asks which higher-order components survive. Scalarization asks which single summary of an edge is retained. A model can be economical in one sense and expensive in another.

This separation changes how graph comparisons should be reported. Edge counts measure support; categorical capacity additionally depends on the maps carried by those edges. Scalar spectra compare one compression of operator action, and contact maps evaluate one structural endpoint. Shared structure may therefore remain in the full operators even when one scalar compression performs poorly.

The same separation controls the physical interpretation. A higher-order ANOVA component certifies irreducibility to lower-order functions in the declared variables and reference measure. The origin of that irreducibility remains an empirical question. It may reflect a genuine many-body term in an effective Hamiltonian, a pair interaction modified by an unobserved solvent or conformational coordinate, the marginalization of latent states, or nonlinear composition inside a predictor. Calling it a three-body or many-body physical force requires the stronger claim that the decomposed function is itself a calibrated energy on a justified coarse-grained state space.

Low rank has a similarly limited physical license. A dominant singular direction is a collective channel of categorical response. It becomes a normal mode only after a self-adjoint physical operator, an inner product with the appropriate metric, and a dynamics or quadratic energy have been specified. The scalar norm of an edge is weaker still: it measures the magnitude of action after gauge projection. Sign, reciprocity, and work around a closed cycle live in additional coordinates. Thus support, rank, order, and norm are possible signatures of physical organization, not physical observables by definition.

So far the geometry has been linear: tables are projected into centered spaces and compared by weighted norms. Probabilities add a nonlinear constraint. A perturbed pair law must remain positive and preserve its declared marginals at finite displacement as well as to first order. The next chapter shows how the linear gauge arises as the tangent geometry of that probability-exact problem.

Chapter 6

Probability-Exact Edge Geometry

A centered interaction table lives in a linear space. A joint probability table occupies a constrained subset: its entries remain nonnegative, sum to one, and often preserve two prescribed marginals. These constraints form a transport polytope. The two geometries describe an infinitesimal interaction coordinate and a finite probabilistic model.

Begin with independent marginals $p_i(a)p_j(b)$ and tilt them by a pair score. At weak coupling, double centering removes directions that only change one marginal, leaving the tangent interaction inside the zero-marginal subspace. At finite coupling, however, linear projection is insufficient: exponentiation bends the geometry and generally changes both marginals.

Exact marginal preservation requires multiplicative row and column factors. Matrix scaling finds those factors and returns the unique information projection onto the transport polytope under the usual positivity assumptions. The nonlinear construction and the linear gauge are complementary resolutions: the gauge-fixed operator is the local tangent coordinate through which the probability-exact family passes.

This relation matters whenever a pair table is given probabilistic language. Tangent-space norms and finite-strength mutual information are different functionals. A centered logit table reaches a joint law with desired marginals through matrix scaling, and separately fitted conditionals reach one global distribution only when their compatibility equations hold. Probability geometry supplies these finite constraints around the local operator coordinate.

The chapter develops that distinction without abandoning the operator view. Exact pair laws, information projections, and tangent operators describe different resolutions of the same local departure from independence. Keeping the resolution explicit prevents a convenient linear score from being mistaken for a fully normalized physical model.

6.1 From tangent coordinates to finite pair laws

The linear gauge identifies a function-space component. Exact preservation of prescribed one-site marginals after exponentiation and normalization additionally requires the transportation polytope

$$\mathcal{U}(p_i, p_j) = \{Q \geq 0 : Q\mathbf{1} = p_i,\ Q^{\mathsf{T}}\mathbf{1} = p_j\}. \tag{6.1}$$

It has dimension $(q-1)^2$, matching the dimension of the categorical tangent-product space. Independence is $Q^0_{ij} = p_i p_j^{\mathsf{T}}$. The standardized residual

$$R_{ij} = D_i^{-1/2}(Q_{ij} - p_i p_j^{\mathsf{T}})D_j^{-1/2} \tag{6.2}$$

has zero constant modes and rank at most $q-1$. It is the correspondence-analysis coordinate of pair association.

6.2 Matrix scaling on the gauge quotient

Given a finite score table S_{ij}, let $K_{ij} = \exp S_{ij}$ elementwise. For strictly positive K_{ij}, p_i, p_j, matrix scaling supplies positive vectors u_{ij}, v_{ij} such that

$$Q_{ij} = \operatorname{diag}(u_{ij}) K_{ij} \operatorname{diag}(v_{ij}) \tag{6.3}$$

has the prescribed marginals [77, 78, 79, 80]. The coupling is unique, while (u, v) are unique up to reciprocal scaling.

Proposition 6.1 (Exact gauge invariance of matrix scaling)**.** *Let $\mathscr{S}_{p_i,p_j}(S)$ denote the coupling obtained by scaling $\exp S$ to marginals (p_i, p_j). For arbitrary vectors a, b and scalar c,*

$$\mathscr{S}_{p_i,p_j}\left(S + a\mathbf{1}^\mathsf{T} + \mathbf{1}b^\mathsf{T} + c\mathbf{1}\mathbf{1}^\mathsf{T}\right) = \mathscr{S}_{p_i,p_j}(S). \tag{6.4}$$

Consequently, the exact nonlinear map $S \mapsto Q$ factors through the same quotient $\mathcal{Q}_{ij}$ as the linear Potts parameter-gauge quotient. The induced map is a bijection from $\mathcal{Q}_{ij}$ onto the relative interior $\operatorname{relint}\mathcal{U}(p_i, p_j)$*, consisting of couplings with strictly positive entries. Finite score tables do not reach the boundary of the transport polytope.*

Proof. Exponentiating the added row, column, and constant terms only left- and right-scales K by positive diagonal matrices and a positive scalar. These factors are absorbed into the Sinkhorn vectors. Uniqueness of the coupling with the prescribed marginals gives Eq. 6.4. Conversely, for any strictly positive $Q \in \mathcal{U}(p_i, p_j)$, choosing $S = \log Q$ returns Q without nontrivial scaling. If two finite score tables return the same positive coupling, their exponentials differ only by the positive diagonal scaling absorbed by Sinkhorn, so their logarithms differ by row, column, and constant terms. This proves bijectivity on the quotient. □

Along $K_\epsilon = \exp(\epsilon S_{ij})$ around independence,

$$\left.\frac{dQ_{ij}(\epsilon)}{d\epsilon}\right|_{\epsilon=0} = D_i(P_i S_{ij} P_j^\mathsf{T}) D_j. \tag{6.5}$$

Thus the reference-centered representative is both the tangent-space linearization of exact marginal-preserving pair geometry and a linear coordinate on the same gauge quotient through which the exact Sinkhorn map factors.

A two-state finite-coupling calculation

The difference between tangent geometry and an exact pair law is already visible for two uniform binary variables. Let

$$p_i = p_j = (1/2, 1/2)^\mathsf{T} \tag{6.6}$$

and choose the centered score

$$S_\theta = \theta \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{6.7}$$

Because symmetry gives equal row and column sums after exponentiation, exact matrix scaling can be written in closed form:

$$Q_\theta = \frac{1}{4\cosh\theta}\begin{pmatrix} e^{\theta} & e^{-\theta} \\ e^{-\theta} & e^{\theta} \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 1+t & 1-t \\ 1-t & 1+t \end{pmatrix}, \qquad t = \tanh\theta. \tag{6.8}$$

Every row and column sums exactly to $1/2$ for every finite θ.

Near independence, $t = \theta + O(\theta^3)$ and

$$Q_\theta - p_i p_j^\top = \frac{\theta}{4}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} + O(\theta^3), \tag{6.9}$$

which is exactly the tangent direction predicted by Eq. 6.5. At finite strength the linear coordinate θ is replaced in the probability table by the bounded coordinate $\tanh\theta$. The transport polytope bends and eventually saturates as probability concentrates on the two matching states.

The mutual information is also exact:

$$I(\theta) = \frac{1+t}{2}\log(1+t) + \frac{1-t}{2}\log(1-t). \tag{6.10}$$

For weak coupling, $I(\theta) = \theta^2/2 + O(\theta^4)$, agreeing with the quadratic tangent norm. At strong coupling, the nonlinear probability geometry matters and the quadratic approximation no longer describes the finite law.

Finally, add arbitrary row and column scores to S_θ. They change the unscaled exponential matrix, but exact marginal scaling absorbs them and returns the same Q_θ. The nonlinear probability model therefore respects precisely the same gauge quotient as the linear operator, while encoding finite displacement on that quotient rather than only its tangent direction.

6.3 Local pair laws and global compatibility

The exact conditional $q_{i|j}(a \mid b) = Q_{ij}(a,b)/p_j(b)$ is normalized for each b, and averaging over $b \sim p_j$ returns p_i. Its mutual information

$$I_{ij} = \mathrm{KL}(Q_{ij} \| p_i p_j^\top) \tag{6.11}$$

is an exact nonnegative scalar association. Near independence,

$$I_{ij} = \frac{1}{2}\|R_{ij}\|_F^2 + O(\|R_{ij}\|_F^3), \tag{6.12}$$

with a remainder controlled only when the smallest marginal probability stays away from zero. Rare and unsupported categories therefore require an explicit smoothing convention.

This construction separates three levels:

1. $P_i S_{ij} P_j^\top$ is a marginal-orthogonal logit contrast;
2. $Q_{ij} \in \mathcal{U}(p_i, p_j)$ is one exact pair law with prescribed marginals;
3. a collection of pair laws is globally probabilistic only if it lies in the marginal polytope of one compatible sequence distribution.

The third property is a global compatibility constraint beyond a sparse pair layer or pairwise Sinkhorn scaling. Marginal log-linear, discrete dependence, and minimum-information models provide global alternatives but introduce a separate estimation problem [9, 10, 11].

Coda: local coordinates and finite laws

The reference-centered operator is now connected to an exact statistical object. Linear double projection gives the tangent direction that preserves marginals to first order. Multiplicative matrix scaling follows the same quotient at finite displacement and enforces the marginal constraints exactly. Information geometry explains both the relation and the difference.

This distinction will recur in every model correspondence. Potts fields may belong to one global Gibbs law. Gaussian precision blocks encode conditional structure under a different sample space. Hopfield and RBM free energies arise by marginalizing different latent variables. Transformer softmaxes usually normalize routes rather than configurations. Similar algebra does not tell us which level of probability geometry a model inhabits.

Matrix scaling also admits an exact variational reading. If $\overline{K} = \exp(S)/\sum_{a,b} \exp S(a,b)$, then the scaled coupling satisfies

$$Q^* = \operatorname*{argmin}_{Q \in \mathcal{U}(p_i, p_j)} \mathrm{KL}(Q \| \overline{K}). \tag{6.13}$$

It is the distribution closest to the score-induced pair law while obeying the prescribed one-site marginals. If $S = -E/\tau$, the same objective can be written as an energy–entropy competition, which explains the formal connection to Gibbs and free-energy language.

The physical scope of that connection depends on what supplied E, τ, and the marginals. They must represent a justified coarse-grained energy, temperature, and ensemble before Q^* can be read as a physical equilibrium pair law. Even then, one pair law is only a marginal; separately valid pair couplings need not be compatible with a single many-residue distribution. Mutual information measures departure from independence in this probability geometry. It is not a binding free energy, an interaction enthalpy, or a mechanical coupling unless an additional thermodynamic construction establishes that identity. When those conditions are absent, the variational statement remains exact as information geometry and should be named that way.

With the common edge coordinate and its probabilistic boundary established, the book can now turn outward. The next part places familiar sequence and neural models into these coordinates, then asks how their operators are estimated and what survives spectral compression.

Part III

Models, Estimation, and Spectra

Chapter 7

Model Correspondences in Operator Coordinates

A common mathematical language is useful only if it preserves the differences among the models being translated. Potts models, Gaussian graphical models, Hopfield systems, restricted Boltzmann machines, and Transformers all combine indexed states through familiar linear maps and nonlinear normalizations. The resemblance is real. So are the differences in what is normalized, what is latent, what is updated, and what qualifies as an edge.

This chapter uses three passes. The model portraits first compare systems horizontally in one set of coordinates. Formal correspondences then state exactly what survives projection. Two longitudinal cases finally follow equations through historical transitions: the 2019 unified reconstruction framework, and the path from spectral graph filters to typed geometric messages. The three passes are complementary views governed by the same questions. What is the state space? Is there a joint energy, a collection of conditionals, or only an update rule? What does an edge carry? Which variable is marginalized or normalized? Is the graph fixed across data, inferred per input, or merely the execution support? What claim remains unavailable even after the algebra has been matched?

The operator coordinate supplies the common frame. A Potts coupling can be projected into a canonical categorical block. A Gaussian precision block supplies a conditional linear analogue under a continuous law. Hopfield–Potts patterns factorize a static pair energy. Marginalized RBM features generate a hierarchy of visible interaction orders. A Transformer exposes routes and can be audited for categorical output response. Typed extraction places these objects in shared coordinates; their native parameters remain model-specific.

The most instructive correspondences and historical transitions therefore come with a remainder. A shared softmax leaves the sample space to be specified. A shared matrix product leaves the energy or update law to be identified. A low-rank latent factor distributes influence across residue pairs, while dense attention support shares parameters across its realized routes. These remainders are the typing information that makes the unification useful.

The framework compares sequence models coordinate by coordinate. A model M enters through the typed extraction

$$\mathrm{Audit}(M) = (h_M, \mathcal{K}_M, E_M, \mathcal{C}_M, \mathcal{O}_M, \mathcal{D}_M, \mathcal{N}_M, \mathcal{E}_{\mathrm{st},M}, \mathcal{V}_{\mathrm{d},M}), \tag{7.1}$$

This full record replaces comparison by one displayed matrix.

The portraits below follow these coordinates model by model and retain the first claim that remains unlicensed after each correspondence.

7.1 One residue pair through six models

Return to the positions (i, j) used throughout the book. Instead of asking which model is most powerful, ask what object each model assigns to this pair and what operation would be required to compare the assignments.

Portrait I: the PSSM has no edge

A position-specific scoring matrix assigns a field $h_i(a)$ to each residue identity at each site. Its reconstruction rule may be written

$$P(X_i = a) = \mathrm{softmax}_a\, h_i(a). \tag{7.2}$$

There is no off-diagonal operator from j to i. Correlation can still appear in the observed family because sequences are phylogenetically related or sampled from subfamilies, but the PSSM does not represent that correlation as a pair interaction. In operator coordinates it occupies the node-field branch and sets

$$\mathcal{K}_{i\leftarrow j} = 0. \tag{7.3}$$

This apparently trivial portrait is useful because it prevents every successful residue predictor from being described as an interaction model.

Portrait II: the Potts model assigns a static categorical operator

A pairwise Potts model attaches a table $J_{ij}(a, b)$ to the unordered pair. After reference projection,

$$G^{\mathrm{P}}_{ij} = P_i J_{ij} P_j^{\mathsf{T}} \tag{7.4}$$

is a static operator shared by every sequence evaluated under that fitted family model. The conditional probability at i changes with the surrounding sequence because all neighboring terms are summed, but the response rule contributed by j is fixed.

The edge belongs to one joint Gibbs law when the pair tables and fields are assembled consistently. That gives the Potts block a stronger probability semantics than an attention route. It does not give it physical units, remove phylogenetic bias, or make every nonzero block a contact. Those are estimation and validation questions.

Portrait III: Gaussian precision changes the state space

For a nondegenerate Gaussian vector, a zero precision block has an exact conditional-independence meaning. The edge object is a linear map between continuous coordinates, not a table between categorical contrasts. If one-hot residue variables are approximated by a Gaussian, the precision matrix may provide a useful estimator, but the approximation does not erase the simplex constraint or reference-centered categorical coordinates.

The correspondence is therefore structural:

$$\text{precision block} \longleftrightarrow \text{conditional linear edge}, \tag{7.5}$$

not literal equality with a Potts coupling. The two models agree that conditional dependence is encoded by an off-diagonal block; they disagree about the node space and probability law in which that block lives.

Portrait IV: Hopfield–Potts compresses edges into collective patterns

A Hopfield–Potts model factorizes static couplings through patterns

$$J_{ij}(a,b) = \sum_{\mu=1}^{p} \lambda_\mu \xi_{i\mu}(a)\xi_{j\mu}(b). \tag{7.6}$$

For fixed p, many edge operators share a small collection of categorical directions. The model is dense in position support but low rank in its global parameterization. One pattern can touch many positions without declaring every touched pair to be a separate physical contact.

This portrait separates support from rank. A sparse Potts graph may carry independent full-rank operators on a few edges. A Hopfield–Potts model may carry correlated low-rank operators on nearly all edges. Both are pairwise energies; their compression assumptions are different.

Portrait V: an RBM stores motifs and generates all visible orders

A restricted Boltzmann machine introduces hidden variables r_μ coupled to visible residues. Conditional on the hidden layer, the visible sites separate. After the hidden variables are marginalized, each softplus term depends jointly on every visible position connected to that hidden unit. Its expansion contains one-body, pairwise, and higher-order components.

An RBM hidden unit is therefore closer to a distributed motif than to one edge. Its pairwise operator can be extracted by functional projection, but that operator is only one shadow of the latent factor. Reading all pairs touched by a hidden unit as direct contacts would confuse latent support with visible interaction order.

The contrast with modern Hopfield retrieval is equally important. Independent Bernoulli hidden variables allow several motifs to be active simultaneously. A categorical Hopfield softmax makes stored memories compete on one simplex. Both use log-sum-exp after marginalization; they sum over different latent sample spaces.

Portrait VI: the Transformer has a route and an audited response

For a Transformer head, the native edge coefficient is

$$\alpha_{ij}^{(h)}(x) = \mathrm{softmax}_j\, s_{ij}^{(h)}(x). \tag{7.7}$$

It tells us how a target distributes routing mass over source positions. The associated frozen value-path map lies in the span of the head value/output maps. Both quantities depend on the current hidden state and belong to the computation.

The operator comparable with a Potts block is obtained elsewhere: vary the source residue, hold a declared background fixed, observe the categorical output at the target, and project the resulting table. This gives $G_{i\leftarrow j}(C)$, a directed and generally background-dependent response. The route and response may correlate, but they answer different questions:

$$\begin{aligned} \alpha_{ij}^{(h)}(x): &\quad \text{where one layer reads,} \\ G_{i\leftarrow j}(C): &\quad \text{how the complete model prediction changes.} \end{aligned} \tag{7.8}$$

The Transformer portrait is consequently double. It is a dynamic graph computation internally and a background-dependent operator field under external audit. Neither object is automatically a joint sequence energy.

7.2 What survives comparison

The six portraits meet only after a typed extraction. PSSM fields, Potts blocks, Gaussian precision, Hopfield factors, RBM projections, attention routes, and pLM responses should not be placed in one matrix merely because they share position indices. What can be compared is more specific:

$$\boxed{\text{state space} + \text{edge operator} + \text{context} + \text{order} + \text{normalization} + \text{estimator.}} \tag{7.9}$$

Agreement in the edge-operator coordinate can then be measured while the remaining differences are kept as part of the result.

7.3 Static fields, Gaussian conditionals, and low-rank patterns

A PSSM retains only the one-body branch $h_i(a)$ and therefore occupies the field coordinate with zero off-diagonal edge operator. A Potts model adds static full categorical edge blocks. We first retain the model-specific coordinates before asking which of them can be collapsed into a shared reconstruction diagram.

For a nondegenerate Gaussian vector with covariance Σ and precision $\Omega = \Sigma^{-1}$,

$$\Omega_{ij} = 0 \iff X_i \perp\!\!\!\perp X_j \mid X_{[L]\setminus\{i,j\}}. \tag{7.10}$$

This is a distributional theorem under a Gaussian law [2]. Leading-mode deletion is a separate spectral operation, while a Gaussian approximation to one-hot protein variables still requires reference-centered Potts coordinates for categorical comparison.

A classical Hopfield–Potts restriction factorizes the static coupling as

$$J_{ij}(a,b) = \sum_{\mu=1}^{p} \lambda_\mu \xi_{i\mu}(a)\xi_{j\mu}(b). \tag{7.11}$$

It is a low-rank parameterization of a pairwise Potts energy [47, 15]. Attention belongs to a separate state-dependent routing construction. Positive and negative pattern weights select different operator directions, and gauge projection is still required. At the pairwise level one may test

$$\mathbb{K}^{(2)} = \mathbb{K}_{\mathrm{LR}} + \mathbb{K}_{\mathrm{sp}}, \tag{7.12}$$

where the two terms represent distributed collective modes and localized typed constraints. This decomposition is a model hypothesis: low-rank and sparse classes overlap, so recovery needs declared penalties, incoherence assumptions, synthetic ground truth, or independent interventions.

7.4 Transformer computation and interaction graphs

A self-attention layer is a message-passing operation whose nodes are tokens. Its candidate support is usually the complete directed graph (or the graph allowed by a mask), whereas its effective weighted route is generated anew from the current representations. For head h,

$$\alpha_{ij}^{(h)}(x) = \mathrm{softmax}_j\left(\frac{Q_i^{(h)}(x)^\top K_j^{(h)}(x)}{\sqrt{d_h}}\right), \qquad y_i = \sum_{j,h} \alpha_{ij}^{(h)} B_h x_j. \tag{7.13}$$

In this computational sense a Transformer instantiates a special dynamic GN block: the same query–key and value rules are shared across token pairs, the positional aggregation is permutation-respecting before positional encodings or masks are supplied, and the effective edge weights depend on the current sequence and layer [33, 50, 51, 52]. Training therefore learns a shared graph-generating rule $X \mapsto P_h(X)$. The resulting route is input-specific, while the statistical operator in Eq. 3.31 is obtained through a separate output audit.

There are at least four non-equivalent symmetry questions inside Eq. 7.13. For a head with shared linear query–key maps and no additional position-dependent score transform, let X collect the current token states and set

$$C_h = W_Q^{(h)} W_K^{(h)\mathsf{T}}, \qquad S_h(X) = \frac{X C_h X^{\mathsf{T}}}{\sqrt{d_h}}, \qquad P_h(X) = \mathrm{softmax}_{\mathrm{row}}\, S_h(X), \tag{7.14}$$

one may ask whether (i) C_h is symmetric, (ii) the token logit matrix $S_h(X)$ is symmetric, (iii) the normalized route $P_h(X)$ is reversible, or (iv) the audited categorical response obeys Eq. 4.28. Symmetry of C_h implies symmetry of $S_h(X)$ before an asymmetric mask, relative-position bias, or position-dependent query–key transform is applied, but row normalization generally gives $P_{ij} \neq P_{ji}$. Stationary flux, rather than entrywise symmetry, determines whether this asymmetry carries nonreciprocal circulation. Indeed, if $K_{ij} = \exp S_{ij} = K_{ji}$, $z_i = \sum_j K_{ij}$, and

$$P_{ij} = \frac{K_{ij}}{z_i}, \qquad \pi_i = \frac{z_i}{\sum_k z_k}, \tag{7.15}$$

then

$$\pi_i P_{ij} = \pi_j P_{ji}. \tag{7.16}$$

Thus a symmetric score followed by row-softmax is generally asymmetric in Euclidean coordinates but reversible in its stationary geometry. An asymmetric query–key kernel or mask can break this property, although either can also produce accidental reversibility on a particular input.

This distinction sharpens recent empirical results. Saponati et al. analyze symmetry and column dominance primarily in C_h, reporting that bidirectional objectives favor symmetric structure whereas autoregressive objectives favor directionality; symmetric initialization improves several encoder-language training runs, while the reported vision gains are not significant and large-scale behavior remains open [54]. Courtois et al. similarly improve BERT-style efficiency with a symmetric dot-product parameterization [53]. Neither result addresses the normalized token route or the end-to-end substitution response directly. Doubly stochastic attention, in turn, balances incoming and outgoing mass while allowing asymmetric directed cycles. Sinkformers implement this constraint by iterative matrix scaling, while ESPFormer uses parallel expected-sliced transport and reports improvements across several modalities [55, 56]. Marginal balance and detailed balance remain two independently testable properties.

At fixed attention coefficients, the linear value-path map is

$$\mathcal{A}_{i\leftarrow j} = \sum_{h=1}^{H} \alpha_{ij}^{(h)} B_h. \tag{7.17}$$

Proposition 7.1 (Shared span of a frozen attention layer)**.** *Every map in Eq. 7.17 lies in the shared subspace* $\mathrm{span}\{B_1, \ldots, B_H\}$*. Dense connectivity therefore distributes a shared H-dimensional operator family across its edges.*

The complete input–output response also contains derivatives or finite differences through queries, keys, earlier layers, residual paths, and nonlinear blocks. This is why the categorical output audit in Eq. 4.6 is the proper model-level object for comparison with a Potts block. The attention map supplies the route graph. Route and response graphs should therefore be reported separately: the first describes where a layer reads, while the second describes how a source substitution changes the model's categorical prediction.

7.5 MSA Transformer as amortized family inference

MSA Transformer makes the family sample an explicit neural state rather than compressing it into fixed counts before the network [63]. For an alignment with R homologous sequences and L positions, a layer acts on

$$H_f \in \mathbb{R}^{R\times L\times d}. \tag{7.18}$$

Column attention compares alignment rows at one fixed position,

$$A^{\mathrm{col}}_{i,rs} = \mathrm{softmax}_s\left(\frac{q_{ri}^{\top}k_{si}}{\sqrt{d}}\right), \qquad y^{\mathrm{col}}_{ri} = \sum_s A^{\mathrm{col}}_{i,rs}v_{si}, \tag{7.19}$$

whereas tied row attention aggregates evidence across homologues to produce one position route shared by all rows, schematically

$$S^{\mathrm{row}}_{ij} = \frac{1}{R}\sum_{r=1}^{R}\frac{q_{ri}^{\top}k_{rj}}{\sqrt{d}}, \qquad A^{\mathrm{row}}_{ij} = \mathrm{softmax}_j\, S^{\mathrm{row}}_{ij}, \qquad y^{\mathrm{row}}_{ri} = \sum_j A^{\mathrm{row}}_{ij}v_{rj}. \tag{7.20}$$

Thus the native computation contains two graph families: an R-node sequence graph for each position and an L-node position graph conditioned on the complete MSA. Axial factorization reduces full attention over RL cells from $O(R^2L^2)$ to $O(LR^2 + RL^2)$ per layer. Alternating the axes still permits information to travel from (r,i) to (s,j) through two-hop paths, but it is an architectural factorization rather than an exact equality with unrestricted joint attention.

The cross-family commonality is most cleanly expressed hierarchically. Let Θ_f denote the family-specific constraint system—fields, typed couplings, structural organization, and functional restrictions—and let $\mathcal{T}_f$ denote phylogeny and sampling. A schematic population model is

$$\Theta_f \sim P_\star(\Theta), \qquad \mathcal{M}_f \sim p(\mathcal{M} \mid \Theta_f, \mathcal{T}_f). \tag{7.21}$$

Families do not share one coupling graph. They share recurring amino-acid response motifs, structural graph organization, conservation and compensation patterns, and the statistical task of separating constraint from finite depth, gaps, subfamilies, and common ancestry. Training across many alignments learns an amortized inference rule

$$\boxed{\mathcal{A}_\theta : \mathcal{M}_f \longmapsto \widehat{\Theta}_f \text{ or directly to family-conditioned predictions.}} \tag{7.22}$$

Here $\widehat{\Theta}_f$ is a conceptual audit coordinate, not necessarily an explicit stored model variable. The current MSA supplies family evidence; the shared parameters supply a cross-family prior over how such evidence should be interpreted.

This gives a precise intermediate position between DCA and a single-sequence pLM:

$$\begin{aligned}
\text{Potts / DCA:}\quad & \mathcal{M}_f \xrightarrow{\text{family optimization}} \widehat{J}_f,\\
\text{MSA Transformer:}\quad & (\theta_\star, \mathcal{M}_f) \xrightarrow{\text{forward inference}} G_f(C;\mathcal{M}_f),\\
\text{single-sequence pLM:}\quad & (\theta_\star, x) \xrightarrow{\text{amortized inference}} G(C;x).
\end{aligned} \tag{7.23}$$

The first estimates one static family field, the second combines an explicit family sample with a cross-family learned estimator, and the third relies on information already amortized into model parameters. Tied row attention can contain contact-predictive statistics, but its scalar route is not a $q \times q$ Potts block. Column attention is likewise a normalized route over homologues, not a phylogenetic tree. Output substitution audits, clade-aware nulls, and row/column interventions are required to distinguish reusable family constraints from ancestry and sampling shortcuts.

7.6 Softmax over different sample spaces

The Potts conditional in Eq. 4.3 normalizes over residue identities at one position. Attention in Eq. 7.13 normalizes over source positions. Modern Hopfield retrieval normalizes over stored memories,

$$q^{\text{new}} = \Xi \,\text{softmax}_\mu(\beta \Xi^\mathsf{T} q), \tag{7.24}$$

and can be derived from the Lyapunov function

$$E_{\text{MH}}(q) = \frac{1}{2}\|q\|_2^2 - \frac{1}{\beta} \log \sum_\mu e^{\beta \xi_\mu^\mathsf{T} q} + C \tag{7.25}$$

under the assumptions of the modern Hopfield construction [49]. One attention readout matches this update when keys and returned values are tied or compatibly mapped. Untied values, changing memories, residual connections, and feed-forward layers generally define a different dynamics whose energy descent must be established separately.

The bridge between latent-variable models and modern Hopfield retrieval becomes exact after the hidden sample space is specified. Replace the independent Bernoulli hidden vector of an RBM by one categorical memory variable $Z \in [M]$ and define

$$E(q, Z = \mu) = \frac{1}{2}\|q\|_2^2 - \xi_\mu^\mathsf{T} q. \tag{7.26}$$

Its posterior is

$$P(Z = \mu \mid q) = \text{softmax}_\mu(\beta \Xi^\mathsf{T} q), \tag{7.27}$$

and marginalizing Z gives Eq. 7.25 up to a constant. The essential distinction is therefore

$$\boxed{\begin{aligned} &\text{Bernoulli RBM:} \quad \log \sum_{r \in \{0,1\}^M} e^{r^\mathsf{T} s} = \sum_\mu \log(1 + e^{s_\mu}), \\ &\text{modern Hopfield:} \quad \log \sum_{\mu=1}^{M} e^{\beta \xi_\mu^\mathsf{T} q}. \end{aligned}} \tag{7.28}$$

The first permits multiple motifs to be active independently; the second makes memories compete on one simplex. Transformer attention applies the same categorical-memory posterior separately at each target, but its memories are generated by the current sequence rather than stored as fixed parameters. Multi-head attention repeats retrieval in several learned memory spaces.

The three energy-like quantities must therefore remain distinct:

$$\boxed{E_{\text{Potts}}(x) \neq E_{\text{Hopfield}}(q) \neq -\log \alpha_{ij}.} \tag{7.29}$$

They score a sequence configuration, a retrieval state, and one normalized routing coordinate, respectively.

7.7 Latent factors and interaction order

For binary hidden variables r_μ, an RBM energy can be written

$$E_{\mathrm{RBM}}(x, r) = -\sum_i g_i(x_i) - \sum_\mu c_\mu r_\mu - \sum_{i,\mu} W_{i\mu}(x_i) r_\mu. \tag{7.30}$$

Marginalizing the hidden layer gives

$$\log \widetilde{P}_{\mathrm{RBM}}(x) = \sum_i g_i(x_i) + \sum_\mu \log \left[1 + \exp\left(c_\mu + \sum_i W_{i\mu}(x_i)\right)\right]. \tag{7.31}$$

The nonlinear softplus generates one-body, pairwise, and higher-order terms. Its weak-coupling quadratic component is low rank across positions; a Potts approximation additionally requires a negligible higher-order remainder. Protein RBMs can therefore represent extended motifs without identifying every pair touched by a hidden unit as a direct contact [46, 48].

The useful unification is consequently typed: Potts couplings, low-rank Hopfield–Potts patterns, RBM-induced pair components, and pLM substitution responses can all be projected into the same edge-operator coordinates. Equality in those coordinates is then an empirical hypothesis tested after the shared matrix multiplication or softmax has been typed.

7.8 A longitudinal case: the 2019 unified framework

In 2019, Dauparas et al. proposed a *Unified Framework for Modeling Multivariate Distributions in Biological Sequences* [32]. The paper placed PSSMs, Potts or Markov random fields, multivariate Gaussian models, and autoencoders inside one reconstruction diagram. Read retrospectively, its importance lies in two achievements. It exposed exact equivalences that had been hidden by disciplinary notation, and it made those equivalences operational enough that a regularization choice could be transferred from one model family to another. It also shows where semantic typing begins after an algebraic unification succeeds.

This distinction marks a genuine progression that builds on the original result. The 2019 question was: how many familiar sequence models can be implemented as nearby versions of one input–output reconstruction system? The question of this book is stricter: after that implementation has been unified, which mathematical object does each system identify, and what must remain different when probability, energy, computation, and experiment are translated?

7.8.1 The common reconstruction diagram

Let an MSA contain N sequences of length L over an alphabet of size q. Flatten the one-hot tensor to $X \in \{0, 1\}^{N \times d}$ with $d = Lq$. A linear reconstruction system has preactivation

$$H = B + XW, \tag{7.32}$$

where B supplies one-site fields and W carries cross-coordinate dependence. The models in the 2019 framework are obtained by changing four choices: the admissible blocks of W, the output loss, the treatment of self-connections, and the regularizer. Adding hidden layers replaces the single linear map by a nonlinear encoder–decoder while preserving the reconstruction view.

For a PSSM, $W = 0$ and a sitewise categorical softmax learns only $B_i(a)$. For a Potts pseudolikelihood model, the logits at target position i are

$$H_i(a; x_{-i}) = B_i(a) + \sum_{j \neq i} \sum_b x_j(b) W_{j \to i}(b, a), \tag{7.33}$$

and categorical cross-entropy is applied over the residue alphabet at i. Because one-hot targets satisfy $\sum_a x_i(a) = 1$, the loss is exactly the negative sitewise Potts log pseudolikelihood:

$$\mathcal{L}_{\mathrm{CCE}} = -\frac{1}{N} \sum_{n,i} \log \frac{\exp H_i(x_i^{(n)}; x_{-i}^{(n)})}{\sum_a \exp H_i(a; x_{-i}^{(n)})} = \mathcal{L}_{\mathrm{PL}}. \tag{7.34}$$

The equality is an exact identity of objectives after the categorical sample space and the targetwise normalization have been matched.

The Gaussian branch begins from a different law. Center the flattened observations to $\widetilde{X}$ and consider a regularized self-reconstruction objective of the form

$$\mathcal{L}_{\mathrm{MSE}}(W) = \frac{1}{N} \|\widetilde{X} - \widetilde{X} W\|_F^2 + \lambda \|W\|_F^2 - 2\gamma \operatorname{tr}(W). \tag{7.35}$$

Writing $C = N^{-1} \widetilde{X}^\top \widetilde{X}$, its stationary solution is

$$W = (\gamma - \lambda)(C + \lambda I)^{-1} + I. \tag{7.36}$$

After the identity map is separated, the learned cross-coordinate map is therefore an affine transform of a regularized precision matrix. With the corresponding choice of γ, the MSE system and Gaussian likelihood share the same global optimum, even though one is written as a neural self-map and the other as covariance inversion.

The resulting family resemblance can be summarized without erasing the differences:

Model	Reconstruction map	Objective	Native probability object	First semantic remainder
PSSM	bias field only	sitewise CCE	independent categorical sites	no represented pair edge
Potts/MRF	linear cross-position blocks	categorical CCE = pseudolikelihood	compatible categorical conditionals, when parameters are tied into one law	raw block retains gauge freedom; contact is a separate endpoint
Gaussian/MG	centered linear self-map	regularized MSE	continuous Gaussian law through precision	one-hot simplex and categorical contrasts are approximated
Autoencoder	factorized or nonlinear reconstruction	chosen output loss	generally a conditional predictor unless a joint law is supplied	hidden paths can induce higher-order final responses

The diagram unified implementation at exactly the moment when neural-network software made that unification useful. Pseudolikelihood Potts fitting could be expressed as ordinary categorical reconstruction; Gaussian inverse-covariance estimation could be expressed through an MSE layer; and either branch could acquire hidden layers without changing the outer input–output task.

7.8.2 The empirical test made the unification scientific

The 2019 work went beyond redrawing four models with the same boxes. Its contact-prediction experiment asked whether a choice that worked in one formulation could improve another. The MRF literature commonly used L_2 regularization, whereas Gaussian DCA was often expressed through a pseudocount applied to the covariance. Transferring the MRF-style shrinkage logic to the MSE formulation improved contact prediction over GaussDCA. Removing the diagonal self-map gave a further small improvement and connected the resulting score to partial correlation. The categorical cross-entropy model still performed better, but the gap narrowed when the regularization comparison was made more like-for-like.

This is the strongest form of useful mathematical unification. An equivalence should permit a controlled transfer of an estimator, regularizer, diagnostic, or computational method. Here it did. The result showed that part of the apparent difference between model families came from training conventions and could be reduced by matching those conventions across graphical forms. It also showed that the remaining difference mattered: categorical cross-entropy respected the amino-acid sample space more faithfully than squared error on a continuous relaxation.

The contact endpoint introduced one more irreversible step. Both fitted matrices were reduced to scalar position scores, corrected by average-product correction, and ranked against structural contacts. Success at that endpoint established predictive utility for contact ranking. Gaussian precision entries, Potts tables, and neural weights retained their native meanings; the common benchmark compared their scalar shadows after model-specific estimation and shared postprocessing.

7.8.3 What was unified, and what remained plural

The phrase “the difference between the models comes down to the loss function” is exact at the level of the single-layer reconstruction diagram under the paper’s stated constraints. It becomes too strong if it is transported unchanged to probability or physics. Four remainders survive.

First, the sample spaces differ. Potts pseudolikelihood normalizes over residue identities at one site while holding a categorical background fixed. Gaussian MSE treats centered coordinates as continuous. The shared outer matrix multiplication therefore sits above two distinct probability geometries.

Second, a collection of categorical reconstructions yields one joint energy when parameter tying, gauge handling, and compatibility of the targetwise conditionals. A generic linear CCE network may have directed blocks $W_{j\to i}$ and $W_{i\to j}$ that fail the reciprocity criterion of Eq. 4.28. Its loss remains well defined, while the joint-Hamiltonian reading stops at the failed compatibility condition.

Third, the matrix W carries several kinds of structure at once. Its diagonal may encode trivial self-reconstruction. Its row and column directions may contain one-body gauge terms. A block norm for contact prediction removes sign and categorical orientation. The 2019 framework made the shared matrix visible; the reference-centered operator construction of Chapter 3 identifies which part of each categorical block is irreducibly bivariate.

Fourth, adding hidden layers changes more than expressive power. Nonlinear composition can turn pairwise first-layer messages into higher-order output responses. The weight connecting two visible coordinates, a hidden computational route, and the audited substitution response of the complete autoencoder are therefore different edge candidates. The reconstruction diagram leaves their selection to a typed output audit.

These remainders can be stated as a progression of audit maps:

$$\begin{aligned} \text{2019:}\quad & M \longmapsto (\text{reconstruction map},\ \text{loss},\ \text{regularizer}), \\ \text{this book:}\quad & M \longmapsto (\text{fiber},\ \text{edge operator},\ \text{context},\ \text{order},\ \text{normalization},\ \text{estimator},\ \text{validation}). \end{aligned} \tag{7.37}$$

The second map extends the first from the point where it succeeds. Once common code and common algebra have removed superficial differences, the remaining differences become sharp enough to type.

7.8.4 From horizontal equivalence to a longitudinal program

The 2019 framework is a horizontal unification across contemporary model families. Its later consequences are longitudinal. PSSM bias fields became the simplest node-state baseline. Potts CCE made masked categorical prediction continuous with pseudolikelihood. Autoencoder hidden layers opened the route from static family models to amortized nonlinear predictors. Modern pLMs greatly enlarged that route, but in doing so separated the training objective, internal communication graph, and query-specific output response more strongly than the one-layer picture had to confront.

The scientific question consequently moved. It was first useful to ask whether PSSM, MRF, Gaussian, and autoencoder fitting could share one computational grammar. It is now necessary to ask which typed relations survive depth, pretraining, context dependence, and geometric supervision. The operator graph is the next unification in that sequence: a coordinate system that records the exact correspondence and the exact remainder together.

This case also supplies a standard for the longitudinal analyses that follow. A historical sequence of models becomes theoretically informative only when each transition identifies four things: the state that was added, the information that was discarded, the objective that licensed the new state, and the scientific claim that still remained unavailable. The GCN lineage below applies the standard to graph computation; Chapter 8 then applies it to the three generations of AlphaFold.

7.9 A second longitudinal case: from spectral filters to geometric messages

The phrase “graph convolutional network” now covers a lineage of spectral filtering, neighborhood aggregation, and typed message passing. Spectral networks begin from functional calculus of a graph Laplacian. The Kipf–Welling GCN retains a first-order normalized propagation rule. GraphSAGE turns propagation into permutation-invariant neighborhood aggregation. GAT makes the route state-dependent. MPNNs allow typed edge messages, GIN asks which neighborhood multisets survive aggregation, and geometric networks constrain how scalar, vector, and coordinate states transform. Each transition changes what an edge does as well as increasing capacity.

This history is especially useful here because the equations look continuous while their meanings change. The audit will follow

$$\boxed{\begin{aligned} \text{spectral function} &\longrightarrow \text{local polynomial}, \\ \text{fixed normalized route} &\longrightarrow \text{adaptive aggregation}, \\ &\longrightarrow \text{typed geometric message}. \end{aligned}} \tag{7.38}$$

At no point does successful propagation alone identify a statistical interaction, a physical force, or a biological mechanism.

7.9.1 Spectral convolution is functional calculus on a declared graph

Let A be a symmetric nonnegative adjacency matrix, $D_{ii} = \sum_j A_{ij}$, and

$$L = I - D^{-1/2} A D^{-1/2} = U \Lambda U^{\mathsf{T}} \tag{7.39}$$

be the normalized graph Laplacian. The columns of U replace Fourier modes: $U^{\mathsf{T}} x$ are the graph-frequency coefficients of a node signal x. A spectral convolution is defined by

$$g_\theta *_G x = U g_\theta(\Lambda) U^{\mathsf{T}} x = g_\theta(L) x. \tag{7.40}$$

The equality on the right is the important one. The native object is a spectral function of a specified Laplacian. Low values of λ describe signals that vary slowly across heavily weighted edges, so a filter concentrated at low graph frequencies acts as a graph-relative smoother [34, 35].

Nothing in Eq. 7.40 discovers what the graph means. If A is a covalent graph, a spatial contact graph, a sequence-neighborhood graph, or a learned similarity graph, the same functional calculus has different semantics. The spectrum belongs to the pair $(L, \mathcal{H}_{\text{node}})$ after the node state and edge weights have been declared.

The original spectral parameterization also has two practical liabilities. Applying an arbitrary $g_\theta(\Lambda)$ appears to require eigendecomposition, and parameters attached to the eigenvalues of one graph transfer only through an additional alignment or parameterization. Locality appears when the multiplier is restricted, since an unrestricted spectral multiplier generally produces a dense operator in node coordinates.

7.9.2 Chebyshev filters convert frequency response into finite-hop propagation

ChebNet replaces the unrestricted spectral multiplier by a polynomial. Rescale the spectrum to $[-1, 1]$ with

$$\widetilde{L} = \frac{2}{\lambda_{\max}} L - I \tag{7.41}$$

and use Chebyshev polynomials T_k:

$$g_\theta(L) x \approx \sum_{k=0}^{K} \theta_k T_k(\widetilde{L}) x. \tag{7.42}$$

Because a degree-k polynomial in L connects only nodes reachable within k graph steps, a degree-K filter is K-hop localized. Spectral approximation has become a spatial propagation rule without ceasing to be a function of the Laplacian.

This step exposes an assumption that the Fourier notation can hide. A polynomial filter shares coefficients across edges: the graph supplies the support and geometry, while the shared coefficients $\theta_0, \ldots, \theta_K$ specify how different path lengths and spectral scales are combined. Two edges with the same local role are processed by the same polynomial.

7.9.3 The classical GCN is a tied first-order filter

The familiar GCN layer follows from a strong truncation of Eq. 7.42 [36]. Take $K = 1$ and approximate $\lambda_{\max} \simeq 2$. Since

$$\widetilde{L} = L - I = -D^{-1/2} A D^{-1/2}, \tag{7.43}$$

the one-channel filter becomes

$$\theta_0 x - \theta_1 D^{-1/2} A D^{-1/2} x. \tag{7.44}$$

Tying $\theta_0 = -\theta_1 = \theta$ gives

$$\theta \left(I + D^{-1/2} A D^{-1/2} \right) x. \tag{7.45}$$

The subsequent renormalization trick adds self-loops before normalization:

$$\widehat{A} = A + I, \qquad \widehat{D}_{ii} = \sum_j \widehat{A}_{ij}, \qquad S = \widehat{D}^{-1/2} \widehat{A} \widehat{D}^{-1/2}. \tag{7.46}$$

For a feature matrix $H^{(\ell)}$, the layer is

$$H^{(\ell+1)} = \sigma\left(S H^{(\ell)} W^{(\ell)} \right). \tag{7.47}$$

Equation 7.47 should be read as two operators with different axes:

$$H \xrightarrow{S} \text{mix node states along the declared graph} \xrightarrow{W} \text{mix feature channels}. \tag{7.48}$$

The propagation matrix S is fixed once the input graph is fixed; the trainable matrix W is shared over nodes. Standard GCN therefore learns channel transformations around a prescribed, degree-normalized node-mixing rule, with edgewise freedom limited by the input graph.

The self-loop has a precise computational purpose: a node's current state participates in the next state. Its meaning is computational. Degree normalization controls the relative scale of messages from differently connected nodes. This normalization belongs to the propagation geometry; probability normalization over molecular conformations is a separate object, and the rows of the symmetric normalization generally have varying sums.

7.9.4 Depth reveals the diffusion assumption

Ignoring nonlinearities and channel maps for a moment, stacking layers repeatedly applies S:

$$H^{(K)} \approx S^K H^{(0)}. \tag{7.49}$$

If $S = U \operatorname{diag}(\mu_r) U^\mathsf{T}$, then

$$S^K = U \operatorname{diag}(\mu_r^K) U^\mathsf{T}. \tag{7.50}$$

Modes with $|\mu_r| < 1$ decay, while dominant smooth modes survive. Under the usual connectivity and aperiodicity conditions, node representations within a component become increasingly aligned with a small stationary subspace. This is the linear skeleton of over-smoothing. Nonlinearities, learned channels, and residual paths alter the exact limit, while the spectral pressure created by repeated propagation remains.

The phenomenon reveals the inductive bias encoded by depth. GCN assumes that neighbors should exchange and often homogenize information. That is useful under homophily. If an edge instead connects complementary types, opposite labels, or directed roles, indiscriminate low-pass smoothing can destroy the signal of interest. Adjacency grants permission to communicate; similarity requires an additional assumption about the signal and edge semantics.

APPNP makes the retained source explicit through a restart term [39]:

$$H^{(k+1)} = (1-\alpha) S H^{(k)} + \alpha H^{(0)}. \tag{7.51}$$

At convergence, when the inverse exists,

$$H^* = \alpha \, [I - (1-\alpha)S]^{-1} \, H^{(0)}. \tag{7.52}$$

The result is a resolvent or personalized-PageRank filter with a persistent contribution from the initial signal. GCNII similarly preserves initial features and identity mappings across depth [40]. Both methods reshape how multi-hop evidence accumulates.

7.9.5 GraphSAGE turns propagation into a set function

GraphSAGE writes the update without committing to one full-graph matrix multiplication [37]:

$$\begin{aligned} m_i^{(\ell)} &= \mathrm{AGG}^{(\ell)}\left(\{h_j^{(\ell)} : j \in \mathcal{N}(i)\}\right), \\ h_i^{(\ell+1)} &= \sigma\left(W^{(\ell)}[h_i^{(\ell)} \| m_i^{(\ell)}]\right). \end{aligned} \tag{7.53}$$

The aggregation must be invariant to an arbitrary enumeration of the neighbors. Mean, sum, pooling, and recurrent aggregators retain different information about the neighborhood multiset. Neighbor sampling makes the computation inductive and scalable, but also turns exact propagation into a sampling estimator.

The mathematical question has shifted. Spectral GCN asks which function of a graph operator acts on the signal. GraphSAGE asks which permutation-invariant function summarizes a local multiset. Both propagate over edges, but their approximation errors and information bottlenecks are typed differently.

7.9.6 GAT makes the route depend on the current state

Graph attention replaces degree-determined coefficients by content-dependent scores [51]:

$$\begin{aligned} e_{ij}^{(\ell)} &= a_\theta\left(Wh_i^{(\ell)}, Wh_j^{(\ell)}, e_{ij}\right), \\ \alpha_{ij}^{(\ell)} &= \frac{\exp e_{ij}^{(\ell)}}{\sum_{k \in \mathcal{N}(i)} \exp e_{ik}^{(\ell)}}, \\ h_i^{(\ell+1)} &= \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(\ell)} W h_j^{(\ell)}\right). \end{aligned} \tag{7.54}$$

Now the graph support may remain fixed while the weighted route changes with input and layer:

$$S \quad \longrightarrow \quad P_\theta(H, E). \tag{7.55}$$

For each receiver i, the coefficients form a categorical distribution over allowed sources. The normalization makes P_θ row-stochastic under the receiver-to-source convention and generally asymmetric. These coefficients type the computational route; conditional dependence, interaction energy, and physical force require separate extraction and validation. The same distinction reappears in Transformer attention on a dense token graph.

GAT replaces the fixed self-adjoint graph-relative filter of a GCN with an input-dependent, generally directed propagation rule. The appropriate spectral geometry, if one is needed, must be reconstructed for that directed state-dependent operator.

7.9.7 MPNNs promote the edge from a coefficient to a typed computation

The message-passing neural network abstraction separates message, aggregation, and update [50, 33]:

$$
\begin{aligned}
m_{ij}^{(\ell)} &= \phi_m^{(\ell)}\left(h_i^{(\ell)}, h_j^{(\ell)}, e_{ij}\right), \\
m_i^{(\ell)} &= \bigoplus_{j\in\mathcal{N}(i)} m_{ij}^{(\ell)}, \\
h_i^{(\ell+1)} &= \phi_u^{(\ell)}\left(h_i^{(\ell)}, m_i^{(\ell)}\right).
\end{aligned} \tag{7.56}
$$

Classical GCN is recovered by choosing a linear isotropic message such as $m_{ij} = S_{ij}h_jW$. General MPNNs can condition on bond type, distance, orientation, molecular identity, or a hidden edge vector. The edge is no longer exhausted by one scalar coefficient; it may implement a map between typed node fibers.

Depth also changes interaction order. A two-layer update at i can depend on paths $k \to j \to i$ and therefore on configurations of several nodes even when every primitive message is pair-indexed. Pairwise execution support can compose into higher-order output dependence across layers.

GIN isolates a separate issue: whether aggregation distinguishes different neighborhood multisets [38]. Its update is

$$
h_i^{(\ell+1)} = \mathrm{MLP}^{(\ell)}\left((1+\epsilon^{(\ell)})h_i^{(\ell)} + \sum_{j\in\mathcal{N}(i)} h_j^{(\ell)}\right). \tag{7.57}
$$

With appropriate injective maps, it reaches the discriminative power of the one-dimensional Weisfeiler–Lehman test. This is an expressivity statement relative to a combinatorial test, not a claim that the recovered features are causal or physical. Mean aggregation, sum aggregation, and scalar edge compression can agree on a downstream benchmark while discarding different structural information.

7.9.8 Geometric networks add transformation law, not molecular energy

For molecular systems the node state may include coordinates $r_i \in \mathbb{R}^3$. A simple invariant message can depend on squared distance,

$$
m_{ij} = \phi_m(h_i, h_j, e_{ij}, \|r_i - r_j\|^2), \tag{7.58}
$$

while an equivariant coordinate update has the schematic form

$$
r_i' = r_i + \sum_j (r_i - r_j)\phi_x(m_{ij}). \tag{7.59}
$$

Under a rigid transformation $r_i \mapsto Qr_i + t$, invariant scalar messages remain unchanged and the displacement in Eq. 7.59 transforms by Q. Consequently

$$
F(QR+t) = QF(R) + t \tag{7.60}
$$

for the coordinate map under the stated architecture [41]. More general geometric networks carry vectors, tensors, or irreducible rotation representations and type every message by its transformation law.

Equivariance licenses a symmetry statement. Energy, momentum conservation, and physical time evolution additionally require integrability, conservation laws, units, and a dynamical model. The same distinction applied to invariant point attention and conditional diffusion in Chapter 8.

7.9.9 Two diffusions that share a word but not a state space

Repeated GCN propagation and a coordinate diffusion model are easily conflated because both use diffusion language. Their operators act on different objects:

$$\begin{aligned} \text{graph-signal diffusion:}\quad & H^{(k+1)} = SH^{(k)}, \quad S : \mathcal{H}_{\text{node}} \to \mathcal{H}_{\text{node}}, \\ \text{generative diffusion:}\quad & \mathrm{d}R_t = f(R_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t, \\ & p_t \text{ is a density on coordinate configurations.} \end{aligned} \tag{7.61}$$

The first smooths features relative to a declared graph. The second smooths a probability density in a configuration space and learns its score for reverse transport. A graph Laplacian can appear inside a generative model. The state space and generator determine which diffusion is meant and which physical interpretation is available.

7.9.10 What actually changed across the lineage

The longitudinal compression is

$$\boxed{\begin{aligned} g_\theta(L) &\longrightarrow \sum_k \theta_k L^k \longrightarrow SHW, \\ &\longrightarrow P_\theta(H)HW \longrightarrow \phi_m(h_i, h_j, e_{ij}), \\ &\longrightarrow \text{typed equivariant edge maps.} \end{aligned}} \tag{7.62}$$

The progression relocates complexity. Spectral GCN places it in the graph filter; classical GCN compresses it into a fixed normalized route plus learned channel maps; attention moves some of it into state-dependent routing; MPNNs place it inside message functions; geometric networks add exact transformation constraints. None of these moves monotonically increases interpretability.

For protein models, this history supplies a clean audit. Is the adjacency observed, constructed, or generated from the current state? Is an edge a scalar route, a hidden vector, or an operator between typed fibers? Does depth merely spread evidence, or does it create a higher-order output response? Is symmetry guaranteed by architecture, and what physical claim remains unlicensed? AlphaFold's persistent pair track and triangle computations go beyond a classical node-only GCN, but they inherit the same central design question: which relations deserve state, and what may be concluded from the way that state is propagated?

7.9.11 One audit ledger across the lineage

The lineage can now be returned to the same coordinates used for the model portraits:

Family	State, support, and edge carrier	Context, dynamics, and normalization	Estimator and first unlicensed claim
Spectral/ChebNet	node signals on a declared graph; edge meaning inherited from A	fixed self-adjoint filter; polynomial locality	supervised filter coefficients; physical meaning remains inherited from the declared graph
GCN/APPNP	node features; fixed degree-normalized scalar route	repeated smoothing, with optional restart to initial state	task loss; propagation defines computational smoothing, with statistical or molecular meaning requiring a corresponding generative law
GraphSAGE and GAT	sampled neighborhoods or state-dependent routes	set aggregation or row-normalized adaptive routing	task loss and neighbor sampling; attention weights describe adaptive routes
MPNN, GIN, and geometric networks	typed node and edge states, optionally coordinates	compositional messages; equivariant variants constrain transformation laws	task-specific loss; force, energy, and mechanism claims additionally require their defining physical structure

The common entry is computation on a graph. What changes is the graph's provenance, the object carried by an edge, the information available to the message, and the transformation law of the state. Those are precisely the coordinates in Eq. 7.1; together they determine the meaning carried by each historical name.

Coda: correspondence with a remainder

The model portraits and the two longitudinal cases reveal a recurring pattern. Two systems often share one exact layer of mathematics and diverge at the next. Potts and Gaussian models share a notion of conditional coupling but inhabit different state spaces. Hopfield retrieval and attention can share a softmax update while differing in memory persistence and value tying. RBMs and low-rank pair models share a weak-coupling component while the marginalized RBM retains higher orders. Frozen attention maps lie in a shared head span, while the full network response also travels through queries, keys, residual paths, and nonlinear blocks.

The productive comparison is an explicit map into common operator coordinates, accompanied by the terms, assumptions, and dynamics left in its remainder. The shared coordinates state the exact correspondence; the remainder states where the systems diverge. Once both are fixed, empirical agreement becomes a meaningful question.

The unified-framework case showed how an exact reconstruction equivalence can transfer an estimator or regularizer while leaving probability semantics plural. The GCN lineage showed a different kind of continuity: functional calculus became local propagation, then adaptive routing, typed messages, and geometric transformation laws. In both cases, progress relocated complexity rather than monotonically increasing physical interpretability.

Chapter 8 applies the same audit to AlphaFold 1, 2, and 3: pair geometry becomes a prediction, then a persistent relational state, then conditioning for generative coordinate transport. This is a protein-scale stress test of the framework. Chapter 9 then returns to the observations and estimators through which operator graphs become visible.

Chapter 8

Phylogeny as Tree-Constrained Transport

Phylogeny is not merely a correction applied after an MSA statistic has been computed. It is a distinct generative graph on the sequence axis. Its vertices include latent ancestors, its edges carry evolutionary transition kernels, and its shared topology makes the observed alignment rows dependent. This graph must be separated from the residue-position graph, whose edges describe statistical or functional relations within a sequence, and from an MSA Transformer's row route, which is generated between observed sequences by a learned computation. The distinction supplies the missing type information behind the informal statement that an MSA contains both constraint and common ancestry.

8.1 Why a tree is a special computational structure

Let $T = (V, E)$ be a tree with observed leaves $\mathcal{L}$, latent internal vertices $\mathcal{I} = V \setminus \mathcal{L}$, and a root chosen to orient the computation. A tree is not only a sparse graph. It has no cycles, every pair of vertices is joined by one path, and deleting any edge separates it into exactly two components. If p is the parent of u, let D_u be the descendant component containing u after (p, u) is deleted. The Markov property on the tree gives

$$X_{D_u} \perp\!\!\!\perp X_{V\setminus(D_u\cup\{p\})} \mid X_p. \tag{8.1}$$

Thus the single categorical variable X_p separates the complete descendant subtree from the rest of the model. In graphical-model language the separator has size one and the tree has treewidth one.

This separator property is the source of the algorithm. All observations below u can be summarized for the rest of the tree by the q numbers

$$m_{u\to p}(a) = P(x_{\mathcal{L}\cap D_u} \mid X_p = a)\,, \qquad a \in [q]. \tag{8.2}$$

The size of this interface does not grow with the number of descendants. Dynamic programming, belief propagation, variable elimination, and contraction of a tree tensor network are four descriptions of this same compression. On a graph with cycles, eliminating a node generally creates larger separators and higher-order intermediate factors; exact cost is then governed by treewidth rather than vertex count alone.

8.2 The classical recursion is exact message passing on a tree

For one aligned site, let Q be a continuous-time substitution generator on q categorical states. An oriented edge $e = (p, u)$ of length t_e carries the Markov operator

$$M_e = \exp(t_e Q), \qquad M_e(a, b) = P(X_u = b \mid X_p = a). \tag{8.3}$$

Let $\phi_u(b)$ be the observation factor: it is one at an unobserved internal vertex and is the indicator or ambiguity distribution of the observed category at a leaf. Conditional independence between child subtrees gives the Felsenstein message

$$m_{u\to p}(a) = \sum_{b=1}^{q} M_{(p,u)}(a, b)\, \phi_u(b) \prod_{v\in \mathrm{ch}(u)} m_{v\to u}(b). \tag{8.4}$$

The site likelihood is

$$p(x_{\mathrm{leaf}} \mid T, Q, t) = \sum_{a=1}^{q} \pi(a)\phi_r(a) \prod_{v\in \mathrm{ch}(r)} m_{v\to r}(a), \tag{8.5}$$

where r is the root and π its state distribution. For a fixed tree this sum–product recursion evaluates the likelihood in $O(|V|q^2)$ time per site rather than summing explicitly over all ancestral assignments [19]. Its age is therefore not evidence of a missing neural replacement: it is already the exact dynamic program for the declared tree model.

Equation 8.4 also locates phylogeny inside the typed framework. The edge map $M_{(p,u)}$ transports a categorical function from the child fiber to the parent fiber, while the product at u combines conditionally independent descendant evidence. This is a Graph Network on an augmented vertex set with latent nodes, but it is not a residue interaction graph and not a message-passing layer over only the observed MSA rows.

8.3 Inference on a tree is not inference of a tree

Two computational problems are often compressed into the phrase ”phylogenetic inference”:

$$\boxed{\begin{array}{rl} \text{inference on a tree:} & (T, Q, t) \longmapsto p(x_{\mathcal{L}} \mid T, Q, t), \\ \text{inference of a tree:} & x_{\mathcal{L}} \longmapsto \widehat{T}, \widehat{Q}, \widehat{t} \text{ or } p(T, Q, t \mid x_{\mathcal{L}}). \end{array}} \tag{8.6}$$

The first problem uses Eq. 8.4 and is exact for the declared model. The second contains a discrete search over topologies and continuous estimation of branch lengths and substitution parameters. Its parameter space has the schematic form

$$\mathfrak{M}_R = \coprod_{T\in\mathcal{T}_R} \left(\mathbb{R}_{>0}^{|E(T)|} \times \mathcal{Q}\right), \tag{8.7}$$

a union of topology-specific continuous regions whose boundaries meet when branches contract. Ordinary gradient descent operates within one continuous region; a topology move, relaxation, or sampling rule is needed to cross between them.

The number of unrooted binary topologies on R labeled leaves is

$$N_R = (2R - 5)!!, \tag{8.8}$$

so maximum-likelihood and Bayesian procedures must search or sample a vast outer space. UPGMA, neighbor joining, parsimony, local NNI or SPR moves, stochastic search, and tree-space MCMC are different responses to this topology problem. Modern implementations such as IQ-TREE 2 improve search, substitution models, model selection, bootstrap approximations, and parallel evaluation while retaining the inner likelihood recursion [20, 21].

Rooting supplies a further type distinction. If the substitution process is reversible,

$$\pi(a)M_t(a,b) = \pi(b)M_t(b,a), \tag{8.9}$$

the likelihood can be evaluated by placing the computational root anywhere on the same unrooted tree. The chosen root orients the dynamic program but is not thereby statistically identified. A molecular clock, an outgroup, temporal samples, or a declared nonreversible process supplies additional information for evolutionary direction. Computational direction and biological time direction must therefore be audited separately.

8.4 Three graphs occupy three different base spaces

For an MSA $X \in [q]^{R\times L}$, three graph constructions must be kept separate:

$$\boxed{\begin{aligned} T_f &: \text{ancestral vertices and evolutionary branches,} \\ \mathcal{G}_{\text{pos}} &: \text{residue positions and typed categorical interactions,} \\ A_i^{(\ell,h)}(X) &: \text{observed MSA rows and learned attention routes.} \end{aligned}} \tag{8.10}$$

The first usually shares one topology across sites under the basic model, although recombination or gene-tree discordance can require local trees. The second acts within a sequence and may encode structural or functional constraint. The third is dense, layer- and input-dependent, generally directed after normalization, and defined only on observed homologues.

The associated operators are also different. A tree Laplacian acts on ancestral and observed vertices. The generator Q in Eq. 8.3 acts on categorical states. A position Laplacian acts on residue positions after a position graph has been declared. A Kronecker-sum construction can combine such axes under special separability assumptions, but their spectra are not interchangeable. In particular, a leading singular vector across alignment rows does not become a residue interaction mode merely because both are low rank.

An additive tree metric d_{rs} satisfies the four-point condition: for every four leaves, the two largest members of

$$d_{rs} + d_{uv}, \qquad d_{ru} + d_{sv}, \qquad d_{rv} + d_{su} \tag{8.11}$$

are equal. A generic attention-derived dissimilarity need not satisfy this condition, share a topology across positions, compose along paths, or correspond to a Markov transition kernel. Sequence similarity or clade-like clustering in column attention is therefore evidence of useful row structure, not identification of a phylogenetic tree.

8.5 Static constraint and evolutionary dynamics are compatible

Calling a Potts model static and phylogeny dynamic refers to two different coordinates. A fitted Potts Hamiltonian

$$E_{\Theta_f}(x) = -\sum_i h_i(x_i) - \sum_{i<j} J_{ij}(x_i, x_j) \tag{8.12}$$

is one family-level constraint field shared across MSA rows. Evolution is a stochastic process that transports sequences along the branches of T_f. A hierarchical generative statement is therefore

$$\Theta_f \sim P_\star(\Theta), \qquad (T_f, t_f) \sim P_{\text{tree}}, \qquad X_f \sim p(X \mid T_f, t_f, \Theta_f). \tag{8.13}$$

The static field and the dynamic genealogy are not competitors: the first specifies which sequence configurations are favored, while the second specifies how correlated observations were produced.

There is nevertheless a missing arrow in common practice. A Gibbs distribution proportional to $\exp[-E_{\Theta_f}(x)]$ is an equilibrium law, not an evolutionary transition kernel. To evolve complete sequences on a tree one must additionally specify rates

$$\mathcal{Q}_{\Theta_f}(x, x') \quad \text{with, for example,} \quad e^{-E_{\Theta_f}(x)} \mathcal{Q}_{\Theta_f}(x, x') = e^{-E_{\Theta_f}(x')} \mathcal{Q}_{\Theta_f}(x', x), \tag{8.14}$$

if reversible evolution with the Potts equilibrium is intended. This generator acts on a q^L-state sequence space and is generally intractable. Classical site-independent phylogenetics regains tractability by factorizing the edge law across sites; a Potts model regains tractability by describing an equilibrium or using pseudolikelihood rather than exact evolutionary histories. Neither approximation contains the other.

This separation exposes the central identifiability problem. Correlated residues can arise from shared constraint, while correlated rows arise from shared ancestry; finite and taxonomically biased sampling couples the two in the empirical MSA. Sequence weighting changes an estimator but does not identify the evolutionary process, and a high-scoring Potts block need not be free of phylogenetic contribution [17, 18].

8.6 What modern phylogenetic learning actually replaces

Recent learning methods can be classified by the coordinate they change rather than by the broad label "deep phylogenetics":

1. *Posterior representation.* Variational Bayesian phylogenetic inference uses subsplit Bayesian networks to represent distributions over topologies; normalizing flows enrich the branch-length posterior. The phylogenetic likelihood remains the target [22, 23].

2. *Tree-space parameterization.* GeoPhy embeds topology distributions into continuous geometric coordinates, while Dodonaphy combines hyperbolic embeddings with a differentiable soft-NJ decoder. These methods make optimization differentiable without turning a generic dense matrix into a tree [24, 25].

3. *Combinatorial sampling.* PhyloGFN trains a generative flow network to sample diverse high-reward or posterior-weighted trees, replacing part of topology exploration rather than the biological definition of likelihood [26].

4. *Likelihood-free amortization.* Phyloformer learns evolutionary distance estimation and reconstruction from simulated alignments. It can be faster and can absorb richer simulated dependencies, but its conclusions inherit the training simulator and distribution-shift problem [27].

Thus the modern frontier is not one universal successor to neighbor joining or pruning. It is a collection of learned proposal distributions, posterior approximations, continuous relaxations, and simulator-trained estimators attached to different stages of the classical pipeline.

8.7 MSA attention is an amortized mixed signal

An MSA Transformer is trained to predict masked residues, not to recover (T_f, Q, t_f). Its column attention may exploit close homologues, clades, subfamilies, gaps, or taxonomic redundancy whenever these improve reconstruction. Its tied row attention may simultaneously collect conservation and coevolutionary evidence across the same rows. Consequently the learned representation is more accurately described as

$$H_f = \mathcal{A}_\theta\big(X_f;\ \text{constraint}, \text{ancestry}, \text{sampling}, \text{gaps}\big), \tag{8.15}$$

where the semicolon lists latent causes, not supplied labels. Successful masked reconstruction does not by itself identify their decomposition.

The useful comparison with classical phylogenetics is therefore not "old tree versus modern attention." Classical phylogenetics declares a tree and substitution law, then performs exact latent-state marginalization conditional on them. MSA Transformer learns a cross-family rule that compresses several latent causes into hidden states and dynamic routes. It is more flexible but less identified. An explicit hybrid would supply a tree-conditioned row operator and reserve a learned residual route for dependence not explained by that tree:

$$H^{(\ell+1)} = \Phi_\theta\Big(H^{(\ell)}, \mathcal{K}_{T_f,Q,t_f} H^{(\ell)}, A^{(\ell)}_{\theta,\text{res}}(X_f) H^{(\ell)}\Big). \tag{8.16}$$

Equation 8.16 is an architectural hypothesis, not a uniquely identified additive decomposition. It becomes scientific only when the tree component and residual are tested by interventions or matched nulls.

8.8 A phylogeny-aware operator audit

A defensible audit should preserve the distinction in Eq. 8.10:

1. infer or sample trees with uncertainty rather than naming leading row principal components "phylogeny" from magnitude alone;
2. simulate an independent-site or declared epistatic process along the fitted tree while matching depth, gaps, composition, and taxonomic sampling;
3. pass real and null MSAs through the identical operator-extraction pipeline and compare full typed blocks before scalar compression;
4. evaluate leave-clade-out transfer and clade-preserving interventions, because arbitrary row shuffling destroys both ancestry and the sampling structure one intends to control;
5. report sensitivity across plausible topologies, branch lengths, substitution models, and product versus empirical background measures.

The resulting null-relative quantity

$$\Delta G_{i\leftarrow j} = G^{\text{real}}_{i\leftarrow j} - \mathbb{E}_{\widetilde{X}\sim Q_{T_f}} G_{i\leftarrow j}(\widetilde{X}) \tag{8.17}$$

asks which operator structure exceeds that generated by the declared genealogy. It does not prove physical contact or causal epistasis; those remain separate validation coordinates. Its advantage is narrower and more rigorous: common ancestry is represented as a generative row-graph rather than removed by an unlabeled spectral heuristic.

Chapter 9

Three AlphaFolds: The Evolution of Relational State

AlphaFold did not advance through three versions of one unchanged algorithm. Each generation reorganized the statistical experiment. The first compressed a family alignment into engineered site and pair features, predicted explicit geometric distributions, and optimized coordinates in the resulting learned potential. The second kept the alignment as an evolving neural state, coupled it repeatedly to a persistent pair representation, and realized geometry through an equivariant structure module. The third shortened the lifetime of the alignment state, enlarged the molecular type system, concentrated relational evidence in a pair representation, and replaced the protein-specific coordinate module with conditional all-atom diffusion.

The progression is often summarized as an improvement in accuracy and scope. For the purposes of this book, it is more revealing to read it as the evolution of an edge object:

$$\begin{aligned}
&\text{AF1:} \quad \text{explicit distance distribution and derived pair potential,} \\
&\text{AF2:} \quad \text{persistent protein pair state coupled to an MSA state,} \\
&\text{AF3:} \quad \text{persistent multimolecular pair state conditioning a generator.}
\end{aligned} \tag{9.1}$$

At every step the pair representation becomes more capable and less self-interpreting. That tradeoff makes the three systems an unusually strong test of the operator-graph framework.

This chapter uses the theory developed here as a retrospective coordinate system. Its conclusions are therefore audit claims: they classify which sample spaces, information bottlenecks, graph states, symmetries, consistency operations, and probabilistic outputs are present in each published architecture. They are not derivations of AlphaFold from the operator framework, nor equivalence theorems among the three generations. The historical question of the designers' explicit theoretical influences would require different evidence. Within this scope, the framework makes the architectural choices comparable in one language and identifies the additional assumptions needed for physical or biological interpretation.

9.1 One audit, three generations

We will track six coordinates through the three systems.

Evidence. What query-specific information is observed at inference: a current MSA, templates, chemical identity, covalent structure, or recycled predictions?

State. Which axes remain explicit inside the network: alignment row, residue or molecular token, pair, rigid frame, or atom coordinate?

Edge carrier. Is a pair represented by a categorical coupling table, a distance distribution, a hidden vector, an attention route, or an audited response operator?

Composition. How do paths and triples update pair information? Is the operation a linear graph diffusion, a nonlinear triangle computation, or a coordinate-equivariant message?

Normalization and dynamics. Which variable is normalized by softmax or diffusion, and is an iterative update guaranteed to decrease any declared energy?

Licensed interpretation. What can be concluded directly from the architecture, and what would require a gauge, a compatible joint law, a null model, or a biological intervention?

The resulting comparison is summarized before it is derived:

System	Main relational state	Irreversible bottleneck	Strongest immediate semantics
AF1	explicit distogram and torsion-related outputs from an $L \times L$ trunk	raw MSA rows removed before the neural trunk	learned pairwise geometric evidence
AF2	evolving z_{ij} coupled bidirectionally to m_{si}	alignment information gradually projected into pair and first-row states	joint MSA–pair computation with equivariant coordinate realization
AF3	heterogeneous token-pair state conditioning all-atom denoising	MSA rows discarded after a short MSA module; provenance deposited into z_{ij}	multimolecular relational conditioning and a conditional coordinate generator

None of the final-column statements is yet a claim of microscopic energy, direct evolutionary coupling, or biological mechanism.

9.2 AlphaFold 1: an explicit geometric edge

9.2.1 The alignment disappears before the network begins

AlphaFold 1 constructs a query-specific MSA and transforms it into engineered one-site and pair features before the main neural trunk [65]. Site features include profiles, deletion statistics, and effective alignment counts. Pair features include gap covariation, pseudolikelihood coupling tensors, and scalar summaries of those tensors. Schematically,

$$X_{sia} \longmapsto \{\phi_i(X), \Psi_{ij}(X)\} \longmapsto F_\theta(\phi_i, \phi_j, \Psi_{ij}), \tag{9.2}$$

where the MSA row index s is no longer present after the first arrow.

This is an irreversible statistical decision. The two-dimensional network can learn how to combine profiles, couplings, templates, and positional context. Individual homologous sequences and new representations of row-level phylogeny are no longer addressable after this compression. Any correction for redundancy, subfamily structure, alignment errors, or finite-depth noise must be performed during feature construction or inferred indirectly from the summaries that remain.

The compression has an advantage that later systems partly surrender: the provenance of the input features is comparatively visible. There is a named profile, a named coupling tensor, and a named gap statistic. What is lost in adaptivity is gained in inspectable interfaces.

9.2.2 Distogram softmax is local geometry, not a global conformation law

The two-dimensional residual network predicts a distribution over distance bins for each residue pair,

$$P_{ij}(d = r_k \mid X) = \operatorname{softmax}_k s_{ij,k}(X). \tag{9.3}$$

The normalization axis is the set of possible distances for one fixed pair. This is already much richer than a binary contact: it distinguishes short, intermediate, and long separations and can encode multimodal geometric uncertainty.

This is a collection of pairwise marginals, while a joint distribution over atomic coordinates requires mutual realizability in three dimensions. Triangle inequalities, chain connectivity, chirality, excluded volume, and stereochemistry couple different pairs. The product

$$\prod_{i<j} P_{ij}(d_{ij} \mid X) \tag{9.4}$$

therefore requires a separate compatibility argument before it can be read as the marginalization of one normalized conformation ensemble.

AF1 addresses realization algorithmically. The predicted distance distribution is compared with a learned background distribution and converted into an effective potential of mean force, schematically

$$U_{ij}(r; X) = -\log \frac{P_{ij}(r \mid X)}{P_{ij}^{\mathrm{ref}}(r)}, \tag{9.5}$$

then combined with torsional and stereochemical terms during coordinate optimization. This is an effective learned geometric objective. The logarithm gives it a legitimate energy coordinate; the background ratio gives it a useful statistical reference. Its dependence on the complete input sequence and learned structural database makes it an effective statistical potential whose pair terms can absorb collective constraints beyond a physical two-body force decomposition.

The distinction is subtle but important. AF1 uses energy language at the level needed for optimization: compatible structures should lie in low-objective regions. Calibrated thermodynamics, conformational populations, kinetics, solvent free energies, and transferable atomic forces would require additional physical calibration.

9.2.3 The operator-graph reading

AF1's native pair object is geometric. Each edge carries a distribution over distance states and related orientation information. The input may contain a $q \times q$-like pseudolikelihood tensor, but after learned mixing the output edge no longer says how one amino-acid substitution acts on another residue's categorical distribution.

In the terminology of this book, AF1 builds a dense scalar-state geometric graph from compressed family evidence. It exposes its edge readout clearly; the alignment sample axis is compressed before the network, and the computation has no identified categorical operator. Its great conceptual move is to replace contact classification with a learned geometric field; its corresponding limitation is that the route from individual evolutionary observations to that field is largely one-way.

9.3 AlphaFold 2: the pair state becomes a dynamical memory

9.3.1 An architectural discontinuity

AlphaFold 2 changes the state space [64]. Persistent single and pair representations replace refinement of a fixed AF1-style potential. The Evoformer maintains an MSA representation

$$m_{si}^{(\ell)} \in \mathbb{R}^{c_m} \tag{9.6}$$

and a pair representation

$$z_{ij}^{(\ell)} \in \mathbb{R}^{c_z} \tag{9.7}$$

through many blocks. The sequence axis s is now an evolving part of the neural computation rather than a feature-engineering prelude.

The two states communicate repeatedly. Pair features bias row-wise attention in the MSA. The MSA updates the pair state through an outer-product mean,

$$\Delta z_{ij} = W \left[\frac{1}{N} \sum_s a(m_{si}) \otimes b(m_{sj}) \right]. \tag{9.8}$$

The expression is an adaptive, learned second-moment-like statistic. The vectors being multiplied have already passed through attention, pair bias, nonlinear transitions, and earlier blocks. Their row average can carry denoised family signals, deletion structure, phylogenetic patterns, and model priors together.

This explains why retaining raw MSA rows can outperform replacing them by fixed profiles and pair frequencies. The result shows that adaptive access to row-level information matters. Attribution of the gain among higher-order coevolution, improved weighting, denoising, and alignment heterogeneity remains an empirical decomposition.

9.3.2 A graph whose edges have memory

The pair state $z_{ij}^{(\ell)}$ persists and changes across the trunk. It is edge memory on a dense position graph, updated jointly across residue pairs rather than emitted as a final score. Information deposited from the MSA at one layer can influence later MSA routing, later triangle updates, the structure module, and recycling.

This is a major shift in graph semantics. In AF1 the pair grid is primarily the domain of a learned geometric predictor. In AF2 the pair grid is an evolving relational workspace: the model infers and computes with edge state.

The channels of the hidden vector z_{ij} are learned coordinates. A categorical interpretation requires a decoder

$$T_{ij}(a, b) = D_\theta(z_{ij}; a, b) \tag{9.9}$$

followed by reference-gauge projection, while a geometric interpretation requires its own readout. Norms, eigenvectors, and individual channels summarize hidden-state geometry before either typed identification has been performed.

9.3.3 Triangle operations as higher-order computation

AF2 updates an edge through other edges that share an endpoint. A schematic triangle multiplication has the form

$$\Delta z_{ij} \sim \sum_k g_{ijk} \odot A(z_{ik}) \odot B(z_{kj}), \tag{9.10}$$

with learned projections and gates. Triangle attention provides a related content-dependent aggregation. The index k makes relational consistency explicit: the state of (i, j) is judged in the context of two-edge paths through k.

Three claims must be separated.

First, the computation is genuinely triple-indexed. It can compare or compose the relations (i, k) and (k, j) while updating (i, j). Second, the stored output remains a pair state. The architecture can realize the dependence without retaining an explicit hyperedge variable h_{ijk}. Third, a three-body Hamiltonian interpretation additionally requires a compatible energy and global law. The update is gated, channel-dependent, input-dependent, and embedded in a deep residual system. An irreducible three-body statistical component must be established at the audited output after lower orders have been removed.

Triangle operations differ from ordinary graph-Laplacian diffusion. A Laplacian applies a fixed or declared linear edge geometry to node signals; AF2 composes learned edge states nonlinearly and changes the effective operator at every layer. Local Jacobians, singular directions, and intervention responses are more faithful diagnostics than the eigenvectors of one scalarized pair matrix.

9.3.4 Invariant point attention separates symmetry from energy

After the Evoformer, AF2 maps the first-row representation and pair state into three-dimensional structure. Invariant point attention represents residues by rigid frames, forms query and key points in local coordinates, compares them in global space, and uses pair features as additional biases. The construction makes the update equivariant to global rigid motion while allowing the attention weights themselves to depend on invariant distances.

This is a precise use of physical symmetry. Rotating or translating the entire protein changes the coordinates in the required way while preserving the intrinsic prediction. The architecture guarantees this transformation law directly.

Equivariance states how the output changes when the coordinate frame changes. A physical-force or potential-gradient interpretation adds a separate dynamical model. The structure module is a learned coordinate map trained with geometric losses. Its attention softmax normalizes source residues for an update; conformation probabilities live in another space. Symmetry controls transformation law, not thermodynamic semantics.

9.3.5 Recycling as iterative inference

AF2 feeds predicted structure and internal representations back into the network. Recycling lets the model refine a hypothesis after seeing its own geometric consequences. It resembles iterative inference and can correct globally inconsistent early states.

But an iterative map

$$y^{(t+1)} = F_\theta(y^{(t)}, X) \tag{9.11}$$

is an iterative inference dynamics. An energy-descent reading becomes available with a declared state function $E(y)$ and evidence that

$$E(y^{(t+1)}) \leq E(y^{(t)}) \tag{9.12}$$

under fixed memories and update conditions. AF2 provides a powerful learned refinement process; the weaker description is also the accurate one.

9.4 AlphaFold 3: relational conditioning becomes multimolecular

9.4.1 The node and edge types expand

AlphaFold 3 moves from protein structure prediction to complexes containing proteins, nucleic acids, small molecules, ions, modified residues, and covalent links [67]. Standard polymer residues can be represented as tokens, while non-polymer components are tokenized more nearly at heavy-atom resolution. The token-pair representation must therefore relate objects with different internal state spaces and different sources of evidence.

This immediately makes the pair track heterogeneous. Two protein residues may receive family evidence from an MSA, whereas a protein–ligand pair draws on chemical identity, covalent structure, spatial context, templates where applicable, and learned priors. The same tensor index type contains several relation types.

The operator-graph view handles this naturally by allowing typed node fibers and typed edge maps. It also warns against treating the hidden z_{ij} as one homogeneous scalar graph. Any analysis must stratify or transport across entity types before norms and spectra can be compared.

9.4.2 The MSA deposits evidence and then disappears

AF3 still constructs MSAs for polymer chains, including paired rows when cross-chain genetic information is available [66, 67]. But the explicit MSA state has a much shorter lifetime than in AF2. A small MSA module updates both row representations and the pair state through outer-product means, pair-weighted averaging, and triangle operations. After that module, the MSA rows are discarded.

The main Pairformer operates on a single-token state s_i and pair state z_{ij}. Pair features bias token attention, while triangle operations continue to update the pair track. Within the Pairformer, information flow is asymmetric: pair state shapes token updates, while the AF2-style token-to-pair write-back loop is absent. Much of the relational evidence has already been deposited into z.

The change can be written as a shift in bottleneck location:

$$\begin{aligned} &\text{AF2:} \quad m_{si} \rightleftarrows z_{ij} \quad \text{throughout the trunk,} \\ &\text{AF3:} \quad m_{si} \longrightarrow z_{ij} \quad \text{near the entrance,} \qquad z_{ij} \longrightarrow s_i \quad \text{in the core.} \end{aligned} \tag{9.13}$$

AF3 gains a common trunk for heterogeneous molecular systems and reduces the cost of maintaining a large alignment axis. It loses direct access to row-level provenance once the compression has occurred.

The pair state becomes the principal relational memory of the model. That makes it more central than in AF2, not more interpretable. Evolutionary evidence, templates, chemical type, covalent relations, recycling information, interface geometry, and confidence-relevant signals can coexist inside the same vector.

9.4.3 Mathematical interlude: one probability flow, two path laws

AF3 replaces the AF2 structure module with a diffusion model over all-atom coordinates. The following interlude isolates the mathematical object introduced by this replacement and the scope of its physical interpretation. Let R_0 denote a clean structure and R_t a noised version. A denoiser or score-like network is trained under a forward corruption process and conditioned on the trunk representations. In score notation, the learned field is associated with

$$s_\theta(R_t, t \mid s, z) \approx \nabla_{R_t} \log p_t(R_t \mid s, z). \tag{9.14}$$

This probability space consists of noisy coordinate configurations conditional on the model inputs and trunk state. Attention, distograms, and masked-token heads normalize over three other sample spaces: source positions, distance bins, and amino-acid identities.

The compact formula hides three different objects: a deliberately chosen corruption process, a vector field estimated from corrupted data, and a numerical transport procedure. Their separation reveals what diffusion contributes and prevents the sampling dynamics from being mistaken for molecular dynamics.

The forward SDE destroys information in a controlled way. A continuous-time idealization of Gaussian corruption is the Itô stochastic differential equation (SDE)

$$\mathrm{d}R_t = f(R_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}W_t, \qquad 0 \le t \le T, \tag{9.15}$$

where f is a prescribed drift, g is a noise amplitude, and W_t is Brownian motion. An SDE specifies infinitesimal transition laws for an ensemble of nonsmooth paths: over a short interval Δt, the deterministic displacement is of order Δt, while the random displacement is of order $\sqrt{\Delta t}$. Its ensemble density obeys the Fokker–Planck equation

$$\partial_t p_t(R) = -\nabla_R \cdot \big[f(R,t)p_t(R)\big] + \frac{1}{2} g(t)^2 \Delta_R p_t(R). \tag{9.16}$$

The first term transports probability; the second smooths it. In the drift-free case this is the heat equation. If the added variance is σ_t^2, then

$$p_t = p_0 * \mathcal{N}(0, \sigma_t^2 I) = \exp\left(\frac{\sigma_t^2}{2}\Delta_R\right) p_0. \tag{9.17}$$

Thus forward diffusion is Gaussian convolution, or heat-semigroup smoothing, viewed at a sequence of resolutions. Fine features of the data density disappear first; at sufficiently large noise, the terminal law is chosen to be close to a simple reference distribution.

The common finite-time parameterization

$$R_t = \alpha_t R_0 + \sigma_t \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{9.18}$$

is the transition law of an appropriate linear SDE. The forward schedule is a design choice, and its clock records noise scale. A physical-time interpretation would require an independently identified molecular dynamics.

Denoising learns the score of every smoothed density. At a fixed noise level, the score is the spatial log-density gradient

$$s_t(R \mid s, z) = \nabla_R \log p_t(R \mid s, z). \tag{9.19}$$

It points toward locally increasing probability density and contains no normalization constant. For the Gaussian corruption in Eq. 9.18, least-squares noise prediction estimates a conditional mean,

$$\epsilon_\theta(R_t, t, s, z) \longrightarrow \mathbb{E}[\epsilon \mid R_t, s, z], \qquad s_t(R_t \mid s, z) = -\frac{1}{\sigma_t}\mathbb{E}[\epsilon \mid R_t, s, z]. \tag{9.20}$$

Equivalently, a minimum-square denoiser estimates the posterior mean of the clean coordinates. Tweedie's identity gives

$$\mathbb{E}[R_0 \mid R_t, s, z] = \frac{R_t + \sigma_t^2 s_t(R_t \mid s, z)}{\alpha_t}. \tag{9.21}$$

Noise prediction, clean-coordinate denoising, and score estimation are therefore different parameterizations of the same conditional object when the Gaussian kernel and scaling convention are fixed. The essential learned quantity is a multiscale family of density-gradient fields from which clean structures are generated.

This also explains how denoising recovers structure through population statistics. The posterior over the particular ϵ used for one corrupted example retains uncertainty given R_t. Squared-error training returns its conditional expectation, which is precisely the quantity needed to reconstruct the score of the noised population.

The reverse SDE transports noise back toward data. For the forward process in Eq. 9.15, the time-reversed diffusion has drift

$$\mathrm{d}R_t = \big[f(R_t, t) - g(t)^2 \nabla_R \log p_t(R_t \mid s, z)\big]\,\mathrm{d}t + g(t)\,\mathrm{d}\overline{W}_t, \qquad t : T \longrightarrow 0, \tag{9.22}$$

where $\overline{W}_t$ is Brownian motion in the reverse-time filtration [68, 69]. The displayed differential is integrated with decreasing t; this convention matters when interpreting its signs. The score correction is the information that turns a generic noisy reference law back toward the learned conditional data law. Replacing the unknown score by s_θ and discretizing the equation yields a sampler.

The reverse SDE is a probability-generating dynamics. Its individual paths are algorithmic transport paths; Brownian increments are introduced by the model, the drift depends on a noise-indexed training density, and time runs backward along an artificial corruption schedule. An exact score recovers the intended data distribution. Molecular kinetics would additionally require a physically calibrated forward-time process.

The probability-flow ODE exposes the deterministic core. The same forward marginals p_t also satisfy the continuity equation generated by the ordinary differential equation (ODE)

$$\frac{\mathrm{d}R_t}{\mathrm{d}t} = f(R_t, t) - \frac{1}{2}g(t)^2 \nabla_R \log p_t(R_t \mid s, z). \tag{9.23}$$

Indeed,

$$-\nabla_R \cdot \left[\left(f - \frac{1}{2}g^2 \nabla_R \log p_t\right) p_t\right] = -\nabla_R \cdot (f p_t) + \frac{1}{2}g^2 \Delta_R p_t, \tag{9.24}$$

which is exactly Eq. 9.16. Run from the terminal reference law at T back to time zero, this probability-flow ODE transports the same one-time marginal distributions as the reverse SDE when the score is exact [69].

An ODE assigns a smooth deterministic path once its initial point is fixed. An SDE continues to inject Brownian randomness along the path. Their one-time marginal densities agree, while their path measures, couplings between times, numerical errors, and exploration behavior remain different. In the usual sampling experiment the ODE still begins from a random terminal point drawn from the reference law. The probability-flow result isolates the common mathematical core shared by the stochastic and deterministic transports:

$$\boxed{\text{multiscale density smoothing} \longrightarrow \text{conditional score estimation} \longrightarrow \text{reverse probability transport.}} \tag{9.25}$$

SDE and ODE samplers are two realizations of the last arrow. With an approximate neural score and a finite step size, each incurs a characteristic law-approximation error; sampler choice is therefore a numerical and statistical decision.

Diffusion changes the epistemic status of the output. Multiple coordinate samples can be drawn from the same conditioning state, allowing the generator to express some multimodality and sampling variability. A deterministic coordinate map and a conditional coordinate distribution are different objects even when their top-ranked structures are similar. Calibration of sample diversity additionally depends on the learned score, the terminal law, the numerical solver, and any selection or ranking imposed after generation.

From statistical energy to molecular energy. The score is the gradient of a learned log density. A physical equilibrium force additionally requires a calibrated density with physical units and temperature, an appropriate measure on configuration space, correct treatment of solvent and chemical state, and a relation such as

$$F(R) = -\nabla_R U(R) \tag{9.26}$$

for the intended physical potential. The diffusion score follows the noised training distribution and depends on the conditioning network. Its native interpretation is a principled generative vector field; molecular force-field or kinetic interpretations enter through the additional physical structure above.

One may always define a noise-dependent statistical energy

$$E_t^{\mathrm{stat}}(R \mid s, z) = -\log p_t(R \mid s, z) + C_t, \qquad s_t = -\nabla_R E_t^{\mathrm{stat}}. \tag{9.27}$$

This identity is mathematical bookkeeping. To identify $k_{\mathrm{B}}T\, s_0$ with a thermodynamic mean force would additionally require the learned density to equal the relevant equilibrium distribution under the correct configuration-space measure and experimental conditions. To identify a reverse trajectory with kinetics would require still more: a physically valid mobility, noise law, and forward-time dynamics. Those identifications require evidence beyond AF3's conditioning variables and database distribution.

Repeated diffusion samples reflect the learned structural distribution, training data, conditioning information, noise schedule, and sampling procedure. Comparing sample frequencies with experimental populations requires separate calibration.

Return to AlphaFold 3. The interlude identifies the coordinate generator, while AF3's protein-specific content enters through the conditioning fields s_i and z_{ij}. The score network is conditional: it reads a token state and a persistent pair state into which MSA, template, chemical, and geometric evidence has already been compressed. Diffusion changes the output law; the pair track still determines which relational evidence is available to shape that law.

9.4.4 Input chemical graph, relational graph, and coordinate graph

AF3 also makes three graphs especially easy to conflate. The input chemical graph records known covalent bonds and molecular identity. The Pairformer graph is a dense or broadly connected token-pair relational workspace. The generated coordinate graph is obtained only after distances, contacts, and interactions are defined on a sampled structure. A covalent edge can condition the pair state without being inferred; a noncovalent pair state can be important without becoming a bond; a spatial contact can emerge in a sample without corresponding to one persistent internal edge channel.

The broader molecular scope therefore validates the need for typed graphs. Once proteins, ligands, nucleic acids, ions, and modifications coexist, "one graph over tokens" is too weak a description. Node type, edge provenance, allowed context, and output interpretation become part of the mathematical object.

9.5 What the three generations mastered

AlphaFold's most impressive theoretical competence is architectural. Several principles recur.

They promote relations to state. AF1 makes pair geometry an explicit prediction target. AF2 and AF3 go further by giving pair representations persistent memory and their own update algebra. Relations become state well before the final layer.

They choose bottlenecks deliberately. AF1 compresses the MSA before learned inference, AF2 delays compression, and AF3 moves it back toward the entrance after a learned deposition step. These are different judgments about the value and cost of provenance.

They use paths to enforce relational consistency. Triangle updates make the state of one edge depend on two-edge paths. This is a powerful way to make dense pair predictions geometrically coherent without storing an explicit variable for every triple.

They separate geometric handling from biological content. AF2 hard-codes rigid-motion behavior in its structure module. AF3 explicitly handles coordinate geometry, rigid transformations, and noisy structures inside a conditional generative procedure, with transformation guarantees that differ from IPA. In both cases geometric transformation is controlled more directly than the biological content, which remains learned from data and supervision.

They match output objects to tasks. AF1 predicts distance distributions. AF2 predicts frames and coordinates with geometric confidence. AF3 samples an all-atom conditional distribution. Across generations, the output state space itself expands.

They iterate hypotheses. Recycling and denoising let later computation condition on an evolving structural proposal. The systems behave more like inference procedures than one-pass regressors, even when no exact energy descent theorem applies.

These principles are close to the concerns of statistical physics: state choice, coarse graining, symmetry, collective consistency, iteration, and probability. They operate first at the level of architecture; physical-observable status is a further identification problem.

9.6 What the operator framework still asks

The three systems solve structure prediction far more successfully than they solve internal identification. The framework adds a set of questions left open by predictive performance.

For AF1, how much of a learned geometric edge comes from categorical coupling rather than profile, gap, phylogenetic, or template structure? Does the background-corrected distance potential behave consistently across families and sequence interventions?

For AF2, can a decoder from z_{ij} recover a reference-centered categorical field? How stable is that field across backgrounds? How much higher-order response is produced by triangle operations and deep composition? Are the two directed substitution responses reciprocal even when the attention routes are not?

For AF3, how should operator fibers be typed across residues, nucleotides, ligand atoms, and ions? Which pair-state components originate in MSA evidence and which in learned chemical priors? Does diffusion sampling reproduce experimentally measured alternate-state populations, or only generate plausible structural modes? Which noncovalent relations persist under matched changes of ligand, sequence, template, and diffusion seed?

Across all generations, attention maps should be separated from output response graphs, hidden pair norms from decoded operators, and coordinate accuracy from physical mechanism. The required experiments are interventions: remove or null the MSA while preserving composition, alter one evidence track at a time, decode typed pair responses, compare backgrounds, test reciprocity, and validate against mutation or binding measurements that preserve the predicted categorical or chemical structure.

9.7 The changing location of interpretability

The three generations reveal a general law of model development. Later representations often integrate more evidence and support more accurate predictions, while preserving less information about where each component came from.

AF1 exposes explicit statistical and geometric interfaces but discards individual MSA rows early. AF2 preserves the row axis and builds a rich pair dynamics, but the pair channels mix evolution, geometry, templates, and learned priors. AF3 expands to heterogeneous molecular systems and a generative coordinate law, but concentrates still more kinds of evidence in one pair state before diffusion.

Integration is often exactly what prediction requires. Interpretation must therefore distinguish computational factorization into MSA, pair, single, and coordinate tracks from biological disentanglement of phylogeny, evolutionary constraint, physical interaction, and function.

AlphaFold's deeper lesson concerns the design of representation: successful molecular inference depends on choosing when evidence should remain separate, when it should be compressed, which relations deserve persistent state, which symmetries should be exact, and which uncertainty should be represented by a distribution. Those are the same design questions posed by the operator framework, answered in three different regimes.

Chapter 10

From Observations to Operator Graphs

An interaction graph is never observed in the same way that a sequence is observed. An MSA is a tensor of residues indexed by sequence, position, and category. A structure is an ensemble of coordinates under an experimental and modeling pipeline. A pLM response is produced by a chosen intervention and a chosen set of sequence backgrounds. Each supplies observations from which a particular operator graph can be estimated.

This distinction becomes critical in an MSA because its three axes support different notions of variation. Modes across rows can reflect ancestry, taxonomy, or subfamilies. Modes across positions can describe collective organization along the sequence. Categorical constant modes are removed by the local gauge at each position. Operations on these three axes generally survive one another because estimation, inversion, pair projection, scalarization, and spectral decomposition have a noncommuting order.

The chapter compares two principal estimation routes. A family Potts model uses many homologous sequences to estimate one static pair field. A pretrained pLM uses amortized information to produce a background-dependent response from one query sequence, after which a family or another declared background law can be used to estimate its mean and variance. Common reference-centered coordinates make the results comparable while the estimand ledger records their distinct targets.

The difference between the two estimates can be decomposed into Potts estimation error, pairwise model misspecification, pLM transfer or response gap, and response-sampling error. This ledger is more informative than one correlation coefficient. It shows where disagreement can enter and which source of disagreement an additional dataset could resolve.

Finally, evidence must be relative to a null that preserves the nuisance structure under discussion. Sequence reweighting, phylogenetic projection, APC, and spectral mode removal each encode a specific nuisance hypothesis. Their meaning comes from a declared estimand and a null ensemble that justifies what is being removed.

10.1 MSA observations and interaction operators occupy different spaces

The one-hot encoding of an alignment with N rows belongs to an observation space with three axes,

$$X \in \mathcal{H}_{\mathrm{seq}} \otimes \mathcal{H}_{\mathrm{pos}} \otimes \mathcal{H}_{\mathrm{cat}}. \tag{10.1}$$

This tensor belongs to the observation space, while $\mathcal{H}^0 = \bigoplus_i \mathcal{H}_i^0$ is the operator domain. The sequence axis indexes correlated samples; position and category index the object estimated from those samples. A phylogenetic mode across MSA rows, a graph eigenvector across residue positions, and the constant categorical mode removed by P_i therefore live in three different spaces.

Let $w_n \geq 0$ with $\sum_n w_n = 1$, and define the weighted site marginal

$$p_i(a) = \sum_{n=1}^{N} w_n \mathbf{1}[X_i^{(n)} = a]. \tag{10.2}$$

With smoothing for unsupported states, the centered whitened one-hot tensor is

$$Y_{nia} = \sqrt{w_n}\, \frac{\mathbf{1}[X_i^{(n)} = a] - p_i(a)}{\sqrt{p_i(a)}}. \tag{10.3}$$

Singular vectors of an unfolding of Y along the sequence axis describe between-sequence variation that may reflect ancestry, subfamilies, taxonomic imbalance, or function. Position interaction modes arise only after an interaction operator has been estimated.

If U_{phy} has orthonormal columns in the square-root-weighted row coordinates of Y and contains sequence directions declared nuisance by an external phylogeny, replicate null, or preregistered rule, define

$$R_{\text{phy}} = I - U_{\text{phy}} U_{\text{phy}}^{\mathsf{T}}, \qquad Y_{\text{res}} = R_{\text{phy}} Y. \tag{10.4}$$

Projection is exact relative to the declared subspace. External evidence licenses the phylogeny label; magnitude alone identifies leading principal components.

10.2 Two estimators in one coordinate system

The weighted marginal in Eq. 10.2, after declared smoothing, provides a family reference. A Potts fit estimates one $\widehat{J}_{ij}$ and hence one static block $\widehat{G}^{\text{P}}_{i\leftarrow j} = P_i \widehat{J}_{ij} P_j^{\mathsf{T}}$ per direction [12, 13, 14]. This estimate depends on the fitting objective, regularization, sequence weights, and finite MSA sample. Gauge projection addresses parameter redundancy after those estimation choices have been made.

A pretrained pLM takes a single query sequence at inference and directly yields the conditional logits needed for Eq. 4.6. It has amortized information across its training distribution and can therefore produce a dynamic response without an inference-time MSA. An MSA remains useful for specifying the family references p_i, the family background distribution μ_{ij}, the static mean $\overline{G}_{i\leftarrow j}$, and uncertainty under finite family sampling. The two estimators answer different questions:

$$\begin{aligned} &\text{MSA route:} \quad \{x^{(n)}\}_{n=1}^{N} \longrightarrow \widehat{G}^{\text{P}}_{i\leftarrow j}, \\ &\text{pLM route:} \quad x \longrightarrow G^{\text{pLM}}_{i\leftarrow j}(x_{-ij}). \end{aligned} \tag{10.5}$$

Their comparison contains several estimands. Let $G^{\text{P},*}_{i\leftarrow j}$ be the block of the population minimizer of the declared Potts fitting risk; let $G^{F,(2)}_{i\leftarrow j}$ be the pair component of the true family conditional log score under the same anchor reference; and let $\overline{G}^{\text{pLM}}_{i\leftarrow j} = \mathbb{E}_{C\sim\mu_{ij}} G^{\text{pLM}}_{i\leftarrow j}(C)$. If $\widehat{\overline{G}}^{\text{pLM}}$ is a sampled response mean, then

$$\begin{aligned}\widehat{G}^{\mathrm{P}}_{i\leftarrow j} - \widehat{\overline{G}}^{\mathrm{pLM}}_{i\leftarrow j} = {} & \underbrace{\widehat{G}^{\mathrm{P}}_{i\leftarrow j} - G^{\mathrm{P},*}_{i\leftarrow j}}_{\text{Potts estimation}} + \underbrace{G^{\mathrm{P},*}_{i\leftarrow j} - G^{F,(2)}_{i\leftarrow j}}_{\text{pairwise objective or misspecification}} \\ & + \underbrace{G^{F,(2)}_{i\leftarrow j} - \overline{G}^{\mathrm{pLM}}_{i\leftarrow j}}_{\text{pLM transfer and response gap}} + \underbrace{\overline{G}^{\mathrm{pLM}}_{i\leftarrow j} - \widehat{\overline{G}}^{\mathrm{pLM}}_{i\leftarrow j}}_{\text{response sampling}}. \end{aligned} \tag{10.6}$$

Common reference-centered coordinates make all four terms comparable; their magnitudes remain empirical. In particular, agreement between a pseudolikelihood Potts optimum and a pLM response mean is a testable transfer statement of the framework.

An estimand ledger with exact cancellation

The four-term decomposition turns an aggregate comparison into a ledger of discrepancies. It also exposes agreement produced by cancellation. Let U be one unit-norm direction in the common reference-centered operator space and suppose the four gaps along that direction are

$$\begin{aligned} \widehat{G}^{\mathrm{P}} - G^{\mathrm{P},*} &= +0.10U, && \text{finite Potts estimation,} \\ G^{\mathrm{P},*} - G^{F,(2)} &= +0.25U, && \text{pairwise objective or misspecification,} \\ G^{F,(2)} - \overline{G}^{\mathrm{pLM}} &= -0.30U, && \text{pLM transfer and response gap,} \\ \overline{G}^{\mathrm{pLM}} - \widehat{\overline{G}}^{\mathrm{pLM}} &= -0.05U, && \text{response sampling.} \end{aligned} \tag{10.7}$$

Their sum is zero. The estimated Potts block and estimated pLM mean agree exactly along U, even though neither estimator equals the family pair component and the pLM has a substantial transfer gap. Agreement arose by cancellation.

The reverse situation is also possible. If the four terms point in orthogonal operator directions, the total discrepancy has squared norm equal to the sum of their squared norms. A single scalar correlation then reports the aggregate disagreement but cannot say whether more MSA depth, a richer statistical model, a different pLM, or more response backgrounds would reduce it.

The ledger suggests an end-to-end comparison protocol.

Fix one anchor geometry. Estimate site marginals from a declared training partition, smooth unsupported categories, and hold those references fixed for both estimators. Transport every block into these fibers before averaging or subtraction.

Separate family sampling from response sampling. Fit the Potts model on weighted or clade-partitioned MSA rows. Estimate the pLM response mean on a separate set of sequence backgrounds. Resample the two axes independently so that each uncertainty term remains identifiable.

Preserve the operator before scalarization. Report block alignment, singular-direction agreement, reciprocity, and dynamic variance before taking norms or contact scores. Two methods can agree in magnitude while disagreeing on which substitutions compensate.

Use a synthetic oracle to identify all four terms. In real protein families the true pair component $G^{F,(2)}$ and Potts population optimum $G^{\mathrm{P},*}$ are unknown. Resampling can estimate the first and fourth finite-sample terms, but misspecification and transfer remain partly confounded. A controlled generative family with a known conditional score is needed to separate all four contributions exactly.

Use biological data only for the claim it can test. Contacts can validate scalar positional association. Double-mutant matrices can test categorical operator alignment. The same mutations across multiple backgrounds can test the dynamic field. No one endpoint identifies every term in Eq. 10.6.

The practical lesson is that a high Potts–pLM correlation is an observation, not an explanation. The estimand ledger identifies which stronger conclusions require additional sampling, a richer family model, a synthetic oracle, or an external intervention.

Finite-sample reporting should include the smoothing convention, regularization path, and uncertainty of full blocks as well as scalar scores. The descriptive quantity $N_{\mathrm{eff}} = (\sum_n w_n^2)^{-1}$ records weight concentration. Independent-sample uncertainty under phylogeny instead requires clade-aware or tree-based resampling rather than an ordinary row bootstrap when estimating confidence intervals for Eq. 10.6 and $\rho^{\mathrm{dyn}}_{i\leftarrow j}$.

Protein language models trained on single sequences and MSA Transformer instantiate different amortization regimes [61, 63]. The former transfers a cross-family prior through fixed parameters and one query sequence; the latter combines that learned prior with an explicit current-family sample as in Eq. 7.23. Both can be audited in the same categorical output coordinates. MSA column attention acts on the sequence axis, tied row attention produces a family-conditioned position route, and neither route is itself the audited categorical residue operator.

10.3 Null-relative evidence

Correlated sampling can bias covariance, inverse-coupling, and sector estimates [17, 18]. Let O^{real} be an estimated operator statistic and let Q_{null} preserve declared nuisance structure, such as one-site composition or a phylogenetic tree, while removing the dependence under test. The null-relative contrast is

$$\Delta O = O^{\mathrm{real}} - \mathbb{E}_{\widetilde{X}\sim Q_{\mathrm{null}}} O(\widetilde{X}). \tag{10.8}$$

Both terms must be produced by the same extraction pipeline and transported to the same reference fibers before subtraction. The contrast updates the evidence for an estimated interaction while leaving the node fiber and Potts parameter-gauge class fixed. Likewise, APC and leading-mode deletion are useful position-level corrections only when the removed direction is supported as nuisance by a declared null or downstream validation [16].

The sequence, position, and categorical axes must be kept separate. Phylogenetic residualization on MSA rows and eigenvector deletion on the final scalar graph produce different results because pair counting, inversion, projection, and norms have a noncommuting order.

For a covariance-like null operator Σ_0 that is positive definite after regularization, relative spectral enrichment can be defined by

$$\Sigma_{\mathrm{real}} u_k = \lambda_k \Sigma_0 u_k. \tag{10.9}$$

Large generalized eigenvalues identify directions enriched relative to the null scale. Contact and mechanism interpretations enter through downstream validation.

10.4 New coordinates, old decision engines

The contrast between an MSA estimator and a protein language model can be generalized. Many recent protein methods preserve the statistical decision engine while replacing the coordinates in which it operates. A useful abstraction is

$$\boxed{x \xrightarrow{\phi_\theta} z \xrightarrow{\mathcal{T}_D} (\widehat{y}, \widehat{\sigma}) \xrightarrow{\mathcal{A}} \text{ranking, retrieval, or experiment.}} \tag{10.10}$$

Here ϕ_θ is a pretrained representation, $\mathcal{T}_D$ is fitted on the task-specific data D, and $\mathcal{A}$ converts its output into a decision. Most of the expensive biological regularity may be amortized into z, while the final estimator remains a random forest, Gaussian process, bilinear similarity, nearest-neighbor rule, or sequence-alignment algorithm.

A better coordinate system can turn a difficult nonlinear problem in raw sequence space into a statistically efficient problem for a simple estimator. Novelty must therefore be located correctly. Improved performance can come from the representation, the downstream estimator, the acquisition rule, or the data loop. Calling the complete pipeline a new foundation model obscures which component actually changed.

10.4.1 EVOLVEpro: a learned coordinate system inside classical active learning

EVOLVEpro uses a protein language model to embed candidate sequences, fits a random-forest regressor to measured activity, and uses that surrogate to choose variants for the next experimental round [43]. Schematically,

$$x \xrightarrow{\text{PLM}} z(x) \xrightarrow{\text{RF}_D} \widehat{y}(x) \xrightarrow{\text{rank}} \text{next experimental batch.} \tag{10.11}$$

The language model supplies a coordinate system in which evolutionary and biochemical regularities may already be easier to separate. The random forest supplies piecewise-constant regression and nonlinear feature selection from a small labeled set. The iterative experiment supplies the most important new information: measured values near the region currently being optimized.

The scientific object is an adaptive estimator whose training distribution changes after every acquisition. Feature importance describes how the forest uses its coordinates; residue interaction and calibrated campaign uncertainty require their own estimators and measurements. The achievement is that amortized sequence coordinates make a classical small-data optimization loop substantially more useful.

10.4.2 VIRAL: modern embeddings, Gaussian-process geometry

VIRAL places viral protein sequences in an ESM3-derived representation and models measured fitness with a Gaussian process, using its predictive distribution to explore the landscape [44]. For a kernel k on the learned coordinates, the posterior mean and variance at a candidate z_* have the familiar form

$$\widehat{m}(z_*) = k_*^\top (K + \sigma_n^2 I)^{-1} y, \widehat{v}(z_*) \qquad = k(z_*, z_*) - k_*^\top (K + \sigma_n^2 I)^{-1} k_*.$$

The same inverse-covariance operation now acts on distances defined by a pretrained representation and a chosen kernel, in place of raw one-hot sequence coordinates. A rational-quadratic kernel can mix several length scales, but those scales are geometric scales in embedding space; physical interaction ranges and evolutionary times require separate calibration.

An upper-confidence acquisition rule such as

$$a(z) = \widehat{m}(z) + \beta^{1/2}\widehat{v}(z)^{1/2} \tag{10.12}$$

then balances predicted fitness against model uncertainty. The exploration term is meaningful only relative to the GP prior, kernel, noise model, and observed region. Other sources of biological failure remain outside that posterior. VIRAL is a particularly clear instance of old probabilistic machinery becoming powerful because the representation changes which sequences count as nearby.

10.4.3 PLMSearch: representation narrows the search, alignment retains the verdict

PLMSearch addresses remote homology by using protein-language-model representations to retrieve promising candidates and then retaining classical sequence comparison in the decision pipeline [45]. A simplified decomposition is

$$q \xrightarrow{\phi_\theta} \text{embedding retrieval candidates} \xrightarrow{\text{sequence alignment}} \text{homology ranking.} \tag{10.13}$$

The learned coordinate system supplies sensitivity and speed at large database scale. Alignment restores residue-level order, gaps, and local correspondence discarded by a pooled vector. The two stages therefore answer different questions: which proteins should be compared closely, and what ordered residue correspondence supports the final score?

This hybrid construction assigns a distinct evidential role to each stage. Proximity in representation space records encoder similarity; homology, structural mechanism, and evolutionary history enter through additional evidence. The learned representation remains consequential because it changes the candidate distribution on which the expensive classical rule operates and can expose remote relationships that raw-sequence screening misses.

10.4.4 The repeated pattern is an estimator factorization

The three systems can be compared in the same ledger:

System	New coordinate	Decision engine	First unlicensed claim
EVOLVEpro	PLM sequence embedding	random-forest regression plus iterative ranking	feature importance describes surrogate use; mechanism requires intervention
VIRAL	ESM3-derived embedding geometry	Gaussian-process posterior and acquisition rule	kernel distance lives in embedding geometry; posterior variance is model-relative
PLMSearch	pLM retrieval representation	similarity screening plus sequence alignment	embedding proximity supports retrieval; alignment restores residue correspondence

The shared mathematical structure is

$$\boxed{\begin{array}{c}\text{pretraining moves the data into new coordinates;}\\ \text{a declared estimator still decides what follows.}\end{array}} \tag{10.14}$$

This factorization is useful for evaluation. Freeze ϕ_θ and compare downstream engines; freeze the engine and compare representations; preserve the acquisition rule while changing the uncertainty model; and evaluate the complete loop prospectively. An aggregated benchmark gain leaves attribution among the biological prior, statistical estimator, and decision policy unresolved.

In the coordinates of this book, these methods mostly change the observation and estimator maps; the interaction-edge definition remains a separate coordinate. Their representations may contain relational information, but extracting an operator graph still requires endpoint-controlled interventions, reference projection, and a declared validation target. A powerful coordinate system can make an old engine newly effective without turning that engine into a new physical theory.

Coda: the estimand survives the pipeline only by declaration

The path from data to graph contains several irreversible operations. Correlated sequences are weighted or projected; a model is fitted; raw tables are gauge-fixed; dynamic responses are averaged; operators may be scalarized; null expectations are subtracted. Changing the order can change the result. A transparent analysis therefore names the target operator before choosing the correction pipeline.

This perspective also clarifies the role of disagreement. A Potts estimate and a pLM response mean can differ because of finite family sampling, pairwise misspecification, transfer failure, or background sampling. The common operator coordinate makes the difference decomposable rather than mysterious. It does not privilege either estimator as the truth by construction.

Their physical interpretations are not interchangeable. A Potts coupling is an effective statistical energy coordinate for variation across a protein family when the fitted joint law is adequate. The sampled objects are sequences shaped by ancestry, selection, and experimental ascertainment, not equilibrium molecular conformations. A pLM substitution response is instead a finite-difference susceptibility of a trained predictor: it measures how one model output changes under a sequence intervention. It becomes a physical susceptibility only if the intervention is conjugate to a physical field, the response is a calibrated observable, and the background law matches the physical ensemble of interest.

Agreement between the two estimators can therefore support a shared relational pattern without identifying its origin. Structural contacts can test geometric proximity; deep-mutational scans can test single-site functional consequences; double-mutant cycles can test nonadditivity in a matched experimental background. Those endpoints supply different missing arrows from a statistical operator toward physical structure or biological mechanism. No correction of the MSA or increase in pLM sample size can replace the endpoint needed for the stronger claim.

Once an operator graph has been estimated, a new temptation appears: diagonalize the first matrix available and interpret its leading eigenvectors. Spectra can be powerful summaries, but only after the space, edge compression, directionality, normalization, and null geometry have been declared. That is a separate stage of reasoning and therefore a separate chapter.

Chapter 11

Spectra After the Graph Has Been Declared

Spectral analysis begins after the graph has been defined, not before. This ordering sounds obvious, yet protein sequence analysis offers many matrices with plausible eigenvectors: an MSA covariance, a Potts norm matrix, an attention route, a dynamic response shadow, a signed coupling graph, or a connection operator acting on categorical fibers. Their spectra live in different spaces and answer different questions.

The first decision is what has been compressed. A scalar Laplacian describes smoothness of functions over residue positions relative to nonnegative edge weights. A signed Laplacian retains one oriented scalar polarity. A connection Laplacian transports vectors between local categorical spaces and can test whether substitution directions remain coherent around the graph. No one spectrum contains the others.

Directionality creates an earlier branch. A row-softmax route is naturally a Markov operator. Its Euclidean asymmetry need not imply irreversible circulation: a symmetric score kernel can produce an entrywise asymmetric but reversible route under its stationary measure. Conversely, a doubly stochastic route can remain nonreciprocal. Reversibility, detailed-balance defect, and directed singular modes should be measured before projecting the route to an undirected shadow.

Context dependence adds another noncommutativity. The spectrum of a mean graph, the mean spectrum of context-specific graphs, and the distribution of spectral projectors are different objects. Near degeneracy makes individual eigenvectors unstable even when their invariant subspace is stable. Spectral claims should therefore be phrased at the level of the appropriate projector and calibrated against replicate null graphs rather than inferred from rank alone.

The guiding question of the chapter is not "what does the leading eigenvector mean?" but "which operator, on which space, relative to which null, produced this eigenvector?" Only then can a spectral mode be connected responsibly to conservation, sectors, communication, allostery, or another biological interpretation.

11.1 Reference-projected spectral transforms

Let $O : \mathcal{H}_R \to \mathcal{H}_L$ be an observed operator, O_0 a null expectation, and R_L, R_R residualizers away from declared nuisance subspaces in the left and right geometries. A general corrected operator has the form

$$\boxed{\mathcal{D}_{R,\Phi}(O) = R_L \Phi(O - O_0) R_R.} \tag{11.1}$$

Here Φ must be typed for the operator space. For a rectangular $A = U\Sigma V^{\mathsf{T}}$, a singular-value transform is $\Phi_f^{\mathrm{sv}}(A) = Uf(\Sigma)V^{\mathsf{T}}$. For a square self-adjoint $A = U\Lambda U^{\mathsf{T}}$, spectral functional calculus instead gives $\Phi_f^{\mathrm{eig}}(A) = Uf(\Lambda)U^{\mathsf{T}}$ and preserves eigenvalue signs when f does. Equation 11.1 is an audit template, not an identification theorem. The meaning of Φ depends on a separate model: covariance inversion, graph diffusion, shrinkage, and deletion of an empirical mode make different assumptions.

Operation	Space and observed object	Transform	Identified object
Potts table redundancy	pair table $T_{ij}(a,b)$	$P_i T_{ij} P_j^{\mathsf{T}}$	reference-centered bivariate contrast
Sequence residualization	weighted MSA rows	$R_{\mathrm{phy}} Y$	residual after a declared sequence nuisance model
Position nuisance removal	scalar graph M	$R_Z(M - M_0)R_Z^*$	residual orthogonal to declared position covariates
Gaussian inversion	covariance Σ	Σ^{-1}	conditional graph under a Gaussian law
Graph diffusion	Laplacian L_G	e^{-tL_G}	smoothing relative to a specified graph
Mode deletion	symmetric statistic M	selected $f(\lambda_k) = 0$	residual after a declared spectral choice

For independently specified position nuisance coordinates $Z \in \mathbb{R}^{L\times r}$ and a positive position weight W, let

$$\Pi_Z = Z(Z^{\mathsf{T}}WZ)^{-1}Z^{\mathsf{T}}W, \qquad R_Z = I - \Pi_Z. \tag{11.2}$$

Assume Z has full column rank and let R_Z^* denote the adjoint in the W-weighted position geometry; then $R_Z^* = R_Z$. The corrected scalar position operator is

$$M_\perp = R_Z(M - M_0)R_Z^*. \tag{11.3}$$

This removes exactly the declared linear nuisance space. APC and leading-mode deletion are special rank-one approximations; improvement in contact precision validates their utility for that task, not a universal entropy or phylogeny label.

11.2 A longitudinal case: one covariance matrix, many inverse graphs

An essay written in 2022 under the deliberately exaggerated title "Inverse covariance is all you need" placed a Gaussian chain, MSE reconstruction, sparse and ridge precision estimation, network deconvolution, balanced network deconvolution, and eigenmode deletion in one sequence [3]. The visual argument was compelling because the methods repeatedly took the same form: diagonalize a matrix, alter its eigenvalues, and reconstruct a cleaner graph. Read through the operator framework, this family resemblance is real and stronger than a metaphor. For a symmetric observed matrix

$$O = U\,\mathrm{diag}(\lambda_1, \ldots, \lambda_d)U^{\mathsf{T}}, \tag{11.4}$$

most of the methods in the essay can be written

$$O \longmapsto U\,\mathrm{diag}(f(\lambda_1), \ldots, f(\lambda_d))\,U^{\mathsf{T}}, \tag{11.5}$$

possibly preceded by centering or scaling and followed by block compression and contact ranking. The eigenvectors are often retained; the scientific decision is concentrated in the scalar function f.

That algebraic statement explains why the works look so similar. Each f nevertheless estimates the graph defined by a different forward story. Covariance inversion assumes a Gaussian probability law. Network deconvolution assumes that indirect paths are matrix powers of a direct network. Balanced deconvolution changes which path lengths exist in that story. Leading-mode deletion treats selected spectral directions as nuisance rather than as indirect paths. A deep contact predictor may use the transformed matrix only as an input feature, so its final edge is produced by the subsequent network rather than by f alone.

The common family can therefore be typed by the chain

$$\boxed{\begin{array}{c}\text{observed statistic} \longrightarrow \text{propagation law} \longrightarrow \text{inverse or spectral transform} \\ \longrightarrow \text{regularization} \longrightarrow \text{edge object} \longrightarrow \text{validation endpoint}\end{array}} \tag{11.6}$$

Two methods can be nearly identical in numerical linear algebra and different at any later arrow. This section rewrites the historical sequence in those coordinates.

11.2.1 The Gaussian chain: when the inverse really is the graph

Let independent unit Gaussian noises generate

$$A = \epsilon_A, \qquad B = A + \epsilon_B, \qquad C = B + \epsilon_C, \qquad D = C + \epsilon_D. \tag{11.7}$$

The covariance matrix is dense because an upstream fluctuation propagates to every downstream variable. For example, A covaries with C and D even though the structural equations contain no A–C or A–D term. The joint negative log density, however, is

$$-\log p(a, b, c, d) = \frac{1}{2}\left[a^2 + (b-a)^2 + (c-b)^2 + (d-c)^2\right] + \text{constant}. \tag{11.8}$$

Its Hessian, and hence its precision matrix $\Theta = \Sigma^{-1}$, is

$$\Theta = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}. \tag{11.9}$$

The tridiagonal support recovers the conditional-independence graph: for a nonsingular Gaussian law, $\Theta_{ij} = 0$ if and only if X_i and X_j are conditionally independent given all other coordinates [2]. Here “direct” has an exact meaning because the probability family, conditioning set, and node variables have all been declared.

This example also gives precision a limited physical interpretation. In a Gaussian equilibrium field, covariance measures fluctuations or susceptibility, whereas precision is the quadratic restoring operator: it is the Hessian of the dimensionless Gaussian free energy. A large eigenvalue of Σ is a soft, strongly fluctuating direction; inversion makes it a small stiffness. A small covariance eigenvalue is a stiff direction and becomes a large precision eigenvalue. This language becomes literal molecular physics only if the variables are physical coordinates, the distribution is an equilibrium ensemble, and hidden degrees of freedom have been handled. For one-hot MSA columns, Θ is instead a Gaussian surrogate for categorical sequence statistics. Its graph is probability-exact inside that surrogate; forces and contacts require an additional correspondence.

11.2.2 MSE reconstruction: the same optimum through another objective

Chapter 7 showed that a regularized linear self-reconstruction problem can have an optimum that is an affine function of $(S+\lambda I)^{-1}$. The 2022 essay emphasized the same fact from the opposite direction: a matrix learned by squared-error reconstruction can land on the same precision-like solution as a multivariate Gaussian fit even though the loss surfaces are not the same. One route differentiates a log determinant and imposes positive definiteness; the other differentiates a quadratic reconstruction loss and can be implemented as an ordinary linear layer.

The exact shared optimum is valuable. It permits solvers and regularizers to move between statistical modeling and neural reconstruction. The Gaussian objective identifies a normalized joint density when its constraints are satisfied. The MSE objective identifies a best linear reconstruction map under a chosen data geometry. A common stationary point makes the fitted matrices equal, while thermodynamic and probabilistic interpretations remain tied to their respective objectives and constraints.

This is the same pattern encountered in the 2019 unified framework [32]. The horizontal equivalence there grouped models by reconstruction loss. The present longitudinal case groups them by spectral action. Together they show why implementation-level unification is scientifically productive: it exposes exactly which remainder must still be typed.

11.2.3 Sparse and ridge precision: regularization chooses the graph

Given an empirical covariance S, regularized Gaussian precision estimation takes the generic form

$$\widehat{\Theta} = \arg\min_{\Theta \succ 0} \{\operatorname{tr}(S\Theta) - \log\det\Theta + R(\Theta)\}. \tag{11.10}$$

PSICOV used sparse inverse covariance estimation for protein contact prediction [4]. An L_1 penalty makes many fitted entries exactly zero, so it performs two jobs: it stabilizes an underdetermined estimate and declares a sparse Gaussian conditional graph. Its zeros belong to that regularized model and express its selected conditional graph. Experimental influence remains a separate endpoint.

ResPRE instead used the Frobenius penalty $R(\Theta) = \rho\|\Theta\|_F^2$ and then supplied the raw 21×21 precision blocks to a deep residual contact predictor [5]. Its stationarity equation is

$$S - \Theta^{-1} + 2\rho\Theta = 0. \tag{11.11}$$

If $S = V\Lambda V^{\mathsf{T}}$, the solution shares the eigenvectors of S and transforms each eigenvalue by

$$k_\rho(\lambda) = \frac{-\lambda + \sqrt{\lambda^2 + 8\rho}}{4\rho}, \qquad \widehat{\Theta} = V\operatorname{diag}(k_\rho(\lambda_i))V^{\mathsf{T}}. \tag{11.12}$$

Thus the L_2 estimator is literally a smooth spectral filter. It usually produces graded shrinkage, while L_1 fitting can select an exact support. ResPRE then continued beyond the precision graph. Its final edge was a contact probability learned by a residual convolutional network from the full categorical blocks. The precision matrix was a representation that made indirect correlations easier for a finite neural predictor to use. Indeed, covariance and invertible precision contain the same information in principle; their empirical difference arose because the downstream learner, sample size, and regularization were finite.

The two papers therefore differ less in their observed statistic than in the composite estimator. PSICOV couples Gaussian fitting to sparse support and a contact score. ResPRE couples ridge spectral shrinkage to a nonlinear, structurally supervised decoder. Calling both "inverse covariance methods" is correct but incomplete in the same way that calling AlphaFold 1 and AlphaFold 2 "distance predictors" is correct but incomplete.

11.2.4 Network deconvolution: inversion of a walk algebra

Network deconvolution begins from a different forward model. It assumes that an observed similarity network contains direct edges and all of their transitive walks:

$$G_{\mathrm{obs}} = G_{\mathrm{dir}} + G_{\mathrm{dir}}^2 + G_{\mathrm{dir}}^3 + \cdots = G_{\mathrm{dir}}(I - G_{\mathrm{dir}})^{-1}. \tag{11.13}$$

Under the convergence and diagonalizability assumptions, inversion gives

$$G_{\mathrm{dir}} = G_{\mathrm{obs}}(I + G_{\mathrm{obs}})^{-1}, \qquad f_{\mathrm{ND}}(\lambda) = \frac{\lambda}{1+\lambda} \tag{11.14}$$

after the paper's required spectral scaling [6]. The expression resembles a precision transform because both contain a matrix inverse. The inverse is solving a different equation. Gaussian precision inverts a fluctuation covariance under a joint probability law; network deconvolution inverts a geometric series of walks under a propagation model. Its off-diagonal entry means "direct relative to the assumed walk closure," not "conditionally independent in a Gaussian law."

The distinction is more than philosophical. Matrix multiplication counts weighted walks only after a node basis, edge composition rule, and treatment of signs have been fixed. A categorical operator graph requires compatible fiber maps before G^2 even has a meaning. For an ordinary scalar symmetric network the composition is automatic, but its biological validity remains an empirical assumption. Network deconvolution improved several biological networks because that assumption was useful; the algebra alone cannot prove that an observed covariance was generated by the full walk series.

11.2.5 Balanced deconvolution: changing which paths exist

Balanced network deconvolution changed the forward model by retaining only odd path lengths:

$$G_{\mathrm{obs}} = G_{\mathrm{dir}} + G_{\mathrm{dir}}^3 + G_{\mathrm{dir}}^5 + \cdots = G_{\mathrm{dir}}(I - G_{\mathrm{dir}}^2)^{-1}. \tag{11.15}$$

Consequently its spectral inverse is

$$f_{\mathrm{BND}}(\lambda) = \begin{cases} \dfrac{-1+\sqrt{1+4\lambda^2}}{2\lambda}, & \lambda \neq 0, \\ 0, & \lambda = 0. \end{cases} \tag{11.16}$$

Removing even powers preserves the sign balance of eigenvalues and maps every real observed eigenvalue to a direct eigenvalue with magnitude below one, avoiding the network-dependent scaling required by the original transform [7]. Applied as a postprocessor, BND improved contact rankings from several coevolutionary methods, with smaller gains for methods such as DCA and PSICOV that had already attempted to suppress transitive effects.

In the language of this book, BND changes the algebra of admissible paths. It should not be described as the same inverse covariance estimator with a technical fix: even-length paths are absent from its generative story. Nor does odd-only propagation have an automatic physical law behind it. Its justification is a combination of spectral behavior and downstream validation. This is a legitimate scientific route, but the validation endpoint, rather than the square root in Eq. 11.16, supplies the biological evidence.

11.2.6 Leading-mode deletion: a nuisance model, not an inverse

Qin and Colwell analyzed why phylogenetic relatedness contaminates covariance matrices built from MSAs [8]. Their random-matrix calculation predicts a power-law tail among the largest covariance eigenvalues. On that model, selected leading directions are dominated by shared ancestry rather than by structural coupling. The corresponding correction is a spectral projection such as

$$C_{\text{clean}} = C - \sum_{k\in\mathcal{P}} \lambda_k u_k u_k^{\mathsf{T}}, \tag{11.17}$$

where $\mathcal{P}$ is chosen from the declared phylogenetic-tail criterion. Removing those modes improved contact prediction in their tests.

This method is especially revealing because it acts on the same eigendecomposition as precision estimation to target a different object from C^{-1}. Inversion applies $f(\lambda) = 1/\lambda$ to every retained direction, suppressing large covariance modes while amplifying small ones. Leading-mode deletion applies $f(\lambda) = 0$ to a selected nuisance set and $f(\lambda) = \lambda$ elsewhere. The first transform follows from a Gaussian conditional model; the second follows from a phylogenetic null model. They can improve the same contact benchmark for different reasons.

This corrects an overstatement in the 2022 essay. The Colwell construction belongs in the same spectral family. Its conceptual advance is to identify which large modes should be removed and supply a model for why. Without that null, deleting leading eigenvectors is only rank-relative cleaning and may erase real subfamily or functional structure. With it, the projection is an estimand-level decision of the kind developed in Chapter 9.

11.2.7 What is genuinely the same

The longitudinal sequence can now be summarized without either exaggerating or dissolving the unification:

Method	Forward claim	Spectral action	Output semantics and endpoint
Gaussian precision	covariance of one Gaussian joint law	$1/\lambda$	conditional graph in the fitted Gaussian family
MSE reconstruction	best linear reconstruction under squared loss	regularized inverse, up to an affine identity term	reconstruction weight; Gaussian meaning only under the equivalence conditions
PSICOV / ResPRE	high-dimensional Gaussian surrogate plus regularization	sparse support or ridge filter k_ρ	sparse coupling score, or full precision feature decoded into contacts
Network deconvolution	observation is the sum of all directed walk powers	$\lambda/(1+\lambda)$ after scaling	direct edge relative to the all-walk propagation model
BND	observation is the sum of odd walk powers	$f_{\text{BND}}(\lambda)$	direct edge relative to an odd-walk model; validated as contact postprocessing
Qin–Colwell	leading covariance tail is enriched for phylogenetic nuisance	delete selected leading modes	phylogeny-corrected covariance used for contact ranking

At the level of Eq. 11.5, the methods really are close. They reuse the same matrix, eigenvectors, rational or thresholded scalar functions, and often the same contact benchmark. This is why one implementation can suggest another and why regularization transfers are fruitful. At the level of scientific objects, their differences are concentrated rather than negligible. The forward law determines what has been inverted; the null determines what has been removed; the regularizer determines what can be identified from finite data; and the downstream decoder determines whether the transformed matrix is the answer or only an input feature.

Formally, let a method be the composite map

$$\mathcal{A}_m = \mathcal{V}_m \circ \mathcal{E}_m \circ \mathcal{R}_m \circ \Phi_m \circ R_m(O - O_{0,m})R_m^{\mathsf{T}}, \tag{11.18}$$

where $O_{0,m}$ is a null expectation, R_m a nuisance residualizer, Φ_m the inverse or spectral transform, $\mathcal{R}_m$ regularization, $\mathcal{E}_m$ edge extraction, and $\mathcal{V}_m$ validation or decoding. Two methods are fully equivalent on a declared class of inputs only when these composite maps agree on that class. Sharing O or even sharing $\Phi_m(O)$ is a weaker correspondence. Agreement of final contact rankings is weaker still, because scalarization can hide different categorical operators.

The durable conclusion is therefore not that inverse covariance is literally everything. It is that a surprisingly large body of protein inference can be organized as competing inverse problems on one second-order statistic. Once written that way, what looked like many unrelated algorithms becomes one design space. The important questions are no longer which paper used an inverse, a square root, or an eigenvalue cutoff. They are: which propagation law was assumed, which modes were declared nuisance, which edge object survived the transform, and which experiment made that declaration credible.

11.3 Graph Laplacians and multiscale operator spectra

11.3.1 Directed routes and stationary geometry

Graph spectral analysis begins only after the edge object and its compression have been declared. This first branch concerns a route matrix whose rows already sum to one. It asks where routing mass settles after repeated application and whether the stationary flow between every pair is balanced. This branch analyzes an iterated computational route. Categorical substitution effects and thermodynamic equilibrium belong to other constructions. For a nonnegative row-stochastic attention route P, the native object is directed. Assume first that P is irreducible, let $\pi^{\mathsf{T}}P = \pi^{\mathsf{T}}$ be its positive stationary law, and write $\Pi_\pi = \mathrm{diag}(\pi)$. Here P_{ij} uses the query-to-source convention $i \to j$; the value message in Eq. 7.13 travels along the oppositely oriented edge $j \to i$. The convention must remain fixed when cycles are interpreted. The adjoint in the stationary inner product is

$$P_\pi^* = \Pi_\pi^{-1}P^{\mathsf{T}}\Pi_\pi, \qquad P^{\mathrm{rec}} = \frac{1}{2}(P + P_\pi^*), \qquad P^{\mathrm{circ}} = \frac{1}{2}(P - P_\pi^*). \tag{11.19}$$

Both P and P_π^* are row-stochastic. The first term is reversible, while the second has zero row sum and records stationary directed circulation. A dimensionless detailed-balance defect is

$$\delta_{\mathrm{db}}(P) = \frac{\|\Pi_\pi P - P^{\mathsf{T}}\Pi_\pi\|_F}{\|\Pi_\pi P\|_F}. \tag{11.20}$$

It vanishes exactly when $P = P_\pi^*$. Equation 7.16 shows why the criterion uses stationary flux: unequal stationary masses can make a reversible route entrywise asymmetric. Doubly stochastic normalization fixes π to the uniform law while still allowing asymmetric directed cycles, so bistochasticity and reversibility remain different constraints [55, 56].

When both directional fluxes are positive, the same departure can be summarized by the entropy production functional

$$\mathrm{EP}(P) = \frac{1}{2}\sum_{i,j}(\pi_i P_{ij} - \pi_j P_{ji}) \log \frac{\pi_i P_{ij}}{\pi_j P_{ji}} \geq 0. \tag{11.21}$$

One-way support, as in a causal mask, requires a separate support diagnostic or an explicit smoothing convention; otherwise Eq. 11.21 can diverge. Reducible routes likewise require restriction to recurrent classes or a declared regularization. These are properties of a computational Markov route. A thermodynamic entropy-production interpretation additionally requires an independently identified physical stochastic dynamics [59]. Context-conditioned Markov-chain and interacting-particle analyses motivate studying route dynamics; output response and biological causation remain separate audits [57, 58].

The directed random-walk Laplacian

$$L_{\mathrm{rw}} = I - P \tag{11.22}$$

is generally non-self-adjoint and can have nonorthogonal modes. The reversible component admits the symmetric representative

$$L_{\mathrm{rec}} = I - \Pi_\pi^{1/2} P^{\mathrm{rec}} \Pi_\pi^{-1/2}, \tag{11.23}$$

which is the additive reversibilization used in a directed-graph Laplacian [60], whereas P^{circ} must be retained separately. Left/right singular subspaces, stationary fluxes, and nonnormality in the π-geometry retain directed transport omitted by an ordinary eigenbasis. For a frozen attention-only rollout across layers, the route is the ordered product $P^{(L)}(x)\cdots P^{(1)}(x)$; the input-dependent factors form a noncommuting product in general. A spectrum of the mean route, a mean of layer spectra, and a spectrum of the product are therefore different statistics.

The practical output of this branch is therefore a three-part report: stationary mass π, reversible routing P^{rec}, and residual circulation P^{circ}. A single symmetric spectrum sees only the second part. A causal residue-to-residue response is tested by a separate intervention on model outputs.

11.3.2 Scalar, signed, and connection Laplacians

The next branch starts from interaction estimates. Its Laplacian depends on what the signal carried by a node is allowed to remember. A scalar Laplacian retains only strength, a signed Laplacian also retains one declared polarity, and a connection Laplacian retains a transported categorical direction. Moving down this list preserves more structure but also requires more modeling choices.

An arbitrary signed or block-valued response has a native operator geometry; a Markov law arises only after an additional nonnegative normalization. Its directionality should be audited through the block reciprocity split in Eq. 4.30 and through typed left/right singular geometry before any nonnegative scalar normalization is imposed. Only if the scientific question deliberately discards edge direction should one transport all directed blocks to common anchor fibers and define a nonnegative symmetric static shadow, for example

$$A_{ij}^{\mathrm{stat}} = \frac{1}{2}\left(M_{i\leftarrow j}^{\mathrm{static}} + M_{j\leftarrow i}^{\mathrm{static}}\right), \qquad A_{ii}^{\mathrm{stat}} = 0. \tag{11.24}$$

Assuming positive degrees, let $D_{\mathrm{deg}} = \mathrm{diag}(A^{\mathrm{stat}}\mathbf{1})$ and

$$L_{\mathrm{sym}}^{\mathrm{stat}} = I - D_{\mathrm{deg}}^{-1/2} A^{\mathrm{stat}} D_{\mathrm{deg}}^{-1/2}. \tag{11.25}$$

Its quadratic form is

$$u^{\mathsf{T}} L_{\mathrm{sym}}^{\mathrm{stat}} u = \frac{1}{2}\sum_{i,j} A_{ij}^{\mathrm{stat}} \left(\frac{u_i}{\sqrt{d_i}} - \frac{u_j}{\sqrt{d_j}}\right)^2. \tag{11.26}$$

Low eigenvectors are therefore position fields that vary slowly across strong static edges, and the zero mode is $D_{\mathrm{deg}}^{1/2}\mathbf{1}$ [73]. This Laplacian is invariant under any replacement of the full blocks that preserves their norms. It can identify positional domains or bottlenecks. Signs, singular directions, substitution-specific coherence, and directed circulation have already been removed by this compression.

A signed positional score requires a different construction. If a declared categorical contraction produces a symmetric $S_{ij} \in \mathbb{R}$, define

$$D_{|S|} = \operatorname{diag}\left(\sum_j |S_{ij}|\right), \qquad L_\pm = D_{|S|} - S. \tag{11.27}$$

Then

$$u^\mathsf{T} L_\pm u = \frac{1}{2}\sum_{i,j} |S_{ij}|\,(u_i - \operatorname{sign}(S_{ij})u_j)^2 \geq 0. \tag{11.28}$$

Positive edges favor aligned position signals and negative edges favor opposing signals [75]. The sign must come from a declared projection of the categorical operator or from a signed routing variable. A block norm has discarded that information. A negative spectral edge encodes opposition in the declared contraction; probability and biological antagonism have their own meanings.

To retain categorical contrast orientation on an undirected support, work in the anchor-whitened fibers $\widetilde{\mathcal{H}}_i^0 = Q_i\mathbb{R}^q$ and choose for each undirected edge an orthogonal transport $U_{i\leftarrow j} : \widetilde{\mathcal{H}}_j^0 \to \widetilde{\mathcal{H}}_i^0$ satisfying $U_{j\leftarrow i} = U_{i\leftarrow j}^\mathsf{T}$. The polar factor of $\widetilde{G}_{i\leftarrow j}$ restricted to the centered fiber supplies one such orientation when that restriction is nonsingular; a rank-deficient fiber map requires a declared completion or restriction to its supported singular subspace. For symmetric nonnegative weights $a_{ij} = a_{ji}$, the connection energy is

$$\mathcal{E}_{\mathrm{conn}}(f) = \sum_{i<j} a_{ij}\,\|f_i - U_{i\leftarrow j} f_j\|_2^2 = \langle f, \mathbb{L}_{\mathrm{conn}} f\rangle. \tag{11.29}$$

The resulting connection Laplacian is positive semidefinite [76]. Its low modes are categorical contrast fields that remain coherent after edge-specific transport. The scalar Laplacian asks where a mode is smooth; the connection Laplacian additionally asks how a residue contrast must rotate between positions. Singular blocks admit several transport completions, so the chosen completion is an audited modeling coordinate.

Thus the connection construction computes smoothness only after neighboring categorical vectors have been aligned by the chosen transports. It can reveal a coherent substitution pattern that a norm graph erases. Uniqueness, allosteric interpretation, and correspondence to microscopic motion remain validation questions.

If $L = V\Lambda V^\mathsf{T}$ denotes either a scalar or connection Laplacian, the graph Fourier coefficients of a compatible signal y are $\widehat{y}_k = v_k^\mathsf{T} y$, and a spectral filter is

$$g(L)y = Vg(\Lambda)V^\mathsf{T} y. \tag{11.30}$$

The heat kernel e^{-tL} probes progressively coarser interaction neighborhoods as t increases; band-pass filters can separate domain-scale modes from localized edge variation. Low frequency means smoothness relative to the inferred graph; conservation, phylogeny, allostery, and function enter through calibrated correspondences.

11.3.3 Wavelets: spectral selection with spatial localization

Fourier modes answer which graph frequencies are present, but each eigenvector is generally spread over the whole graph. A wavelet asks a more localized question: at what scale, and around which

typed edge operators $\longrightarrow$ declared scalarization or transport $\longrightarrow$ L $\xrightarrow{g(L)}$ multiscale coefficients,
$g(\lambda) = e^{-t\lambda}$: heat smoothing $\quad g(\lambda) = \psi(s\lambda)$: wavelets $\quad g(\lambda) = \lambda^k$: polynomial propagation.

Figure 11.1: Spectral processing is downstream of edge identification. Different choices of g act on the same declared operator geometry but answer different multiscale questions.

node, does a change occur? On a self-adjoint graph Laplacian $L = V\Lambda V^\mathsf{T}$, choose a band-pass generating function ψ and define

$$\Psi_s = \psi(sL) = V\psi(s\Lambda)V^\mathsf{T}, \qquad \psi_{s,i} = \Psi_s\delta_i. \tag{11.31}$$

The vector $\psi_{s,i}$ is a graph-wavelet atom centered at node i and scale s [74]. Its spectrum is selected by $\psi(s\lambda)$, while its node-space profile records how that band is expressed around i. Graph wavelets couple spectral selectivity to spatial localization.

The scale convention depends on the chosen generator. For the common dilation $\psi(s\lambda)$, increasing s shifts a fixed band-pass profile toward smaller eigenvalues and usually produces a broader node-space atom. Small scales emphasize finer graph variation; large scales emphasize coarser organization. This is a statement relative to L. Atomic-motion and protein-domain interpretations require a separately calibrated correspondence between graph scale and the biological geometry.

Localization follows from the spectral construction. If $\psi(s\lambda)$ is approximated by a degree-K polynomial,

$$\psi(sL) \approx \sum_{k=0}^{K} c_k(s)L^k, \tag{11.32}$$

then $\psi(sL)\delta_i$ is supported, up to approximation error, within K graph steps of i. This is the same spectral-to-spatial passage used by Chebyshev graph filters. The wavelet family organizes several band-pass scales around localized centers; a task-specific graph filter instead learns the propagation profile required by its objective.

A low-pass scaling function ϕ_J retains the coarsest component. A stable wavelet transform requires more than an appealing collection of bands. If constants $0 < A \le B < \infty$ satisfy

$$A \le |\phi_J(\lambda)|^2 + \sum_{s\in\mathcal{S}} |\psi(s\lambda)|^2 \le B \tag{11.33}$$

on the spectrum of L, then

$$A\|y\|_2^2 \le \|\phi_J(L)y\|_2^2 + \sum_{s\in\mathcal{S}} \|\psi(sL)y\|_2^2 \le B\|y\|_2^2. \tag{11.34}$$

The transform is then a stable frame: it can be redundant without losing control of the signal energy. This distinguishes multiscale analysis from an arbitrary collection of spectral plots.

Heat diffusion, wavelets, and graph convolution can now be placed in one functional calculus:

$$\begin{aligned} e^{-tL} :&\quad \text{low-pass smoothing at diffusion scale } t,\\ \psi(sL) :&\quad \text{localized band-pass analysis at scale } s,\\ \sum_{k=0}^{K} \theta_k L^k :&\quad \text{finite-hop polynomial filtering,}\\ S^K H :&\quad \text{repeated propagation with possible over-smoothing.} \end{aligned} \tag{11.35}$$

Their common mathematics is $g(L)$; their state, objective, and interpretation remain different. A heat kernel describes smoothing relative to a graph, a wavelet transform analyzes a signal, and a GCN layer is trained as part of a predictor.

For an operator-valued protein graph, the analyzed signal must still be declared. It might be a scalar conservation score, a static interaction norm, a mutational response field, or a categorical section $f \in \bigoplus_i \mathcal{H}_i^0$. Scalar signals can be analyzed with a scalar Laplacian. Categorical sections require the connection Laplacian so that neighboring contrasts are transported between fibers before they are compared:

$$f \longmapsto \psi(s\mathbb{L}_{\mathrm{conn}})f. \tag{11.36}$$

The resulting coefficient measures scale-dependent coherence in the declared transport geometry. Recovery of the original edge operator would require an additional invertible representation and its associated reconstruction map.

Classical spectral graph wavelets also assume a self-adjoint operator with real spectral calculus. A directed attention route P is generally nonnormal and therefore calls for a different spectral construction. One may analyze the reversible Laplacian, a singular-value construction, or a declared symmetrization, but each discards or reorganizes directed circulation differently. Thus the order of reasoning remains

$$\boxed{\begin{array}{c}\text{declare the graph and signal} \longrightarrow \text{choose the operator geometry}\\ \longrightarrow \text{choose the scale filters} \longrightarrow \text{interpret localized coefficients.}\end{array}} \tag{11.37}$$

Low-, band-, and high-frequency coefficients are mathematical facts about that construction. Calling them conservation, allostery, physical vibration, or mechanism requires an external endpoint that licenses the name.

11.3.4 Context averaging and null-relative spectra

Static and dynamic shadows define different Laplacians. Construct A_{ij}^{dyn} from $M_{i\leftarrow j}^{\mathrm{dynamic}}$ by the same declared symmetrization as Eq. 11.24, and compare the resulting spectra as subspaces. If V_r^{stat} and V_r^{dyn} contain a chosen r-dimensional spectral band, one basis-invariant distance is

$$d_r^{\mathrm{spec}} = \frac{1}{\sqrt{2r}} \left\| V_r^{\mathrm{stat}} V_r^{\mathrm{stat}\mathsf{T}} - V_r^{\mathrm{dyn}} V_r^{\mathrm{dyn}\mathsf{T}} \right\|_F. \tag{11.38}$$

Individual eigenvectors are unstable inside a nearly degenerate band, whereas the spectral projector remains well defined. For context-specific graphs, spectral averaging is nonlinear: in general

$$L_{\mathrm{sym}}(\mathbb{E}_C A(C)) \neq \mathbb{E}_C L_{\mathrm{sym}}(A(C)), \qquad \lambda_k(\mathbb{E}_C L(C)) \neq \mathbb{E}_C \lambda_k(L(C)). \tag{11.39}$$

When averaging graphs creates a mode that no context has

Consider three positions and two equally likely sequence contexts. In context A, only positions 1 and 2 are connected; in context B, only positions 2 and 3 are connected. Their unnormalized Laplacians are

$$L_A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad L_B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix}. \tag{11.40}$$

Each graph consists of one edge and one isolated vertex, so both spectra are

$$\mathrm{spec}(L_A) = \mathrm{spec}(L_B) = \{0, 0, 2\}. \tag{11.41}$$

The mean Laplacian is

$$\overline{L} = \frac{1}{2}(L_A + L_B) = \begin{pmatrix} 1/2 & -1/2 & 0 \\ -1/2 & 1 & -1/2 \\ 0 & -1/2 & 1/2 \end{pmatrix}, \tag{11.42}$$

with spectrum

$$\mathrm{spec}(\overline{L}) = \{0, 1/2, 3/2\}. \tag{11.43}$$

The mean graph is connected even though no context contains both edges simultaneously. Averaging has created a path 1−2−3 that exists only across contexts. Correspondingly, the second zero mode present in every individual context disappears from the mean.

This tiny example separates three claims. The mean context spectrum is $\{0, 0, 2\}$. The spectrum of the mean graph is $\{0, 1/2, 3/2\}$. The distribution of zero-mode projectors records which vertex is isolated in each context. None determines the others.

For protein responses, the distinction can be biological. Two alternative sequence backgrounds may support mutually exclusive communication paths. Their mean graph suggests a connected route through all three positions, while no individual protein realizes that route. A spectral analysis of $\mathbb{E}_C A(C)$ therefore answers a population-level aggregation question, not a statement about a typical sequence. Context-specific projectors or their distribution must be retained when that difference matters. The mean graph, mean Laplacian, mean spectrum, and distribution of spectral projectors answer different questions. Because this construction begins after symmetrization, it compares undirected shadows only; context-dependent detailed-balance defect, circulation, and directed singular subspaces must be compared before that projection.

Finally, a null-relative Laplacian spectrum must account for zero modes. For unnormalized Laplacians one may restrict to $\mathbf{1}^\perp$; for normalized graphs with unequal degree modes or unequal null spaces, use an explicitly declared ridge $\epsilon > 0$. Generalized modes then extremize

$$\gamma(u) = \frac{u^\mathsf{T} L_{\mathrm{real}} u}{u^\mathsf{T} L_0 u + \epsilon \|u\|_2^2}. \tag{11.44}$$

Small and large values identify directions that are respectively smoother or rougher in the real graph than under the null geometry. Significance should be calibrated from replicate null graphs for eigenvalues, gaps, localization, and spectral projectors. The position Laplacian, signed Laplacian, connection Laplacian, categorical projector P_i, and Potts pair-table redundancy therefore occupy distinct coordinates of the framework.

Coda: spectra inherit every earlier choice

A spectrum inherits the meaning of the graph from which it was built. Gauge projection, scalarization, symmetrization, stationary weighting, normalization, context averaging, and null subtraction all change the operator being diagonalized. Reporting those choices is therefore part of the spectral result.

Used with that discipline, spectral analysis becomes a genuine multiscale extension of the operator framework. Scalar modes describe positional smoothness, signed modes encode frustrated polarity, connection modes probe transported categorical coherence, and directed decompositions separate reversible flow from circulation. Comparing their projectors can reveal organization that no single edge statistic exposes.

The word “mode” carries a physical temptation that must be resisted until a dynamics has been declared. An eigenvector of a scalar graph Laplacian is a field that is smooth relative to chosen edge weights. It becomes a vibrational mode only when the operator is a stiffness matrix with a mass metric; it becomes a relaxation mode only when the operator generates a justified dissipative dynamics; and it becomes a diffusion mode only when the state variable and transition rates have physical meaning. Without those structures, low frequency means graph smoothness and nothing more.

The distinction is sharper for the other spectra. A connection-Laplacian mode transports amino-acid contrast directions, not atomic displacement vectors. A signed mode records coherence or frustration under a declared scalar contraction, not automatically favorable or unfavorable thermodynamic coupling. Entropy production of an attention route quantifies irreversibility of a computational Markov chain; it is literal thermodynamic entropy production only if that chain has been identified with a physical stochastic process. Spectral resemblance to sectors, allosteric pathways, or collective motions is therefore a hypothesis to test against structural ensembles, perturbation responses, or time-resolved dynamics, not a conclusion supplied by diagonalization.

The remaining question is constructive. If the desired interaction object is now clear, can a trainable architecture realize it while preserving the measurability conditions that gave it meaning? Sparse support, dynamic routing, endpoint exclusion, persistent edge state, and efficient execution jointly define a genuine design problem. The next part turns from interpretation to realizability.

Part IV

Realizability, Evidence, and Interpretation

Chapter 12

Architectural Consequences and Realizability

Once an interaction object has been defined, it is tempting to design an architecture that emits it directly. The difficulty is that semantic requirements become information constraints. If an edge is claimed to be invariant under substitutions at its endpoints, then the mechanism choosing that edge cannot depend on those endpoint values. If a zero edge is claimed to save computation, the implementation must actually skip its work. If an update is claimed to descend an energy, its memories, state variables, and parameter tying must satisfy the assumptions of the descent proof.

These constraints are easy to blur because neural architectures combine several roles in one tensor. An attention coefficient can select a route, scale a message, and contribute to an output response. A pair representation can store persistent state or merely cache a transient feature. A soft gate can express uncertainty without producing exact support. The operator framework separates these roles before asking which implementation is efficient.

The chapter develops two kinds of result. The first kind is semantic: endpoint-removable context, strict targetwise support, reciprocity, and state–energy compatibility state what an architecture must satisfy for a particular interpretation to be valid. The second kind is constructive: signed routing, affine deletion, persistent pair tracks, sparse execution, and optional MSA supervision are possible ways to meet or approximate those requirements. The constructions are examples, not definitions of the framework.

Two limitations are especially revealing. A deterministic graph router that must remain globally invariant to all active endpoint identities can collapse to a fixed graph. Relaxing the claim to a targetwise directed conditional operator avoids that collapse but gives up the semantics of one globally identified undirected graph. Separately, a router forbidden from observing two endpoint residues faces an irreducible Bayes error whenever the desired edge label depends on information contained only in those residues. Training data and teacher supervision can approach that floor; they cannot remove it at inference.

Realizability is therefore not an afterthought. It marks the boundary between a mathematically well-defined interaction and a trainable mechanism that can infer or execute it under finite information and compute.

The operator coordinates distinguish a semantic interaction object from an implementation that computes it. Support selection, routing normalization, persistent state, endpoint measurability, and update dynamics are independent architectural decisions. Endpoint measurability, support invariance, and state–energy compatibility below are semantic requirements. Signed routing, affine deletion, persistent pair tracks, and optional-MSA supervision are constructive examples, not required components of the master object.

12.1 Routing geometry and persistent relational state

Standard attention chooses a nonnegative row-stochastic route

$$\alpha_{ij} = \frac{e^{z_{ij}}}{\sum_k e^{z_{ik}}}, \qquad y_i = \sum_j \alpha_{ij} V_j. \tag{12.1}$$

For fixed values, y_i lies in $\mathrm{conv}\{V_j\}$. A more general interaction operator may use signed coefficients with bounded total variation. If $\sum_j |a_{ij}| \le 1$, then

$$y_i = \sum_j a_{ij} V_j \in \mathrm{conv}(\{0\} \cup \{V_j, -V_j\}_j), \tag{12.2}$$

which permits source-specific contrasts and cancellation. As one illustrative design, a continuous signed sparse map is

$$w_{ij} = \mathrm{sign}(z_{ij})(|z_{ij}| - \lambda)_+, \qquad a_{ij} = \frac{w_{ij}}{\epsilon + \sum_k |w_{ik}|}. \tag{12.3}$$

It has exact zeros and bounded total variation, but superiority to softmax, sparsemax, entmax, or other signed attention rules is an empirical question [70, 71, 72].

Routing geometry is separate from edge-value capacity. A node-only message-passing layer retains $O(Ld)$ persistent states even if it evaluates dense transient connectivity. A dense pair track retains $O(L^2 d_p)$ relational state, as in structure models with explicit pair representations [64]. A degree-k operator graph retains $O(Lkd_p)$ edge state once support is known. Categorical rank and interaction order are independent capacity coordinates.

State dynamics is another coordinate. The same $\mathbb{K}_z$ may define a one-pass readout, an iterated map, or part of a descent dynamics. A Hopfield claim combines a specified state space, fixed memories, update schedule, and Lyapunov decrease in addition to graph support.

12.2 Inductive bias as structural alignment

An inductive bias is the preference that makes a learning procedure choose some solutions over other solutions that fit the available data equally well. It may enter through a prior, a regularizer, a function class, or the architecture itself. Battaglia et al. use this idea to argue that entity–relation structure should be built into neural computation when the world is usefully described by entities, relations, and reusable composition rules [33]. A structural assumption that matches the data-generating organization can reduce sample complexity and support combinatorial generalization.

In concrete terms, choosing an inductive bias means writing a function whose organization can correspond to regularities internal to the data. That correspondence is a hypothesis about how the problem is structured. A representation ϕ_θ chooses which distinctions and relations become easy to express, while a function class $\mathcal{H}$ determines which maps of those representations can be learned efficiently. Let

$$m(x) = \mathbb{E}[Y \mid X = x] \tag{12.4}$$

be the population response. A structurally aligned representation–function pair makes

$$\inf_{f \in \mathcal{H}} \mathbb{E}\Big[\big(m(X) - f(\phi_\theta(X))\big)^2\Big] \tag{12.5}$$

small while keeping $\mathcal{H}$ simple enough to estimate from the available data. The representation exposes a regularity; the function class is organized so that this regularity can be expressed without relearning the structure from scratch. This is the precise content of "matching the function to the data structure." It is an architectural claim to be tested, not a property guaranteed by scale, pretraining, or a graph-shaped input alone.

One-body structure is one special inductive bias, not a synonym for inductive bias. An additive class $f(x) = c + \sum_i h_i(x_i)$ prefers explanations assembled from independent site effects; a pairwise Potts class enlarges that preference to include $G_{ij}(x_i, x_j)$; low-rank constraints and sparse support impose further biases inside the pairwise class. Other architectures may instead prefer locality, symmetry, relational composition, hierarchy, or low-rank structure. Whether one-body, pairwise, local, equivariant, compositional, or low-rank structure is appropriate is an empirical claim about the endpoint in the chosen coordinates.

For coordinate models, symmetry supplies a particularly explicit example. A global rigid motion $g = (R, t)$ acts on every atomic coordinate by

$$g \cdot r_i = Rr_i + t, \qquad R \in SO(3), \quad t \in \mathbb{R}^3, \qquad g \in SE(3). \tag{12.6}$$

A coordinate-valued map is $SE(3)$-equivariant when $F(g \cdot r) = g \cdot F(r)$; a scalar readout is invariant when $q(g \cdot r) = q(r)$. Restricting the function class to obey these identities builds the physical irrelevance of the laboratory frame into the model before any data are observed. This is a group-theoretic inductive bias. Reflections belong to the larger Euclidean group $E(3)$, but are not in $SE(3)$; because protein stereochemistry is chiral, a mirror reflection is not generally a physically equivalent coordinate change.

Finally, low approximation error on one benchmark establishes task alignment. Alignment with the scientific mechanism requires separating alternatives such as phylogeny, experimental batch, database composition, or another nuisance regularity. Prospective tests, clade-aware splits, uncertainty calibration, and endpoint-specific interventions are what distinguish useful structural alignment from shortcut alignment.

12.2.1 Pallatom: a case study in structural alignment

Pallatom makes this idea unusually concrete [42]. Its coordinate convention, relational architecture, denoising objective, and sequence readout are designed together around one hypothesis: sequence identity, backbone shape, and side-chain packing can be generated coherently by modeling a suitably represented all-atom coordinate distribution.

The first bias is introduced before the neural network sees any data. A residue with variable chemical topology is embedded into fourteen coordinate slots. Native heavy atoms occupy their standard slots; absent atoms are represented by virtual points at the C_α coordinate:

$$(x, a) \xmapsto{\Psi_{14}} r_0 \in \mathbb{R}^{N \times 14 \times 3}. \tag{12.7}$$

This map converts transdimensional generation into ordinary Euclidean point-cloud diffusion. It asserts that a residue is the stable compositional unit, that fourteen slots suffice for the canonical amino acids, and that the artificial coincidence of unused slots at C_α is a learnable nuisance. These assumptions make the problem computationally regular; their status is representational, while molecular cardinality and atom existence belong to the physical ontology.

The second bias is relational and multiscale in the sense emphasized by Battaglia et al. [33]. Pallatom uses local atom-level attention for fine geometry and a globally communicating residue track with pair features for fold-scale organization. Residue features are broadcast to atoms; atom features are aggregated back to residues; denoised coordinates are recycled into pair features; and triangle updates encourage relational consistency. The architecture therefore assumes

$$\text{local atomic geometry} \longleftrightarrow \text{residue identity and state} \longleftrightarrow \text{global residue–residue organization.} \tag{12.8}$$

This hierarchy is well matched to folded proteins and avoids global attention over every atom pair. Long-range atom–atom effects are therefore mediated through a residue-level bottleneck, which defines the model's relational capacity at fold scale.

The slogan "learn P(all-atom)" should be read with care. Pallatom supplies the denoiser with a reduced side-chain labeling, and several amino acids share geometrically degenerate atom14 patterns. Coordinates alone therefore leave sequence ambiguous. The trained object combines coordinate score matching with supervised sequence cross-entropy, residue- and atom-level distogram losses, smooth-LDDT loss, and intermediate supervision. Schematically,

$$\mathcal{L} = \lambda\mathcal{L}_{\text{atom}} + \alpha_0\mathcal{L}_{\text{seq}} + \alpha_1\mathcal{L}_{\text{LDDT}} + \alpha_2\mathcal{L}_{\text{dist,res}} + \alpha_3\mathcal{L}_{\text{dist,atom}} + \alpha_4\mathcal{L}_{\text{intermediate}}. \tag{12.9}$$

Thus the representation, architecture, and auxiliary targets together make the sequence–structure relation learnable beyond the information in an unsupervised coordinate density alone. Pallatom encourages the $SE(3)$ identity statistically through coordinate centering, random rigid-motion augmentation, and a Kabsch-aligned coordinate loss. The function class still admits departures from exact equivariance. This distinction is substantive: symmetry learned from augmentation is an approximate empirical regularity, whereas symmetry imposed by the architecture holds by construction.

The training distribution contributes a third, unusually strong bias. The reported model uses short monomers, crops of length at most 128, 7,459 filtered PDB structures, and 27,697 heavily curated AFDB structures. AFDB candidates are required to have high predicted confidence and are then filtered for packing density, core-residue fraction, loop content and length, radius of gyration, secondary-structure segmentation, and redundancy. Consequently the learned density is better written as

$$p_\theta(r \mid r \in \mathcal{F}_{\text{compact,designable}}(\text{PDB} \cup \text{AFDB})) \tag{12.10}$$

than as a universal distribution over proteins. This is a legitimate design choice: it aligns the training set with the desired endpoint of compact, prediction-compatible monomers. It also limits claims about disordered proteins, complexes, alternate states, membrane environments, and functional negative examples.

The ablations support this alignment interpretation. In the reported setup, atom14 achieves 87% all-atom designability, compared with 5% for a sequence-guided hybrid14 representation; removing differentiable recycling lowers it to 21%. These results identify the computational representation that works with this denoiser and objective. A claim about physical ontology would require independent physical evidence. Indeed, changing the sampling step scale raises reported designability while reducing diversity and shifting the fraction of all-helical outputs. The sampler is therefore part of the inductive bias and changes which region of the learned distribution is visited.

The evaluation closes the same loop. Pallatom reports 85.03% all-atom designability in its primary comparison, where a sample passes when ESMFold assigns mean pLDDT above 80 and the all-atom RMSD between the generated structure and the structure predicted from its generated sequence is below 2 Å. ProteinMPNN, ESMFold, Foldseek, and MMseqs2 provide useful tests of self-consistency, backbone redesignability, novelty, and diversity. Their own training distributions and structural priors also enter the measurement. High designability therefore licenses the statement that the generated sequence and coordinates are mutually recoverable under current predictors. Stability, catalytic activity, binding specificity, equilibrium weights, and experimental success remain separate validation endpoints.

Pallatom remains context-dependent even though its final sequence head is residuewise. The head pools atomic embeddings and applies a linear classifier after those embeddings have received global residue, pair, triangle, and recycled-coordinate information. The output logit at residue i is therefore a function of the generated molecular context. A local readout can therefore express pairwise and higher-order context already mixed into its input representation.

Pallatom's main methodological achievement is therefore an alignment result. A fixed-cardinality all-atom representation, a local–global relational network, curated compact-protein data, a composite geometric objective, and predictor-based designability metrics all point toward the same target. That alignment explains the model's strong in-silico performance more precisely than saying that it simply "learned chemistry." The remaining scientific question is whether the aligned target is broad enough for the intended use; answering it requires prospective synthesis and measurements under functional and negative-control conditions.

Structural alignment and conditional identifiability answer successive questions. Pallatom shows how a representation, architecture, objective, and evaluation loop can point coherently toward one endpoint. The next sections ask a different question: when an edge or router is interpreted as a conditional relation, can its context be held fixed while the endpoint categories are varied? That requirement leads to endpoint-removable context and exposes an information tradeoff that good architectural alignment alone leaves unresolved.

12.3 Endpoint-removable context

Definition 12.1 (Endpoint-removable context)**.** For a deleted set $A \subseteq [L]$, a context map $C_{-A} : [q]^L \to \mathcal{S}$ is endpoint-removable if $x_{-A} = x'_{-A}$ implies $C_{-A}(x) = C_{-A}(x')$.

Endpoint-removable context must be constructed before information propagates from the deleted positions; post hoc subtraction leaves their traces in retained states. The following construction establishes realizability for associative token-local recurrences as one concrete design.

Theorem 12.2 (Associative affine deletion)**.** *Let token k define the affine transition $F_k(h) = A_k(x_k)h + b_k(x_k)$ and augmented matrix*

$$M_k(x_k) = \begin{bmatrix} A_k(x_k) & b_k(x_k) \\ 0 & 1 \end{bmatrix}. \tag{12.11}$$

For a deleted set A, replace M_k by the identity when $k \in A$. Then

$$M_{-A} = M_L^{(A)} M_{L-1}^{(A)} \cdots M_1^{(A)} \tag{12.12}$$

is endpoint-removable and preserves the order of every retained transition.

Proof. Every deleted factor is the identity and is independent of the deleted category. All remaining token-local factors occur in their original order, while associativity permits arbitrary parenthesization. □

The theorem applies exactly to the affine recurrence core. Surrounding convolutions, normalizations, and context-dependent parameter generators require their own deletion laws. A segment-product data structure can evaluate a cavity with $|A| = m$ by composing at most $m + 1$ retained intervals.

12.4 Strict targetwise conditional operators

For a target i and selected source set N_i, the shared strict context is the whole-neighborhood cavity

$$\mathcal{C}_i^E = \sigma\{X_k : k \notin \{i\} \cup N_i\}. \tag{12.13}$$

Every reference, route decision, coefficient, and pair table directed into i must be measurable with respect to this same cavity. Only the final categorical lookup may observe a selected source state x_j.

Definition 12.3 (Strict targetwise support)**.** A directed router is strict for target i if $N_i(x) = N$ implies $N_i(x') = N$ whenever x' differs from x only on $\{i\} \cup N$. For a randomized router the condition is pathwise after fixing a seed independent of the sequence.

Conditioned on c_i^E, define

$$\widehat{G}_{i\leftarrow j}(c_i^E) = P_i(c_i^E) S_{i\leftarrow j}(c_i^E) P_j(c_i^E)^{\mathsf{T}}, \tag{12.14}$$

$$\ell_i(a) = b_i(a; c_i^E) + \sum_{j\in N_i} a_{i\leftarrow j}(c_i^E)\widehat{G}_{i\leftarrow j}(a, x_j; c_i^E). \tag{12.15}$$

Proposition 12.4 (Conditional pairwise operator)**.** *If N_i is strict and every object in Eq. 12.15 is measurable with respect to $\mathcal{C}_i^E$, then the interaction branch is exactly pairwise in the active endpoint variables under the declared product reference. Its only access to a selected source category is the final lookup $\widehat{G}_{i\leftarrow j}(a, x_j; c_i^E)$.*

Proof. Condition on the cavity and the realized support. All tables, references, and coefficients are fixed while (x_i, x_{N_i}) vary. Theorem 3.2 centers each target–source term, and additivity over j excludes components involving two active sources. Theorem 5.1 identifies the branch as the pairwise truncation. □

Projection with an ordinary full-context table remains a useful local gauge. The stricter global functional semantics comes from the shared cavity construction above.

12.5 Dynamic support is directed and targetwise

Let a deterministic router $E : [q]^L \to \mathcal{G}_L$ output an undirected graph. Call it globally endpoint-invariant if changing the categories of any active vertices leaves the entire realized graph unchanged.

Theorem 12.5 (Spanning-support no-go)**.** *If a globally endpoint-invariant router has one input for which its output graph has no isolated vertices, then the router is constant on $[q]^L$. Equivalently, every output of a nonconstant globally endpoint-invariant router contains at least one isolated vertex.*

Proof. For a realized graph with no isolated vertices, every sequence position is active. The complement of the active set is empty, so every other input agrees outside the active set. Global endpoint invariance therefore forces the same graph for every input. □

Targetwise routing remains adaptive under the theorem. $N_i(x)$ stays fixed when its own target and realized sources vary, while other residues may still alter routes for targets to which they are unselected. The resulting predictive object is a directed cavity operator; an undirected contact graph requires a separate identification step.

12.6 Single-sequence recoverability and optional-MSA supervision

Let $T_{ij} \in \{0,1\}$ be a training-only support label, $Y_{ij} = X_{[L]\setminus\{i,j\}}$, and $Z_{ij}(X) = \mathbf{1}\{j \in N_i(X)\}$ for a strict router. Strictness implies that Z_{ij} is a function of Y_{ij} alone. Indeed, invariance to x_i is immediate. If j is selected, invariance to x_j follows from the definition. If j is not selected but changing x_j made it selected, applying strictness at the modified input while reverting x_j would force the modified support at the original input, a contradiction. Define

$$e_{ij}^{\mathrm{cav}} = \mathbb{E}\min\{\eta_{ij}(Y_{ij}), 1-\eta_{ij}(Y_{ij})\}, \qquad \eta_{ij}(y) = \Pr(T_{ij} = 1 \mid Y_{ij} = y). \tag{12.16}$$

Proposition 12.6 (Cavity Bayes floor)**.** *Every strict deterministic or independently randomized router satisfies*

$$\Pr(Z_{ij} \neq T_{ij}) \geq e_{ij}^{\mathrm{cav}}. \tag{12.17}$$

Proof. Conditional on $Y_{ij} = y$, endpoint invariance restricts the prediction to a label that observes neither endpoint. The minimum conditional zero-one risk is the smaller posterior mass. Independent randomization has the same or larger linear conditional risk. Taking expectation proves the claim. □

The exact price of endpoint exclusion

The cavity Bayes floor can be maximal even in a two-bit example. Let X_i and X_j be independent uniform binary variables and define a desired edge label

$$Z_{ij} = \mathbf{1}[X_i = X_j]. \tag{12.18}$$

An endpoint-visible router predicts the label perfectly: it compares the two bits. A strict cavity router observes only

$$Y_{ij} = X_{[L]\setminus\{i,j\}}, \tag{12.19}$$

which is independent of Z_{ij} in this construction. Consequently,

$$\Pr(Z_{ij} = 1 \mid Y_{ij}) = \frac{1}{2}, \qquad e_{ij}^{\mathrm{cav}} = \frac{1}{2}. \tag{12.20}$$

No training procedure, teacher graph, model scale, or randomized decision rule can reduce the strict inference error below one half without restoring endpoint information or adding a correlated side channel.

The example exposes the semantic tradeoff cleanly. The endpoint-visible router has perfect label accuracy, but its support changes when either endpoint is intervened upon. It is a valid dynamic computational route and an invalid fixed conditional pair selector. The cavity router preserves the edge definition during endpoint substitution, but the desired label is not identifiable from its allowed information.

Real proteins lie between these extremes. Neighboring sequence context, predicted secondary structure, templates, MSA-derived family evidence, or other modalities may make an edge partly predictable without direct endpoint access. The cavity posterior then moves away from one half and the Bayes floor falls. That gain should be attributed to the added observable evidence, not to an architecture overcoming an information theorem.

This is why router performance and interaction semantics must be reported together. A useful comparison includes at least three variants:

$$\begin{aligned} q_{ij}^{\mathrm{visible}} &= q(X_i, X_j, Y_{ij}), && \text{adaptive computational support,} \\ q_{ij}^{\mathrm{cavity}} &= q(Y_{ij}), && \text{strict endpoint-invariant support,} \\ q_{ij}^{\mathrm{fixed}} &= q(i, j), && \text{static candidate geometry.} \end{aligned} \tag{12.21}$$

The first measures the attainable predictive frontier, the second the price of strict conditional meaning, and the third the value of content-dependent routing itself. No one variant is universally correct; each licenses a different claim.

As an illustrative training design, an optional MSA teacher can estimate family-level edge importance while the inference-time cavity floor remains fixed. Let E_c be a declared candidate set, $q_{ij}^{\theta} \in [0,1]$ a student support confidence, and $\widehat{G}_{ij}^{\theta}$ its reference-centered block. Teacher and student must use the same detached categorical reference:

$$G_{ij}^{T} = P_i^{[\bar{p}]} T_{ij}^{\mathrm{int}} P_j^{[\bar{p}]\mathsf{T}}, \qquad T_{ij}^{\mathrm{int}} = T_{ij}^{\mathrm{real}} - \mathbb{E}_{\widetilde{X} \sim Q_{\mathrm{null}}} T_{ij}^{\widetilde{X}}. \tag{12.22}$$

A schematic objective is

$$\begin{aligned} \mathcal{L} = \mathcal{L}_{\mathrm{task}} &+ \lambda_m \sum_i \mathrm{KL}(p_i^{\mathrm{MSA}} \| p_i^{\theta}) \\ &+ \lambda_G \sum_{(i,j)\in E_c} \| q_{ij}^{\theta} \widehat{G}_{ij}^{\theta} - G_{ij}^{T} \|_{\bar{p}_i, \bar{p}_j}^2 + \lambda_0 \sum_{(i,j)\in E_c} q_{ij}^{\theta}. \end{aligned} \tag{12.23}$$

MSA availability changes supervision, not the single-sequence inference contract. This objective is one architecture consistent with the framework, not a convergence or consistency theorem.

Coda: information constraints survive architecture

The architectural results expose a tradeoff that is otherwise hidden by flexible notation. Content-dependent routing gains adaptivity by reading the sequence, but strict endpoint invariance removes some of that information. Global undirected semantics demand compatibility that targetwise directed decisions may violate. Soft gates are differentiable, whereas hard zeros are needed for exact support and realized sparse execution. Persistent edge state increases relational capacity but changes the memory and scaling budget.

No single architecture optimizes all coordinates. The correct design depends on the claim being made. A high-performance dynamic route may be entirely appropriate when the object is computational. A cavity construction is required only for the stronger conditional interpretation. A symmetric pair track may be valuable for structural prediction without proving that its state is a physical pair potential.

Architecture can reproduce the algebra of physics without inheriting its ontology. A Lyapunov function proves monotone behavior of a specified update, but it is a thermodynamic free energy only when the state, measure, temperature, and exchange with an environment have been identified. Reciprocity of directed categorical responses is the integrability condition that permits one pair-energy block; failure of reciprocity leaves a legitimate directed computation with no single Hamiltonian representation. Persistent circulation or nonzero detailed-balance defect is correspondingly nonequilibrium-like. A physical-dissipation interpretation additionally requires coupling the network dynamics to measurable currents and entropy flow.

The cavity Bayes floor is an information-theoretic measure of the uncertainty remaining after endpoint identities have been withheld. More side information or a weaker invariance contract can change this inference floor. Activation barriers and free-energy costs instead depend on temperature, kinetics, and molecular pathways. The architecture chapter therefore licenses statements about computability, measurability, and integrability; physical meaning still arrives through calibration and experiment.

The remaining arbiter is evidence. Mathematical calibration verifies whether an implementation realizes its declared object. Separate estimators, nulls, and interventions then test its correspondence to evolutionary constraint, structure, fitness, or mechanism.

Chapter 13

Falsification and Biological Interpretation

Interpretability claims are often evaluated by accumulation: more correlations, more visual examples, more downstream benchmarks. A typed framework suggests a different standard. Every semantic upgrade should have an observation that could reject it while leaving weaker claims intact.

The first level is mathematical calibration. A gauge projection should satisfy its zero-marginal identities. Reference transport should preserve the underlying equivalence class. A Potts audit should recover the known coupling in the additive limit. A claimed Hopfield realization should show Lyapunov descent under the conditions of the proof. These tests establish whether the named mathematical object has actually been implemented; biological validation begins at a later rung.

The second level compares estimators. Static MSA fields and mean pLM responses should be evaluated as full categorical blocks in a common reference, with their estimand gap and uncertainty exposed. Dynamic residuals should be tested across natural, mutational, and held-out backgrounds. Route reversibility should be separated from response reciprocity. Null ensembles should preserve the nuisance structure that the analysis claims to remove.

The third level is biological and must match the type of the prediction. A scalar position score can be compared with contacts or distances. A typed edge operator calls for a gauge-aligned double-mutant matrix. A background-dependent response calls for the same perturbations across multiple sequence contexts. A higher-order residual calls for combinatorial interventions beyond pairs. Validation becomes stronger when the endpoint retains the structure of the inferred object instead of compressing it prematurely.

Falsification is constructive here. Failure of static agreement can reveal transfer or misspecification. Failure of reciprocity can expose directed computation. Failure of pair truncation can quantify higher-order dependence. A theory earns scope not by surviving every possible test, but by saying precisely what each failure means.

13.1 A falsifiable static–dynamic program

The framework is most useful when its objects are forced to fail separately. The following three cases move from a system whose truth is known, through a family comparison with partially known estimands, to biological interventions where the endpoint itself limits the conclusion.

Case I: the additive limit must be boring

Begin with sequences sampled from a known pairwise Potts model under a product or exactly controlled background law. The ground-truth fields, pair blocks, support, and gauge class are available. A correct substitution audit must recover

$$G_{i \leftarrow j}(C) = P_i J_{ij} P_j^{\mathsf{T}} \tag{13.1}$$

for every background C. Dynamic variance must vanish up to numerical and sampling error, and the two directed responses must satisfy reciprocity after transport to the same reference.

This case is intentionally uninteresting biologically. Its purpose is to make implementation errors visible. A nonzero dynamic residual can reveal that endpoint identities leaked into the reference or router, that logits were compared in inconsistent gauges, or that the source intervention changed more than one coordinate. A reciprocity defect can reveal transposed endpoint conventions or incompatible reference transport. A mismatch between tangent norm and exact mutual information at weak coupling can reveal a scaling error.

The same synthetic system can then be perturbed one assumption at a time. Add a pure third-order term while keeping the pair component fixed. The mean pair response should remain the projected second-order component under the product reference, while dynamic variance increases by the exact higher-order mass. This test must assess localization, not merely nonzero variance: rank audited pairs or source–modulator tuples by their recovered order-specific mass and report AUC or enrichment against the known synthetic hyperedges, not against ordinary contact pairs: on a pairwise system, approximation error can scale with the static operator norm and spuriously enrich contacts. If one pair belongs to several planted hyperedges, its truth is the full modulator set; test set recovery or whether the top recovered modulator belongs to that set rather than assigning one arbitrary label. Apply the nontriviality threshold in Eq. 4.23 before ranking ρ^{dyn}. Train the same deep architecture on a column-wise shuffled alignment that preserves one-site marginals while destroying the planted dependence. A large ρ^{dyn} under this null is expected from generic nonlinear context sensitivity; failure to localize the planted hyperedges above the shuffled control rejects the claimed higher-order recovery even when dynamic variance is positive.

Introduce phylogenetic sampling without changing the generative Hamiltonian. The underlying operator stays fixed, but its estimator becomes biased and its uncertainty ceases to follow an independent-row bootstrap. Replace the product background by a correlated one. The mean–residual Pythagorean identity remains exact under the new law, while the decomposition of dynamic variance into order-specific ANOVA masses acquires cross terms. Report both $\rho^{\mathrm{dyn,prod}}$ and $\rho^{\mathrm{dyn,emp}}$ from Eq. 4.24, together with $\Delta\rho^{\mathrm{geom}}$. The product quantity tests the exact ANOVA identity but may probe off-manifold backgrounds; the empirical quantity stays closer to the sampled manifold but has no orthogonal order decomposition. Also repeat the additive calibration under multiple full-support endpoint references: positive magnitudes may change, whereas the zero-dynamic verdict must obey Eq. 3.30.

Each modification targets a different coordinate. Higher-order structure changes the estimand, phylogeny changes the sampling law, and background correlation changes the geometry of the decomposition. Distinguishing them requires operator and sampling diagnostics alongside contact precision.

Case II: family field and pLM response may agree for the wrong reason

Choose a protein family with sufficient effective depth to fit a range of regularized Potts models. Partition the MSA by clade rather than by random rows. Use the training clades to define an anchor reference and fit the family field; use held-out natural or designed sequences as backgrounds for the pLM substitution audit.

The first comparison is operator-valued:

$$\widehat{G}^{\mathrm{P}}_{i\leftarrow j} \quad \text{versus} \quad \widehat{\overline{G}}^{\mathrm{pLM}}_{i\leftarrow j}\,. \tag{13.2}$$

Report alignment of full blocks, singular directions, reciprocity, and the pLM dynamic fraction. Only then compress to contact scores. A high norm correlation with poor singular-direction agreement means the models locate strong pairs but disagree about substitution chemistry. Good mean-block agreement with large dynamic variance means the pLM contains a reusable family-like component plus substantial context dependence. Poor mean agreement with useful contact ranking means the scalar benchmark has hidden an operator-level discrepancy.

The second comparison separates routes from responses. Measure attention or another internal route using its own stationary geometry, detailed-balance defect, and directed support. Then replace the route by its stationary reversibilization or a matched doubly stochastic control while holding value and output maps as fixed as the architecture permits. If masked-token loss and the categorical response change, the model functionally uses the removed circulation. If contact scores change while output responses remain stable, the intervention localizes the fragility to the attention interpretation rather than the model behavior.

The result can fail in several informative ways. Potts–pLM disagreement may reflect finite MSA sampling, pairwise misspecification, transfer failure, or response sampling. Off-manifold substitutions may dominate the pLM variance. MSA removal may alter confidence without altering the decoded edge operator. The estimand ledger determines which additional control is relevant; one aggregate correlation does not.

Case III: biological evidence should retain the predicted type

Suppose the model predicts a strong static operator on (i, j), a particular compensatory substitution direction, and a dynamic modulation by a third position k. Three increasingly rich experiments test different parts of that statement.

A structure supplies a distance or contact endpoint. It can test whether the position pair is geometrically associated, but not whether the predicted amino-acid direction is correct. A linked double-mutant panel supplies an epistasis table. After its own one-body terms are removed and its reference is aligned, it can test the categorical shape of the static operator. Repeating that panel in multiple backgrounds that vary k can test the predicted dynamic component.

The experiments need not agree. A spatially distant pair can carry a reproducible functional or compensatory operator. A close contact can show weak epistasis under one phenotype. A static mean can agree with double-mutant data while the predicted background modulation fails. Together these outcomes identify which edge semantics survive the intervention, replacing an abstract choice of the one “correct” graph with a typed empirical comparison.

Scale and units require their own care. Model logits, evolutionary energies, binding free energies, and measured fitness occupy different numerical scales. A link function between an inferred operator and an experimental table must be fitted on one subset and evaluated on held-out mutations. The word “energy” names their formal role only after the relevant physical or statistical construction has been specified.

What a clean failure looks like

A falsifiable theory permits each layer to succeed or fail independently. The outcomes can be read locally:

- failure in the additive synthetic limit rejects the implementation;
- recovery in the additive limit but failure under controlled third-order terms rejects the claimed higher-order audit;
- nonzero dynamic variance without planted-hyperedge localization above a dependence-destroying null demonstrates generic nonlinearity, not recovered higher-order structure;
- stable estimation with Potts–pLM disagreement rejects transfer, not the common coordinate;
- contact agreement with operator disagreement supports a scalar positional signal, not typed mechanism recovery;
- operator agreement in one background with failed replication rejects background invariance;
- route ablation without response change weakens the proposed internal mechanism while leaving the output association intact.

This locality is the practical value of the claim ladder. Evidence can stop at the strongest rung it supports without forcing the weaker results to disappear.

13.2 One residue pair, end to end

The opening chapter introduced five lines that can be drawn between the same positions (i, j). The following worked audit carries one hypothetical pair through the complete framework. Its deliberately synthetic outcome exposes the logic of interpretation without making an empirical claim about a particular protein.

Assume that one protein family provides a clade-partitioned MSA, a query structure, a pre-trained protein language model, and linked double-mutant measurements in two sequence backgrounds. The pair (i, j) is selected before any of these endpoints are compared.

13.2.1 Declare the comparison before fitting it

Training clades define smoothed anchor marginals p_i, p_j. They are then held fixed. A Potts model is fitted on the training MSA, and its block is projected into the anchor fibers. The pLM is audited on held-out natural and designed backgrounds by changing the source residue at j while measuring the target distribution at i. Every response table is transported to the same fibers before its mean, residual, norm, or singular directions are computed.

This order matters. If each method supplied its own reference, a difference in block norm could come from the reference rather than the relation. If blocks were scalarized first, agreement of strength could conceal disagreement about which amino-acid substitutions compensate.

13.2.2 A hypothetical outcome ledger

Suppose the audit produces four observations. The Potts block and the mean pLM block have similar dominant singular directions. Their scalar norms both rank (i, j) highly, and the query structure places the sites in contact. The pLM nevertheless has a substantial dynamic residual associated with backgrounds that vary a third site k, and its two directed mean responses retain a measurable reciprocity defect. Finally, the double-mutant table agrees with the predicted compensatory direction in one background but changes when k is altered.

Observation	Licensed conclusion	Conclusion still unavailable
Potts and pLM mean blocks align in categorical direction	the pLM contains a reusable family-like response component	equality of training laws or microscopic energies
Both scalar shadows rank the pair highly and the sites contact	the two estimators share a contact-predictive positional signal	equality of the full operators or a direct physical pair force
The pLM residual changes with site k	one static edge is insufficient under the audited backgrounds	one identified three-body physical mechanism
Directed means fail reciprocity	the response is not integrable into one additive pair energy under the tested reference	absence of useful directed computation
Double-mutant shape changes with k	background-dependent biological nonadditivity survives intervention	universal validity outside the tested assay and backgrounds

No single verdict summarizes this ledger. The pair is simultaneously a structural contact, a stable statistical relation to first approximation, a context-dependent model response, and a background-dependent experimental interaction. The failed reciprocity test blocks one stronger claim: these directed responses cannot be compressed into a single additive pair energy without discarding information.

13.2.3 Where the spectral analysis enters

Only after the operator graph has been assembled across many pairs is (i, j) placed in a larger network. A scalar norm may contribute to a contact-weighted Laplacian; a signed contraction may enter a signed Laplacian; the full categorical orientation may enter a connection Laplacian. Heat kernels or wavelets can then ask whether the pair belongs to a localized or domain-scale pattern. Those transforms inherit the reciprocity defect and the categorical resolution retained during scalarization.

The vertical audit therefore closes the path opened in Chapter 1:

$$\boxed{\begin{array}{l}\text{shared endpoints} \longrightarrow \text{typed observations} \longrightarrow \text{common operator coordinates} \\ \longrightarrow \text{declared compression} \longrightarrow \text{matched structural and mutational evidence.}\end{array}} \tag{13.3}$$

The framework places the five original edges in common coordinates, making their exact level of agreement and the first failed upgrade in interpretation visible in one report.

13.3 Biological validation is typed

The scalar graph and the full operator make different predictions. If $F_{ij}(a, b)$ is a measured log-fitness or quantitative phenotype for a double-mutant panel around reference residues (a_0, b_0), define experimental epistasis

$$\epsilon_{ij}(a, b) = F_{ij}(a, b) - F_{ij}(a, b_0) - F_{ij}(a_0, b) + F_{ij}(a_0, b_0). \tag{13.4}$$

The operator-level comparison is between $G_{ij}(a, b)$ and the corresponding projected epistasis table in a common reference, rather than only between M_{ij} and a binary contact label [28, 29]. Model logits, evolutionary energies, and measured fitness generally occupy different units and may be related nonlinearly. Their comparison therefore uses a scale or link function fitted on a training subset and evaluated on held-out mutations, rather than direct numerical equality.

Inferred quantity	Primary hypothesis	Independent endpoint
Declared symmetrization of $M^{\text{static}}_{i\leftarrow j}$	persistent positional association or contact candidate	held-out structure and distance-stratified contact precision
Full $\overline{G}_{i\leftarrow j}(a, b)$	substitution-specific compensation	linked, gauge-aligned double-mutant matrix
$M^{\text{dynamic}}_{i\leftarrow j}$ or $\rho^{\text{dyn}}_{i\leftarrow j}$	background-dependent response	the same mutations across multiple sequence backgrounds
P^{circ} or $\delta_{\text{db}}(P)$	nonreciprocal computational routing	masked-loss and response change under stationary reversibilization
Low scalar or connection eigenspace	coherent positional domain or transported substitution sector	held-out domain boundaries and multi-position mutational coherence
Higher-order residual	non-pairwise epistasis	designed triple-mutant or combinatorial assay

Evolutionary constraints can arise from folding, stability, binding, catalysis, regulation, and phylogenetic sampling. A structurally distal operator may therefore encode reproducible function or compensation. Contact prediction supports positional association; causal epistasis requires matched perturbational evidence [30, 31]. The validation endpoint must match the mathematical object.

Coda: evidence should preserve type

Validation is strongest when the endpoint matches the inferred object's degrees of freedom. Binary contact labels are appropriate for a scalar support hypothesis. Substitution-specific action calls for a linked mutational table. One double-mutant panel tests an operator near a reference sequence; replication across backgrounds tests its invariance. A downstream prediction gain establishes utility, while targeted intervention addresses the internal mechanism that produced it.

This discipline keeps several legitimate results from competing for one label. A model may learn useful evolutionary statistics, reproduce a family interaction field, exploit directed routing, and predict experimental epistasis to different degrees. Reporting those degrees separately is more informative than deciding whether the model has or has not "learned interactions."

The framework is now complete enough to return to the question that opened the book. When five descriptions draw a line between the same two residues, what kind of object, if any, do they share? They share a structured relation whose precise meaning is given by the coordinates and evidence that survive the translation.

Chapter 14

What Kind of Object Is a Protein Interaction?

The phrase *protein interaction* names a family of claims organized around perturbation, dependence, and constraint. At the weakest level, one part of a computation can influence another. At a stronger level, a function contains a nonadditive component after lower-order effects have been removed. Stronger still, compatible components may assemble into a probability law or effective Hamiltonian. Structural and biological interpretations require their own coarse graining and interventions.

This plurality is the reason a typed theory is needed.

14.1 What the operator graph contributes

The reference-centered operator graph provides a carrier that is richer than a scalar network and weaker than a universal physical Hamiltonian. Each residue position carries a centered categorical fiber. Each directed pair carries an operator from source contrasts to target contrasts. Support, context, interaction order, dynamics, normalization, estimation, and validation remain additional coordinates rather than being hidden inside the edge matrix.

This carrier is broad enough to compare Potts couplings with substitution responses, and precise enough to localize their agreement and disagreement. It distinguishes a stable mean operator from its background-dependent fluctuation. It separates reciprocity of an effective pair energy from reversibility of a route. It shows where scalar graphs, signed contractions, and connection Laplacians discard or preserve categorical structure. It connects linear gauge projection to finite probability geometry without treating a tangent coordinate as a complete joint law.

The resulting unification is coordinate-wise. Models meet in some coordinates and remain different in others. That is exactly the level at which comparison becomes scientifically useful.

14.2 What remains irreducibly plural

Several pluralities are structural.

Graph identity begins with an edge criterion. Route, statistical support, operator response, structural contact, and experimental epistasis graphs arise from different constructions. Their overlap is a result.

Softmax supplies a local negative-log coordinate for whichever variable is normalized. Site conditionals, memory posteriors, routing distributions, and graph-state posteriors therefore carry different local energies. Additional compatibility and equilibrium structure support a global Hamiltonian.

Low complexity has several axes. Sparse support, low categorical rank, pairwise interaction order, and a low-dimensional spectrum describe different reductions, each with its own scaling law and approximation error.

A deep sequence model supplies a distribution of responses across backgrounds. Its mean is one useful static representation, while its dynamic residual records the behavior left outside that summary.

14.3 The open mathematical problems

The framework leaves several problems deliberately unresolved.

From factorized edges to graph ensembles. Independent edge-state distributions provide local Gibbs coordinates but omit global topology. A coupled theory must determine which degree, reciprocity, loop, geometry, and chemical constraints belong in a graph Hamiltonian, and whether the resulting ensemble remains learnable at protein scale.

From directed responses to integrable dynamics. Reciprocity characterizes additive pair compatibility, but deep models generate context-dependent operators and higher-order residuals. The appropriate integrability conditions for these fields, and the meaning of their nonintegrable circulation, remain open.

From pair operators to higher-order bundles. Functional ANOVA identifies the mass beyond pairs, but an economical representation of typed hyperedges must balance support, categorical rank, gauge, and realizable computation. A pair graph records the pairwise projection of higher-order behavior, while typed hyperedges retain the remaining information.

From family inference to transferable mechanism. Agreement between a family Potts field and a pLM response mean would show transfer in common coordinates. Whether such agreement obeys a cross-family scaling law in MSA depth, alphabet, degree, phylogenetic distortion, or model scale is an empirical question with the shape of a statistical-physics problem.

From predictive operators to biological intervention. The most decisive evidence will preserve categorical type across multiple sequence backgrounds. Large, linked double- and higher-order mutant panels remain rare. Experimental design and operator estimation must therefore be developed together; reducing assay data to a final scalar benchmark would discard the structure the theory aims to test.

14.4 The edge revisited

Return to the pair (i, j) from the opening chapter. The structural contact, Potts block, attention route, substitution response, and experimental epistasis may eventually converge on one coherent account. Such coherence comes from a sequence of explicit correspondences: common categorical coordinates, controlled contexts, compatible directions, matched nulls, and interventions that retain the predicted structure.

The operator graph treats unity among the five edges as a hypothesis of correspondence. It makes agreement precise enough to be discovered and disagreement precise enough to be informative. That is the role of the framework, and the larger methodological claim of the book: unification becomes stronger when translation preserves the differences that matter.

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